

Staging as a Monitoring Device under Endogenous Deadlines*

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Abstract

We study a principal-agent problem where the principal dynamically chooses between two methods for solving a problem: a direct execution route and a multi-stage training route with an observable milestone. To mitigate moral hazard, the principal commits to an endogenously determined deadline. The optimal contract is shaped by the interplay of three forces: the milestone effect from the training route’s monitoring advantage, the deadline effect that favors the simpler execution route as time runs short, and the relative efficiency of each path.

A principal contracting for a project must decide not only how to reward its completion but how to organize the work. Splitting the project into stages creates a verifiable interim milestone and so makes the agent’s progress observable, but it changes the technology: two successes are now required instead of one, and completion may be slower in expectation. Monitoring is therefore not a signal purchased alongside the

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project but a property of the route chosen to complete it, and it is paid for in efficiency rather than in cash. We characterize the optimal contract when the principal chooses the route at every date alongside an endogenous deadline. Because the only use for observed progress is to extend that deadline, the value of monitoring is denominated in time: the principal stages a project when there is time to exploit the milestone and abandons staging as the deadline nears, so the recommended route reverses at most twice over the horizon. Staging is used throughout precisely when the project return exceeds $(e - 1)/(e - 2)$, whatever the arrival rates, the operating cost, and the agent's temptation to shirk.

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1 Introduction

A principal who contracts for a project chooses more than the terms on which she will pay for it. She also chooses how the work is to be organized, and that choice determines what she will be able to see. A research funder may award a single grant against a final result, or may write a milestone plan in which the performer is paid on completing a defined intermediate artifact. A software buyer may procure a system against one delivery, or against a schedule of demonstrations. An investor may release the whole of a commitment at closing, or hold part of it back against a benchmark. In each case the intermediate output is manufactured: it exists because an agreement asked for it, and the project could have been completed without ever producing it.

This makes monitoring a production decision. Splitting a project into stages generates a verifiable signal of progress, but it does so by changing the technology—two successes are now needed where one would have sufficed, and the divided route may be slower in expectation. The signal is not bought with cash on top of the project; it is bought with the efficiency of the route chosen to complete it.

One implication of this framing organizes everything that follows, and it is worth stating before the model. Under the undivided route the interim milestone is not concealed; it does not exist. There is no partially completed project that the agent might observe and decline to report, because a single success ends the game. Staging does not disclose progress that was there all along—it creates the state that can be disclosed. What the principal is choosing between is not two information structures over a fixed technology, but two technologies, one of which happens to be observable partway through.

The value of that observability is bounded, and by something specific. The principal's instrument is a deadline: limited liability means she cannot punish, so she must be able to stop, and every incentive-compatible contract abandons the project in finite time. A verified milestone is worth exactly the additional time it justifies granting. Monitoring's value is therefore denominated in time, and time runs out. *When intermediate progress*

exists only if the project is designed to produce it, monitoring is a production decision whose value is bounded by the deadline it can buy.

The model. A principal hires an agent to complete a project worth Π , incurring a flow cost c while it remains alive. At each instant she directs him along one of two routes. Under the *direct* route he works on the project as a whole and delivers it finished, at Poisson rate λ_D . Under the *staged* route he first produces a verifiable interim milestone and only then completes the project, each stage arriving at rate $\lambda_S > \lambda_D$. Effort is private: shirking yields the agent a flow benefit $\phi < c$, and limited liability confines the principal to nonnegative payments. She commits at date zero to a deadline, a route recommendation at every date, a schedule of completion payments, and—crucially—a continuation contract to take effect if the milestone arrives. Nothing restricts that continuation: whether the deadline is extended on a milestone, and by how much, is an outcome of the analysis rather than an assumption.

We first study the case in which the two routes are equally efficient, $\lambda_S = 2\lambda_D$, so that a principal who could dictate effort would be indifferent between them. Staging is then *technologically innocuous*, and any staging that survives in the optimal contract is there for incentive reasons alone. Section 5 relaxes this and allows staging to be slower.

Two forces. The routes differ in the shape of their completion times, and this matters before any question of monitoring arises. Under equal efficiency the two distributions share a mean, but the direct route’s is a mean-preserving spread of the staged route’s: it places more mass both near zero and far out, while the staged route concentrates completion around the middle. A deadline truncates the distribution, and near one, dispersion is what a principal wants—the probability of one arrival is first-order in the time remaining while the probability of two is second-order. The direct route therefore wins at short horizons and the staged route at long ones, on shape alone and with no signal in play. We call this the *deadline effect*, and it is the force that makes staging something a principal grows out of.

Against it runs the *milestone effect*. A verified first stage tells the principal that the agent

has worked, and the extension she can grant in response relaxes his incentive constraint. Under binding incentives that extension is exactly one expected stage, $1/\lambda_S$ —independent of the project’s value and of its running cost. The reason is that the extension is priced in incentive units rather than in value units: a longer runway raises what the principal must eventually pay to buy out the agent’s remaining right to divert, and lowers the premium she must attach to completion, and under innocuous staging these offset exactly. The agent buys time with a smaller bonus, at par.

Results. Under innocuous staging the optimal contract is determined by a single statistic, the project return $\pi \equiv \lambda_D \Pi / c$ (Theorem 1). Above a threshold π_S the principal stages throughout; below a lower threshold π_D she never stages; and in between the contract begins staged and switches to the direct route as the deadline approaches, an arrangement that reads as probation followed by conversion. The upper threshold is a pure number, $\pi_S = (e - 1)/(e - 2) \approx 2.392$, independent of the arrival rates, the operating cost, and the agent’s temptation to divert.

That constant is also the point at which this paper meets the closest existing one. [Green and Taylor \(2015\)](#) study contracting for a project that must pass through two stages, and their condition for incentives to bind throughout reduces, under innocuous staging, to exactly $\pi > \pi_S$. Above the threshold the two contracts coincide. Below it they diverge, and the divergence is informative: where their binding-incentive value function turns convex near zero, they restore concavity by slackening incentives, whereas here the direct route supplies it—the two branches paste smoothly, each is concave, and incentives bind everywhere. The same trigger, the same region, a different instrument. What separates the papers is not the contracting problem above π_S but the margin of choice: they take the two-stage structure as given and ask how to contract on progress, while the question here is whether to produce progress at all.

When staging is less efficient than the direct route, the design decision reverses at most

twice over the horizon, and it does so at every efficiency level (Theorem 2). The bound is not a tractability restriction; it is a property of the technology, and it is what makes the resulting taxonomy of six contract forms exhaustive. Two of the six are new. In the two-switch contract the principal begins on the direct route because the horizon is long and efficiency dominates, turns to staging once the deadline is near enough that concentrated completion times are worth their cost, and returns to the direct route as the deadline closes—staging occupies the middle of the relationship and neither end. In the reversed one-switch contract she begins direct and turns to staging only as the deadline nears.

What the results rest on. Two extensions locate the boundaries of the framing. Impatience acts on the same margin as inefficiency—both penalize the route whose completion lies further off—and generates the same taxonomy, so the two-switch form is not an artifact of the particular cost we assign to staging.** YG: Written as robustness, per the note in the chat. If you want claim 4 promoted to a contribution, this sentence moves up into the Results paragraph and gets its own two sentences. Decide once and keep the abstract consistent.** And what the analysis requires of the milestone is falsifiability rather than observability. If the principal cannot see which route the agent is working, nothing changes: binding incentives price a delivery on either route at the inverse of its arrival rate, so the flow value of working is the same on both and route choice is incentive-irrelevant. If she cannot see the milestone but the agent cannot fabricate one, nothing changes either. Only if he can fabricate one does the extension become constrained by what a liar could shirk out, and at that point the problem becomes Green and Taylor (2016)’s. The design framing holds exactly as far as verifiability does.

1.1 Applications

The environment just described is demanding, and it is worth being explicit about what it demands. Four conditions must hold before the analysis has anything to say. The principal

must have a genuine choice of route, so that staging is a decision and not a description. The interim milestone must be verifiable, so that it can be contracted on rather than merely believed. The direct route must have no interim state at all—not a hidden one, but none, so that staging *creates* the observable rather than disclosing it. And transfers must be contingent on the timing of delivery, since that is the instrument the principal uses.

The fourth condition is the restrictive one, and it is what rules out the staged clinical trial. Phase I, II and III exist whether or not a sponsor wishes them to; the sponsor does not choose to manufacture an interim state, and cannot elect to complete the drug in one step. The examples below are chosen because in each the interim output is manufactured—it exists because someone wrote it into an agreement, and it would not otherwise exist. Each is strongest on a different condition, and we present them that way rather than claim all four for any one of them.

Research contracting. The clearest instance of a principal choosing whether to buy a signal by restructuring the work is an agency that holds both instruments on its own menu. DARPA funds research through Other Transactions and through assistance instruments, and the two differ in exactly the dimension the model isolates. A proposer requesting an Other Transaction must supply a milestone plan, in which each milestone carries a description, completion criteria, a due date, and an associated payment.¹ “Completion criteria” is verifiability written into the contract’s own vocabulary, and the payment schedule is the timing contingency. The alternative is a cooperative agreement, where the agency may structure a program into phases with defined objectives and a period of performance, which it describes as enabling it to monitor progress and creating programmatic decision points.²

What makes this the leading case is the fourth condition. A research problem has no natural interim state. There is no partially completed result that a program manager could

¹<https://www.darpa.mil/about/offices/contracts-management/proposer-transactions>. Named programs make the same requirement: solicitations under Disruptioneering (DARPA-PA-25-07) call for fixed payable milestones, with payment released on successful completion of the accomplishments agreed in the plan.

²<https://www.darpa.mil/about/offices/contracts-management/proposer-grants>.

observe if only the performer would report it; the milestone comes into existence when the solicitation defines one, and defining one commits the performer to producing an intermediate artifact that is not itself the object of the program. Monitoring here is plainly a property of how the work is organized rather than a signal purchased alongside it.

Software delivery. Large software contracts are routinely staged around demonstrations of prototypes and presentations of work in progress, and the frequency of those intermediate outputs is a design variable rather than a technological constraint—the same system can be procured against a single final delivery, or against a schedule of gated releases. This is the setting in which verifiability and the efficiency cost of staging are both most concrete.

Verifiability is contractual: a milestone is specified by a deliverable, acceptance criteria, and a date, together with a review window after which an unanswered submission is deemed accepted.** YG: Still no citable source for the acceptance-criteria and deemed-accepted conventions. Green--Taylor cite DOD-STD-2167A for waterfall phases, which is precedent for citing a standard rather than a paper---worth asking whether anyone has a procurement text or standard we can point to. Leaving this sentence uncited is survivable; leaving it out weakens the paragraph.** The extension mechanism is contractual too. United States federal guidance on software procurement recommends award-term arrangements, under which a contractor earns an extension of the period of performance when the government determines that performance has been excellent, and fixed-price incentive arrangements that reserve part of the fee against the achievement of key milestones.³ An extension of the horizon granted on demonstrated progress is the instrument our optimal contract turns on, under another name.

The efficiency cost is concrete here in a way it is not elsewhere. A working prototype at month three exists only because the contract asked for one, and the work of packaging it into something demonstrable—stabilizing interfaces, writing throwaway scaffolding, preparing the demonstration—does not advance the final system. Staging is bought with delay, which is

³<https://techfarhub.usds.gov/solicitation/contract-design/>.

the efficiency loss we introduce in Section 5; the question there is how much delay is worth buying.

Venture financing. Venture capital is the setting in which both contractual forms are observably used, and in which the choice between them is most clearly a choice. [Kaplan and Strömberg \(2003\)](#) document that in roughly fifteen percent of financings the investors release only part of the committed capital at closing, with the remainder contingent on subsequent performance; the split is sharper in first rounds, where staged deals release about half of the commitment up front while the others release all of it. Staging is thus a minority practice, and one that investors select into rather than inherit. Their account of why it is used is that releasing less in a given round shortens the time until the venture must return for further financing, and it is the shortened horizon that preserves the investor’s ability to discontinue—so that a milestone-contingent release of capital is, in substance, a milestone-contingent extension of a deadline. Section 6.1 develops that correspondence once the optimal contract is in hand.

Two features of this literature bear on assumptions we make rather than on results we obtain. Staging intensity responds to the severity of the agency problem: [Gompers \(1995\)](#) finds that the interval between financing rounds contracts where assets are intangible, growth options large, and research intensity high. More useful to us, he confronts the natural rival explanation—that high-technology ventures simply pass through more milestones for technological reasons, so that observed staging reflects the production process rather than any incentive motive. Splitting his sample by technology cohort, he finds the estimated relationships close to identical across the two and cannot reject their equality. Variation in staging is not accounted for by variation in the underlying milestone structure of the technology. That is the closest empirical support available for treating staging as something a principal chooses for incentive reasons within a given technology, which is the benchmark of Section 3.

A caveat belongs in the text rather than in a footnote. What this evidence establishes is a choice of *contractual form*—staged against unstaged—holding the venture’s technology fixed. It does not establish that an investor reorganizes the underlying work so as to generate an observable interim output. Venture financing is therefore our strongest case that both forms are used and that a milestone buys time, and our weakest on the first of the four conditions; research contracting is the reverse. What the model supplies, in each of the three settings, is the missing half of an account that treats staging as a monitoring device and stops there. [Kaplan and Strömberg \(2003\)](#) close by observing that the choice between ex ante staged, longer-term contracts and short-term, ex post staged ones is an open question.

1.2 Related Literature

This paper belongs to a literature on contracting for projects that proceed in stages ([Hu, 2014](#); [Green and Taylor, 2016](#); [Wolf, 2018](#); [Moroni, 2022](#)). The closest study is [Green and Taylor \(2016\)](#), who consider a project requiring several successes and an agent whose effort cannot be observed. Our staged route is their baseline technology, and [Section 4](#) shows that above a threshold project return the two optimal contracts coincide; [Remark 1](#) takes up what separates them below it, and distinguishes the mechanism in their published model from the one in the working paper. The difference in question is not how progress is contracted upon but whether progress is produced: their two-stage structure is a feature of the project, whereas here it is an instrument the principal may or may not deploy, at every date. The nearest existing exercise is their treatment of stages of unequal difficulty, which asks how hard each stage should be made while holding the two-stage structure fixed. The margin here is prior to that one.

The value of the interim milestone is that it reveals the agent has worked, which connects this paper to work on monitoring in dynamic contracts ([Orlov, 2022](#); [Piskorski and Westerfield, 2016](#); [Dilmé and Garrett, 2019](#); [Marinovic and Szydlowski, 2022](#); [Varas et al., 2020](#); [Marinovic and Szydlowski, 2023](#); [Chen et al., 2020](#); [Wong, 2023](#)). The relationship

is also where this paper departs from that literature most sharply. There, the production technology is held fixed and the principal chooses how much of a separate signal to acquire, paying for it in cash or in real resources; the information exists and the question is how much of it to buy. Here there is no separate signal to buy. Under the direct route the agent’s progress is not merely unobserved, it is undefined, and the principal obtains a signal only by adopting a technology that generates one—which she pays for in expected completion time rather than in cash. The milestone is also more than a record of past effort: it changes what remains to be done, since a project one stage from completion faces a different arrival rate than one that has not begun. Monitoring here is thus a property of the route rather than an instrument attached to it.

[Carnehl and Schneider \(2023\)](#) study a two-armed bandit in which one arm requires a single success and the other several, so that a decision-maker chooses between the same two shapes of completion time. The agent there knows the arrival rates of the multi-stage arm and must learn the feasibility of the single-stage one through experimentation, and the horizon is exogenous; ours is a contracting problem in which the deadline is chosen. Despite these differences the two papers arrive at the same bound: the preferred approach reverses at most twice.

Staging is a long-standing subject in corporate finance, where its rationale is usually the preservation of an option to abandon. Capital is committed in tranches so that the financier may withdraw once the news turns bad ([Admati and Pfleiderer, 1994](#); [Bergemann and Hege, 1998](#)), or so that the entrepreneur’s inability to commit is mitigated as assets accumulate ([Neher, 1999](#)); [Cornelli and Yosha \(2003\)](#) show that the resulting incentive to overstate interim progress can be blunted by convertible securities. Our mechanism is distinct, and the distinction is worth stating because the settings overlap. In our environment the principal may already abandon at any moment—the deadline is an object she chooses, not one imposed on her—so the option to stop is fully priced before staging is considered. Staging nonetheless appears in the optimal contract, and appears in a regime where it confers no technological

advantage at all. What the milestone buys is not the right to stop but the ability to continue: it is proof of work, and its value is the additional runway that proof justifies. The concern of [Cornelli and Yosha \(2003\)](#) does reappear, in [Section 6.2](#), where an agent who can fabricate progress caps the extension the principal can safely grant.

The two stages of the staged route are perfect complements—the first produces no value on its own and the project is worth Π only when both are done—which relates the paper to work on complementary innovation and the sequencing of research ([Green and Scotchmer, 1995](#); [Gilbert and Katz, 2011](#); [Bryan and Lemus, 2017](#); [Poggi, 2021](#)).

[Kim and Poggi \(2026\)](#) study the same dual-route technology in a very different environment. Two firms race to develop a product, each dividing resources between developing with an established technology—a single step, at a lower rate—and researching a faster one, which must be discovered before development can begin. A firm that privately makes the interim discovery chooses whether to patent it or conceal it, and concealment is attractive because disclosure lets the rival re-optimize: a rival who learns the frontier technology exists abandons duplicative research and falls back to direct development, which raises its outside option and erodes the discoverer’s licensing rents.

No such motive arises here, and the reason is the identity of the audience. Their discoverer conceals from a rival who will act on what he learns. Our agent would be concealing from a principal who committed at $t = 0$, so her response to a milestone is written into the contract in advance and is favorable to him—the extension—rather than something he would prefer she not make. [Proposition 7](#) makes this precise: revealing dominates concealing at every history. Disclosure incentives turn on whether the observer can re-optimize, and a principal with full commitment cannot.

The two settings share a further feature. In their single-firm benchmark, a firm facing a high termination rate prefers the single-step route and a firm facing a low one prefers the two-step route—the deadline effect of [Section 3](#), with the rival’s success rate in place of a deadline. What is an exogenous constraint here is an equilibrium object there.

2 Setup

Environment. A principal (she) hires an agent (he) to complete a project. Work takes place in continuous time over an unbounded horizon, $t \in [0, \infty)$. Completing the project yields the principal a gross payoff $\Pi > 0$ and ends the game; until the project is completed or abandoned she incurs an operating flow cost $c > 0$. Both players are risk neutral, do not discount the future, and have outside options normalized to zero.⁴ The principal has unlimited resources with which to fund the project; the agent is protected by limited liability, so every transfer to him must be nonnegative.

Effort. The agent's action at time t is his effort $\beta_t \in [0, 1]$, where $\beta_t = 1$ is full effort and $\beta_t = 0$ is full shirking.⁵ Effort is his private information. Withholding the fraction $1 - \beta_t$ leaves him a private flow benefit $(1 - \beta_t)\phi$, and we assume $c > \phi > 0$, so that the cost of keeping the project alive exceeds the agent's gain from shirking.⁶

Routes. At each instant the principal directs the agent along one of two routes to completion. Under the **direct route**, the agent works on the project as a whole and delivers it in finished form; completion arrives at Poisson rate $\lambda_D \beta_t$. Under the **staged route**, he first produces an *interim milestone*—a prototype, a demonstration, a report on work in progress—and only then completes the project; each stage arriving at rate $\lambda_S \beta_t$. Figure 1 illustrates the two paths. We assume $\lambda_S > \lambda_D$: otherwise, the staged route would be slower stage for stage as well as in total, and would never be used.

Staging is not a technological necessity: the same project can be finished either way, and the interim milestone exists only because the contract calls for it. Instead, it generates

⁴** YG: discounting assumption will be discussed in the discussion**

⁵Both the arrival rate and the private benefit are linear in β_t , so the agent's payoff is linear in his action and a corner is optimal. It is therefore without loss to restrict attention to $\beta_t \in \{0, 1\}$. ** YG: TBA: See Appendix ?? for details.**

⁶The friction is familiar from models of cash diversion, e.g. DeMarzo and Sannikov (2006): the principal outlays c per unit of time, and during an instant of full shirking the agent privately captures ϕ of that outlay. In that reading ϕ/c is the diversion rate and $1 - \phi/c$ the resulting deadweight wedge.

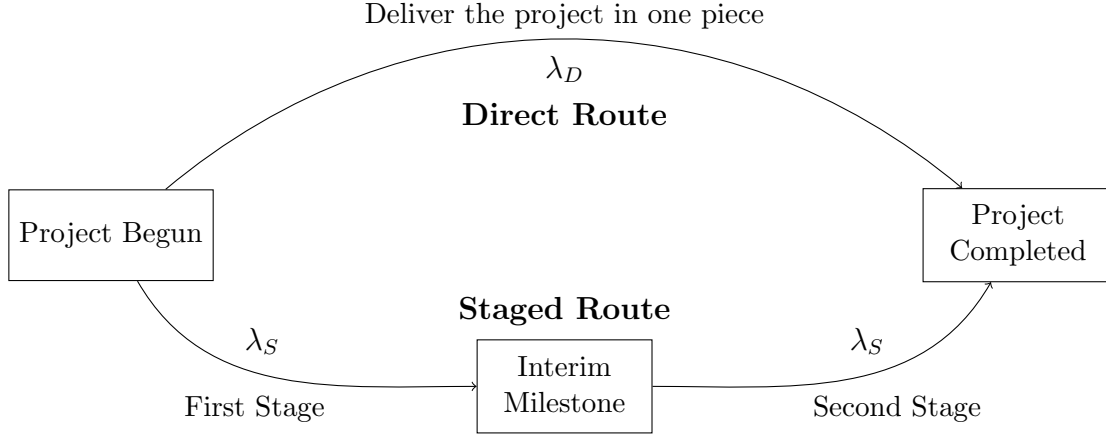


Figure 1: Routes to Project Completion

a verifiable intermediate output along the way. The milestone and the completed project are publicly observable and verifiable; the agent's effort is not. Payments may therefore be conditioned on which deliveries have occurred and when, and, absent any delivery, calendar time is the only payoff-relevant public information.

Efficiency. Absent any agency friction, and with no deadline in force, the choice between routes would turn on expected completion time alone. The direct route completes the project in expected time $1/\lambda_D$, at expected cost c/λ_D ; the staged route requires two successes and so takes $2/\lambda_S$, at expected cost $2c/\lambda_S$. The first best is therefore $\Pi - \min\{c/\lambda_D, 2c/\lambda_S\}$.

Sections 3 and 4 focus on the case in which the two routes are equally efficient, $\lambda_S = 2\lambda_D$. Staging is then *technologically innocuous*: a principal who could dictate effort would be indifferent between the routes, so any staging that appears in the optimal contract is there for incentive reasons alone. Section 5 relaxes this assumption, allowing staging to be costly: $\lambda_S < 2\lambda_D$.⁷

We also assume the project is worth undertaking:

$$\Pi > \frac{c}{\lambda_D} \quad \Longleftrightarrow \quad \pi \equiv \frac{\lambda_D \Pi}{c} > 1, \quad (2.1)$$

⁷If $\lambda_S > 2\lambda_D$, staging would be the more efficient route. We exclude that case because our interest is in the tension between the two routes: if the direct route drops out of the optimal contract entirely, the problem reduces to the single-route analysis of Green and Taylor (2015).

where π denotes the *project return*.

Deadlines. Limited liability makes deadlines indispensable. If the contract were never to terminate, the agent could shirk at every date and still be retained, collecting ϕ per unit of time in perpetuity; because payments cannot be negative, no finite reward for completion would deter him.⁸ Every incentive-compatible contract must therefore abandon the project in finite time, trading the surplus lost when a viable project is abandoned against the rents the principal must otherwise concede.

Contracts. At $t = 0$ the principal offers a contract and commits fully to its terms; if the agent rejects, both parties receive zero. We restrict attention to contracts that recommend full effort at every date, which is without loss of generality: because $c > \phi$, any interval on which the agent is meant to shirk can be excised, advancing all subsequent events and saving $c - \phi$ per unit of time.

A (deterministic) contract is an object $\Gamma \equiv \left\{ T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \leq t \leq T} \right\}$, whose components are defined at calendar time t as follows.⁹

1. $T \in \mathbb{R}_+$: the deadline at which the project is abandoned if no delivery has occurred.
2. $a_t \in \{0, 1\}$: the route recommended at t , where $a_t = 1$ denotes the *direct* route and $a_t = 0$ the *staged* route.
3. $R_t \geq 0$: the payment to the agent if he completes the project at t without having delivered an interim milestone.¹⁰
4. $\hat{\Gamma}^t \equiv \{T^t, \{R_s^t\}_{t \leq s \leq T^t}\}$: the continuation contract taking effect if the milestone arrives at t , consisting of a revised deadline $T^t \geq t$ and payments $R_s^t \geq 0$ for completing the

⁸Absent limited liability the principal could implement the first best by selling the project to the agent upfront at a price equal to its first-best value derived above, and leaving him the residual. Limited liability implies that the agent lacks the resources to make such a payment.

⁹Randomization cannot improve upon this class, though the argument requires care; see Remark 2.

¹⁰Since both parties are risk neutral and do not discount, all payments may be backloaded to the moment of a delivery without loss of generality (see, e.g., Ray, 2002). Nonnegativity is limited liability.

project at s .

Any delivery inconsistent with the recommended route—a milestone under the direct route, or a completed project with no milestone on record under the staged route—terminates the contract without payment, so the agent follows the recommendation. Note that T^t is unrestricted: that the principal extends the deadline, and by how much, is a result rather than an assumption.

3 Observable Effort with Exogenous Deadlines

Before turning to the contracting problem under agency, we examine how the principal deploys the two routes when the agent’s effort is observable and contractible, focusing on the benchmark case where staging is technologically innocuous ($\lambda_S = 2\lambda_D$). Because effort can be directly mandated, no obedience (incentive) constraint arises, and route choice is governed purely by the technological trade-offs between the two paths. Absent a deadline, observable and contractible effort delivers the first-best value, and under equal efficiency, the principal is indifferent between the two routes.

An interesting trade-off arises in the presence of an exogenous deadline T . Suppose further that the deadline is extended by $\Delta \geq 0$ upon delivery of the interim milestone prior to T .¹¹ The principal thus faces two effective deadlines: T if no milestone has been delivered, and $T + \Delta$ if one has. These deadlines introduce inefficiency: whenever the project is worth undertaking (i.e., when (2.1) holds), efficiency demands running it until completion without truncation.

Given (T, Δ) , the principal chooses a route policy $a : [0, T] \rightarrow \{0, 1\}$ specifying which route to use at each calendar date. Her goal is to select an optimal policy that maximizes her expected payoff. This analysis provides key intuition for the optimal contract under

¹¹The case $\Delta = 0$ is permitted, so that staging confers no additional runway. One could also allow $\Delta < 0$; we show in Section 4 that under agency it is optimal to extend, not curtail, the deadline upon delivery of the milestone.

moral hazard and is also of independent interest given the ubiquity of deadlines in project management.

3.1 Deadline and Milestone Effects

To build intuition for route choice, this subsection elaborates two fundamental forces that govern the principal's trade-off under deadlines: the *deadline effect* and the *milestone effect*. To illustrate these mechanisms, we compare two benchmark policies that never switch: the *direct-only* policy ($a_t = 1$ for all $t \in [0, T]$), under which the agent works on the project as a whole throughout, and the *staged-only* policy ($a_t = 0$ for all $t \in [0, T]$), under which he works toward the milestone and then, if delivered, toward project completion. While the principal's expected payoff also reflects operating costs incurred prior to completion or abandonment—which depend on the chosen policy—the probability of completing the project in time plays a central, decisive role in shaping her preference.

Under the direct-only policy, a single success is needed by date T , yielding the completion probability

$$\mathcal{P}^D(T) \equiv \int_0^T \lambda_D e^{-\lambda_D \tau} d\tau = 1 - e^{-\lambda_D T}. \quad (3.1)$$

Under the staged-only policy, the milestone must arrive at some date $\tau_m \leq T$, and the project must then be completed by the extended deadline $T + \Delta$, yielding the completion probability

$$\mathcal{P}^S(T, \Delta) \equiv \int_0^T \left[\int_{\tau_m}^{T+\Delta} \lambda_S e^{-\lambda_S(\tau-\tau_m)} d\tau \right] \lambda_S e^{-\lambda_S \tau_m} d\tau_m = 1 - (1 + \lambda_S T e^{-\lambda_S \Delta}) e^{-\lambda_S T}. \quad (3.2)$$

Each route possesses a distinct advantage. Setting $\Delta = 0$ isolates the technological trade-off by stripping the staged route of any deadline extension.

Lemma 1 (Deadline effect). *Suppose $\Delta = 0$. There exists $\check{T} > 0$ such that $\mathcal{P}^D(T) > \mathcal{P}^S(T, 0)$ for all $T < \check{T}$, and $\mathcal{P}^S(T, 0) > \mathcal{P}^D(T)$ for all $T > \check{T}$.*

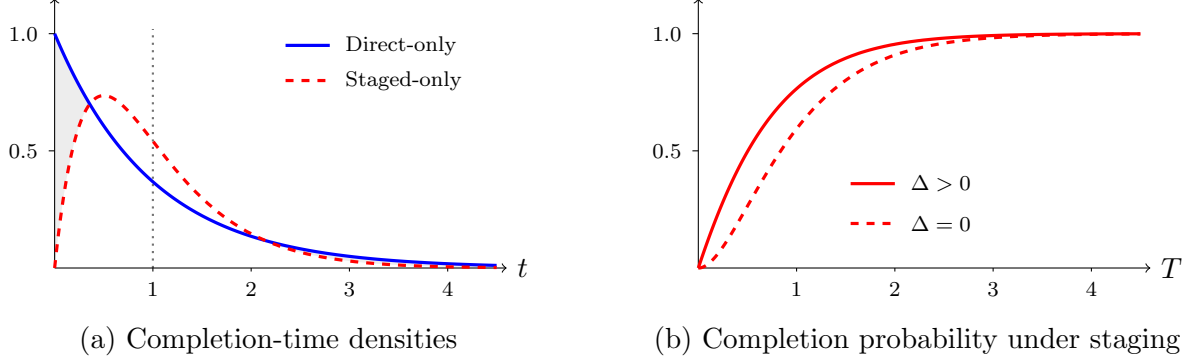


Figure 2: Completion times under equal efficiency ($\lambda_D = 1$, $\lambda_S = 2$)

The direct route's advantage lies in requiring only a single arrival. Over a short horizon, the probability of one arrival is first-order in remaining time, whereas the probability of two arrivals is second-order; when time is short, needing one slow success dominates needing two fast ones. Conversely, the staged route's advantage is that its two individual stages are faster. Over a long horizon, achieving two fast successes is more likely than achieving one slow one.

Panel (a) of Figure 2 illustrates the completion-time densities for both routes. Under equal efficiency ($\lambda_S = 2\lambda_D$), both densities share the same mean (marked by the dotted line). However, the direct route places more mass both near zero and in the far tail (shaded area), whereas the staged route concentrates completion tightly around the mean. Lemma 1 formalizes the observation that the direct route's completion time is a mean-preserving spread of the staged route's: a tight deadline penalizes dispersion (favoring the direct route), whereas a generous deadline rewards concentration (favoring the staged route). We refer to this force as the *deadline effect*.

The deadline extension Δ introduces a second, distinct advantage for the staged route. Because $\mathcal{P}^S(T, \Delta)$ is strictly increasing in Δ while $\mathcal{P}^D(T)$ is independent of it, any $\Delta > 0$ shifts the payoff balance in favor of staging. Panel (b) plots the staged completion probability with and without an extension. This advantage stems not from technology, but from information: the milestone reveals to the principal that concrete progress has occurred, and the extension Δ represents the value of that information. We refer to this force as the *milestone*

effect.

3.2 Optimality of One-Switch Policy

Building on the intuition of the deadline and milestone effects developed in Section 3.1, we now introduce a class of policies that dynamically leverages the respective advantages of both routes. The previous discussion suggests that the staged route is highly effective when the deadline is distant (capitalizing on the milestone effect), whereas the direct route is preferable when the deadline is nigh (capitalizing on the deadline effect). This naturally motivates a policy that begins with staging and falls back to the direct route if time begins to run short.

Definition 1. A policy is called a *one-switch policy* if there exists an intermediate deadline $S \in [0, T]$ such that: (i) the principal chooses the staged route up to S ($a_t = 0$ for $t \leq S$); (ii) if the interim milestone is delivered before S , she proceeds with the second stage until the extended deadline $T + \Delta$; and (iii) if the milestone is not delivered by S , she switches to the direct route until the original deadline T .

Note that the benchmark policies of the previous section also takes a form of one-switch policy: the direct-only policy sets $S = 0$ and the staged-only policy sets $S = T$.

The following proposition establishes that the optimal policy invariably takes the form of a one-switch policy. We prove this result utilizing a guess-and-verify approach to solve the principal's dynamic optimization problem.

Proposition 1. *Suppose that the two routes are equally efficient ($\lambda_S = 2\lambda_D$). When the principal faces a deadline T and a deadline extension Δ contingent on milestone delivery, the optimal policy is characterized as follows:*

- (a) (**Long extension**) If $\Delta \geq \bar{\Delta} \equiv \frac{1}{\lambda_S} \log \left[\frac{2\pi-1}{\pi-1} \right]$, the staged-only policy is optimal.
- (b) (**Short extension**) If $\Delta < \bar{\Delta}$, set $\hat{T} \equiv 2(\bar{\Delta} - \Delta) > 0$. Then:

(i) when $T < \hat{T}$, the direct-only policy is optimal;

(ii) when $T > \hat{T}$, the one-switch policy with the intermediate deadline $S = T - \hat{T}$ is optimal.

This proposition implies that the direct route is never used when the deadline extension is sufficiently generous. On the other hand, when the deadline extension is relatively short, there exists a critical time threshold $\hat{T}$ such that the principal switches to the direct route when fewer than $\hat{T}$ units of time remain before the deadline.

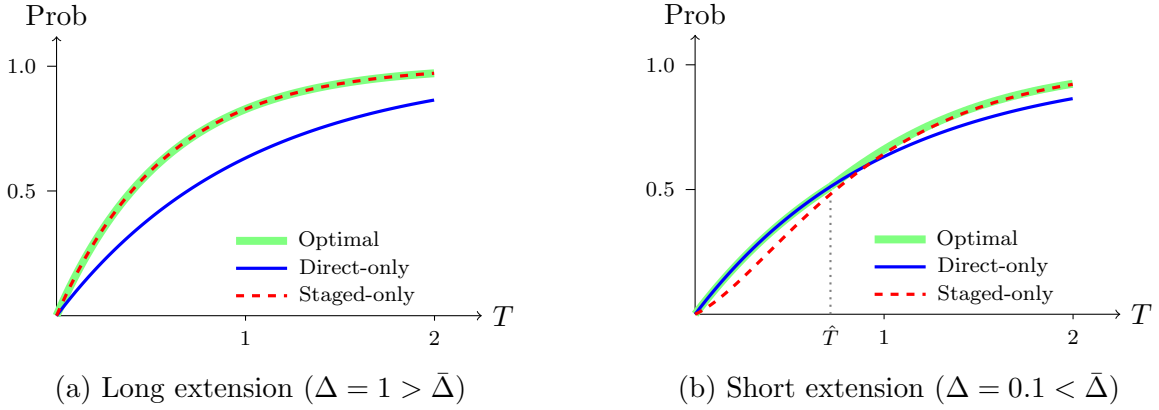


Figure 3: Completion probabilities ($\lambda_S = 2$, $\lambda_D = 1$, $\Pi = 3$, $c = 1$; $\bar{\Delta} \approx 0.46$, $\hat{T} \approx 0.72$)

Figure 3 plots completion probabilities against the deadline for the two benchmark policies and for the optimum. When the extension is long (Panel 3a), the milestone effect outweighs the deadline effect at every horizon: the staged-only policy completes with higher probability for all T , and the optimum coincides with it. When the extension is short (Panel 3b) the two benchmarks cross. The direct route is more likely to deliver when the deadline is near and the staged route when it is distant, and the optimum follows whichever is better—running the direct route alone when $T < \hat{T}$, and staging first when $T > \hat{T}$.

The cutoff $\hat{T}$ is an amount of time remaining, not a fraction of the horizon: the principal stages until $\hat{T}$ units are left, whatever the deadline. It also has a simple form, $\hat{T} = 2(\bar{\Delta} - \Delta)$, so it shrinks to zero as the extension approaches $\bar{\Delta}$ and the two cases of Proposition 1 meet

continuously. Note finally that the optimum is chosen to maximize surplus rather than completion probability; when the project return is low the principal accepts a lower chance of completion in exchange for a shorter expected run, and the optimal policy then completes less often than staging throughout.

Comparative statics of the extension cutoff Proposition 1 establishes a critical cutoff for the deadline extension, $\bar{\Delta}$, such that the staged-only policy is optimal whenever the extension exceeds this threshold; otherwise, the direct route is employed as the deadline draws near.

Observe that taking the derivative of this cutoff with respect to the project return yields

$$\frac{\partial \bar{\Delta}}{\partial \pi} = \frac{1}{\lambda_S} \left[\frac{2}{2\pi - 1} - \frac{1}{\pi - 1} \right] = -\frac{1}{\lambda_S(2\pi - 1)(\pi - 1)} < 0,$$

which is strictly negative since $\pi > 1$ by assumption (2.1). This establishes the following lemma.

Lemma 2. *Suppose that $\lambda_S = 2\lambda_D$ and $\pi > 1$. The extension cutoff $\bar{\Delta}$ is strictly decreasing in the project return π .*

The comparative static reflects how the two terms in the principal's objective scale with the project return. Expected surplus is Π times the completion probability less c times the expected time in operation, and the staged route trades one against the other: over a long horizon it completes more often, by the milestone effect, but it also runs longer, since a first success that is never followed by a second consumes time and delivers nothing. Raising π raises what the completion probability is worth without changing what the time costs, so the balance tilts toward the route that completes more often. A smaller extension then suffices to make staging worthwhile at every horizon, and $\bar{\Delta}$ falls.

4 Agency

Section 3 dispensed with the agency problem: with effort observable and contractible, the principal simply mandated it and route choice was governed by technology alone. We now restore moral hazard. Effort is private, so the principal must concede rents to deter shirking, and by the argument in Section 2 those rents can be controlled only through a deadline.

The central finding is that the contracting problem inherits the structure of the benchmark exactly. The optimal contract is the benchmark policy run with a deadline of u/ϕ and an extension of $1/\lambda_S$, where u is the utility the principal has promised the agent. Route choice under agency is therefore driven by the same deadline and milestone effects as before, and the one genuinely new question is how much time the principal chooses to buy.

4.1 The Optimal Contract

Throughout this section staging is technologically innocuous, $\lambda_S = 2\lambda_D$. As in the tangible-progress case of [Green and Taylor \(2015\)](#), the optimal contract rests on three pillars: (i) the contract carries a finite deadline; (ii) the reward for completing the project declines linearly in calendar time; and (iii) the deadline is extended by exactly $1/\lambda_S$ when the interim milestone is delivered.¹²

The reward schedules below anticipate that derivation. Both decline at rate ϕ , the agent's flow benefit from shirking, and both exceed the agent's continuation utility by a premium— ϕ/λ_D on the direct route, ϕ/λ_S on each stage of the staged route—which is what makes an instant of work at the relevant Poisson rate worth as much to him as an instant of shirking. They are, in other words, the unique schedules leaving the agent exactly indifferent at every date.

Since the contract carries a deadline and extends it upon delivery of the milestone, Proposition 1 suggests that the recommended route should switch at most once, from staged

¹²Section 4.3 derives all three. What distinguishes the present analysis from [Green and Taylor \(2015\)](#) is the presence of two routes and the principal's choice of which to recommend at each date.

to direct. We accordingly define the following class of contracts.

Definition 2. A contract is a *one-switch contract* with final deadline T and intermediate deadline $S \in [0, T]$ if

- (i) the agent is recommended to work on the staged route up to S ;
- (ii) if the interim milestone is delivered before S , the deadline is extended by $1/\lambda_S$ and the reward for completing the project at t is $R_t^S \equiv \phi(T - t + 2/\lambda_S)$;
- (iii) if the milestone is not delivered by S , the direct route is employed until T , and the reward for completing the project at t is $R_t^D \equiv \phi(T - t + 1/\lambda_D)$; and
- (iv) the contract is terminated if the project is not completed by the relevant deadline— $T + 1/\lambda_S$ in case (ii) and T in case (iii).

When $S = T$ we call the contract a *staged-only contract*, and when $S = 0$ a *direct-only contract*.

Written this way, a one-switch contract looks unlike the arrangements one observes in practice, but the resemblance is closer than it appears: Section 6.1 re-expresses it as a probation period ending in an up-or-out decision, and as a runway that a verified milestone extends. The following theorem shows that the optimal contract indeed takes one of these forms, and that which one obtains is determined by the project return alone.

Theorem 1 (Optimal contract). *Suppose staging is technologically innocuous ($\lambda_S = 2\lambda_D$), and define*

$$\pi_F \equiv 1 + \frac{\phi}{c}, \quad \pi_D \equiv \frac{1}{e} \left[e + \frac{\phi}{c} + \sqrt{\frac{\phi}{c} \left(e + \frac{\phi}{c} \right)} \right], \quad \pi_S \equiv \frac{e-1}{e-2} \approx 2.392. \quad (4.1)$$

Then $\pi_F < \pi_D < \pi_S$, and the optimal contract can be implemented as follows:

- (a) (**Stage throughout**) when $\pi > \pi_S$, a staged-only contract is optimal;

- (b) (**Stage, then switch**) when $\pi_S > \pi > \pi_D$, a one-switch contract is optimal;
- (c) (**Never stage**) when $\pi_D > \pi > \pi_F$, a direct-only contract is optimal; and
- (d) (**Abandon**) when $\pi < \pi_F$, the project is infeasible.

Two features of (4.1) are worth drawing out. First, the thresholds depend on the primitives λ_D , λ_S , c , Π and ϕ only through the project return π and the diversion rate ϕ/c . The upper threshold π_S is moreover a constant: staging is used throughout precisely when the project return exceeds $(e - 1)/(e - 2)$, whatever the arrival rates, the operating cost, and the agent's temptation to shirk. Second, the ordering $\pi_F < \pi_D < \pi_S$ is not an additional assumption but follows from $\phi < c$: the diversion rate is below one, which caps π_D at $(e + 1 + \sqrt{e + 1})/e \approx 2.077$ and keeps π_F below 2.

The same constant appears in Green and Taylor (2015). Their condition (C.2) for the incentive condition to bind everywhere is $\Pi \geq (2c/\lambda_S)(e - 1)/(e - 2)$, which reduces to exactly $\pi > \pi_S$ when staging is technologically innocuous. Above that threshold the two contracts coincide—the staged route is used throughout, the deadline is extended by one expected stage, and the direct route is never recommended—and below it they diverge, for reasons taken up in Remark 1.

The comparative statics match the benchmark. Lemma 2 showed that the extension needed to sustain staging falls as the project return rises, so staging should be used more the more valuable the project is; Theorem 1 confirms this, and sharpens it by locating the two thresholds at which the contract changes form.

The remaining subsections derive the result. Section 4.2 introduces promised utility and the incentive constraints, Section 4.3 characterizes the principal's value function, and Section 4.4 completes the proof.

4.2 Incentives and the Shape of the Contract

We track the agent's *promised utility*—the expected payoff he is owed under the contract from a given date onward—and use it as a state variable (Spear and Srivastava, 1987). Write u_t for the promised utility at t when the milestone has not yet been delivered, and u_M^t for the promised utility that takes effect if it is delivered at t .

Incentive compatibility Consider the agent at date t with promised utility u_t , and suppose the direct route is recommended. Working over $[t, t + dt)$ completes the project with probability $\lambda_D dt$ and pays R_t , but completion ends the relationship and so forfeits u_t ; the expected gain is $\lambda_D(R_t - u_t) dt$. Shirking pays ϕdt and retains u_t . Working is therefore optimal if and only if

$$R_t \geq u_t + \frac{\phi}{\lambda_D}. \quad (\text{IC}_D)$$

Under the staged route the agent is compensated not in cash but by a jump in promised utility, from u_t to u_M^t , and the milestone arrives at rate λ_S . The same comparison gives

$$u_M^t \geq u_t + \frac{\phi}{\lambda_S}. \quad (\text{IC}_S)$$

Both conditions say the same thing: whatever the agent is given for succeeding must exceed what he already holds by the premium ϕ/λ . The slower the route, the larger that premium, because success is rarer.

Promised utility is a time budget Following the standard approach in dynamic contracting, we proceed by guess-and-verify: we impose equality in (IC_D) and (IC_S) , derive the contract and value function that follow, and confirm in Section 4.3 that binding incentives are indeed optimal. The verification is not a formality. In Green and Taylor (2015) the corresponding condition is slack over a range of promised utilities; Remark 1 explains why the availability of the direct route rules this out here. Imposing equality pins down the entire

contract, and the mechanism is worth isolating.

When incentive conditions bind the agent is indifferent between working and shirking at every date, so his promised utility must fall at exactly the rate he could earn by shirking:

$$\dot{u}_t = -\phi, \quad \text{or equivalently} \quad u_t = u_0 - \phi t. \quad (4.2)$$

Promised utility is therefore not merely a state variable but a clock. It runs down at the constant rate ϕ , and the contract must end when it reaches the agent's outside option of zero, which happens at

$$T = \frac{u_0}{\phi}.$$

The principal is thus not really choosing a promised utility at all: she is choosing how long to run the project, and u_0 is that length denominated in utils rather than in time. One unit of promised utility buys $1/\phi$ units of the agent's time.

Three features of the contract follow. First, *the deadline is finite*, since u_t declines at a constant rate from a finite starting point; it is not an instrument the principal adds to the contract but what promise-keeping leaves behind once incentives bind. Second, *rewards decline linearly*. Substituting $u_t = \phi(T-t)$ into (IC_D) at equality gives $R_t^D = \phi(T-t+1/\lambda_D)$, the schedule of Definition 2, with the post-milestone schedule $R_t^S = \phi(T-t+2/\lambda_S)$ derived in Appendix B; rewards fall because the agent's outside option, shirking out the remaining clock, shrinks as the deadline approaches. Third, and least obviously, *the extension is exactly $1/\lambda_S$* . Binding (IC_S) requires promised utility to jump by ϕ/λ_S on delivery, and converting that jump into time at the exchange rate $1/\phi$,

$$t + \frac{u_M^t}{\phi} = t + \frac{u_t}{\phi} + \frac{1}{\lambda_S} = \frac{u_0}{\phi} + \frac{1}{\lambda_S} = T + \frac{1}{\lambda_S},$$

so the deadline is pushed back by one expected stage, whenever the milestone arrives.

That the extension is independent of the value of the project and the cost of running

it is initially surprising. Incentive compatibility prices the milestone against the agent's alternative, and the only rates entering that comparison are λ_S , at which effort produces the milestone, and ϕ , at which shirking produces private benefit; since ϕ is also the exchange rate between utility and time, it cancels. What Π and c determine is not the price of a milestone but how much time the principal buys, namely u_0 . A more valuable project is run longer, not rewarded more generously per milestone. Section 4.3 exploits this separation.

Remark 1 (Incentive conditions always bind). When $\pi < \pi_S$, the first-stage incentive condition in Green and Taylor (2015) goes slack over a range of promised utilities near zero. Under binding incentives the value function is convex there, because a first-stage delivery is worth nothing to the principal without its counterpart and with little time left she preserves it for the second stage rather than the first. Slack incentives supply the missing concavity.

Here the second route does that work. As promised utility falls the remaining horizon shrinks, and a route requiring two successes becomes unattractive relative to one requiring a single success; at the switching point $\hat{u}_1$ of Section 4.3 the principal turns to the direct route. This is the deadline effect of Section 3, operating at the point where their principal instead preserves time for the second stage. The two branches, each constructed under binding incentives, join smoothly at $\hat{u}_1$ and each is strictly concave, so the pasted function is concave without slack. Concavity is supplied by a second technology, not by slack incentives, and the incentive conditions bind everywhere.

4.3 The Principal's Value Function

Section 4.2 established that a contract is described by the promised utility the principal starts with, and that this quantity is a time budget. We now ask what she earns from a given budget, and show that the answer is the benchmark surplus of Section 3, net of what she pays for it.

For a contract Γ , let $P_0(\Gamma)$ and $U_0(\Gamma)$ denote the expected payoffs of the principal and

the agent at $t = 0$ when the agent works throughout. Define the principal's value function

$$\mathcal{V}(u) \equiv \max \{P_0(\Gamma) : U_0(\Gamma) = u, \Gamma \text{ incentive compatible}\},$$

the most she can earn while delivering the agent exactly u ; a contract attaining this maximum is said to *implement* the pair $(\mathcal{V}(u), u)$. Once $\mathcal{V}$ is known the remaining problem is one-dimensional:

$$\bar{u} \equiv \arg \max_{u \geq 0} \mathcal{V}(u), \tag{MP}$$

and the optimal contract is whichever contract implements $(\mathcal{V}(\bar{u}), \bar{u})$. Section 4.4 solves (MP); this subsection characterizes $\mathcal{V}$.

Aligned incentives Once incentive conditions bind, promised utility evolves mechanically—declining at rate ϕ and jumping by ϕ/λ_S upon delivery—whichever route is recommended. Route choice therefore affects only the probability and timing of completion, and so the surplus the relationship generates, not what the principal owes for it. Route choice under agency is thus the same as under observable effort. This is not obvious: one might expect the principal to favor whichever route lets her extract more. She cannot, because (IC_D) and (IC_S) fix the price of the agent's cooperation without discriminating between routes.

Two things are still not hers to choose: binding incentives tie the deadline to u/ϕ and the extension to $1/\lambda_S$. But that is precisely the problem of Section 3, which took both as given and asked how the routes should be used.

Proposition 2 (Value function). *Let $\mathcal{W}^*(T, \Delta)$ denote the optimal expected surplus in the benchmark of Section 3 with deadline T and extension Δ . Then*

$$\mathcal{V}(u) = \mathcal{W}^*\left(\frac{u}{\phi}, \frac{1}{\lambda_S}\right) - u, \tag{4.3}$$

and $\mathcal{V}$ is concave.

Proposition 2 also discharges the guess made in Section 4.2. For each promised utility, Appendix B.2 computes the principal’s gain from deviating to a reward, or to a post-milestone promise, more generous than incentives require, and Lemma 10 shows that this gain is never positive on either route. Leaving slack is therefore never profitable, and (IC_D) and (IC_S) bind at every u .

The principal’s problem is thus the benchmark problem in disguise: u/ϕ converts promised utility into a deadline, $1/\lambda_S$ is the extension that binding incentives purchase, $\mathcal{W}^*$ is the surplus the relationship then generates, and subtracting u leaves the principal her share. Moral hazard does not change what the routes are good for. It changes the terms on which she may buy time, and fixes the extension at a level she would not otherwise have chosen.

Two consequences follow. First, Proposition 1 transfers wholesale: because $\mathcal{W}^*$ is generated by a policy with at most one switch, the optimal contract involves at most one switch as well, as Definition 2 anticipated; Appendix B.2.1 makes this explicit by exhibiting, for each u , a one-switch contract that implements $(\mathcal{V}(u), u)$. Second, the deadline effect survives agency intact. Little time remaining now means low promised utility, so the direct route is used precisely when the agent is owed little—late in the relationship, or throughout when the project return is low.

4.4 Choosing the Deadline

It remains to solve (MP): how much time should the principal buy? Because $\mathcal{V}$ is concave, the problem has a unique solution $\bar{u}$, and the deadline is $\bar{u}/\phi$.

Extending the deadline by one instant costs the rent the principal must concede to keep the agent working over it, which by (4.2) is ϕ per unit of time whatever the state of the project; the cost of time is linear. The benefit is the added chance of completing in the extra instant, and it falls as the deadline lengthens, because a project still unfinished at a distant date is one on which several arrivals have already failed to materialize. A linear cost against a diminishing benefit gives an interior optimum.

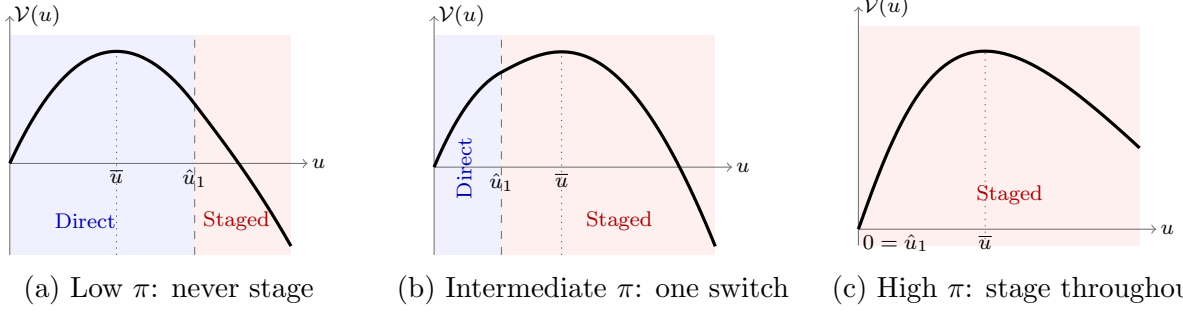


Figure 4: The principal's value function under technologically innocuous staging

Two quantities then determine the form of the contract: the optimum $\bar{u}$, and the switching point $\hat{u}_1 \equiv \phi \hat{T}$ inherited from the benchmark, below which the direct route is used. Since the contract begins at $\bar{u}$ and runs down to zero, it visits the staged region if and only if $\bar{u} > \hat{u}_1$. Figure 4 plots $\mathcal{V}$ for three levels of the project return, the shaded regions recording the route recommended at each promised utility. A contract starting at $\bar{u}$ traverses its panel from right to left, and the routes it uses are those it passes through.

By Proposition 1(a) the direct route is never used when the extension is at least $\bar{\Delta}$, and binding incentives fix the extension at $1/\lambda_S$. The direct route therefore disappears when $\bar{\Delta} \leq 1/\lambda_S$, which by Lemma 2 happens for high project returns. Appendix B.3 solves $\bar{\Delta} = 1/\lambda_S$ for the threshold π_S of (4.1); the arrival rate appears on both sides and cancels, which is why π_S is a constant. For $\pi > \pi_S$ we have $\hat{u}_1 = 0$, the staged region is everything, and a staged-only contract is optimal. This is part (a), illustrated in panel (c).

For $\pi < \pi_S$ the switching point is positive and the comparison with $\bar{u}$ becomes substantive. Panels (a) and (b) both have a kink at $\hat{u}_1$ and differ only in whether the maximum lies to its right. In panel (b) it does, so the contract starts in the staged region and crosses into the direct region as the deadline approaches; in panel (a) it does not, and the milestone is never sought. Because $\bar{u}$ is increasing in the project return while $\hat{u}_1$ is decreasing in it, the two cross exactly once, at the threshold π_D derived in Appendix B.3. For $\pi \in (\pi_D, \pi_S)$ the contract involves a switch, which is part (b); for $\pi < \pi_D$ only the direct route is used, which is part (c).

Finally, the project is worth undertaking only if $\bar{u} > 0$, and the condition has a transparent reading at the deadline. As u approaches zero the direct route is in use and the reward approaches ϕ/λ_D , so the principal's flow profit over the final instant is positive if and only if $\pi > \pi_F \equiv 1 + \phi/c$. She must cover both the operating cost c and, because incentive conditions bind, the shirking benefit ϕ for every instant the contract runs. Agency thus raises the hurdle for undertaking the project from $\pi > 1$ to $\pi > 1 + \phi/c$, and the increment is exactly the diversion rate. Below π_F the project is abandoned, which is part (d).

These four cases exhaust the possibilities. By Lemma 11 the thresholds are ordered $\pi_F < \pi_D < \pi_S$, so every project return falls into exactly one of them, and Theorem 1 follows.

Remark 2 (Randomization). The optimal contract is deterministic, and Theorem 1 is established against the larger space in which the principal randomizes over the terms and need not disclose the realization to the agent. Concealment is the substantive case: an agent who does not know which contract he faces cannot tailor a deviation to it, so a mixture of contracts that are individually not incentive compatible may be incentive compatible as a whole. That channel is closed here. The likelihood of any public history factors into a part governed by the agent's effort and a part governed by the device, so his beliefs are unaffected by his own deviations, and incentive compatibility continues to constrain only the conditional means of the contractual terms—in the same linear form as (IC_D) and (IC_S) . Randomization is then a mean-preserving spread of promised utility, which concavity of $\mathcal{V}$ makes costly. Appendix OA.1 gives the argument, and Section 6.5 discusses its scope.

5 Efficiency Loss from Staging

Section 4 assumed staging to be technologically innocuous. We now relax that assumption and ask what survives. Write

$$\eta \equiv \frac{\lambda_S}{\lambda_D} - 1, \tag{5.1}$$

so that the rate restriction $\lambda_D < \lambda_S \leq 2\lambda_D$ maintained in Section 2 is exactly $\eta \in (0, 1]$. Staging is technologically innocuous at $\eta = 1$; when $\eta < 1$ the two stages take longer in expectation than a single direct attempt, and η measures the loss.

The loss is invisible near the deadline and decisive far from it. With no deadline at all the two routes deliver $\Pi - c/\lambda_D$ and $\Pi - 2c/\lambda_S$, and the second is smaller precisely when $\eta < 1$. Efficiency therefore enters as a third force, operating on a part of the horizon where neither the deadline effect nor the milestone effect does.

Section 5.1 establishes what this implies under observable effort, and Section 5.2 carries it into the contracting problem.

5.1 Route Choice under an Exogenous Deadline

The three forces sort themselves along the horizon. Close to the deadline the deadline effect dominates and the direct route is used: needing one slow success beats needing two fast ones when little time remains. Further out the milestone effect takes over and staging becomes worth its cost. But the deadline effect washes out as the horizon lengthens while the efficiency loss does not, so far enough out the direct route returns. Route choice can therefore change twice, and in that order.

Definition 3. A policy is called a *two-switch policy* if there exist intermediate deadlines $0 \leq S_1 \leq S_2 \leq T$ such that (i) the principal chooses the direct route up to S_1 ($a_t = 1$ for $t \leq S_1$), (ii) she chooses the staged route on $(S_1, S_2]$ ($a_t = 0$), (iii) if the interim milestone is delivered before S_2 , she works toward completion until the extended deadline $T + \Delta$, and (iv) if the milestone is not delivered by S_2 , she switches back to the direct route until the deadline T .

This class nests every policy in Proposition 1: setting $S_1 = 0$ gives a one-switch policy with intermediate deadline S_2 , setting $S_2 = T$ as well gives the staged-only policy, and $S_1 = S_2$ gives the direct-only policy. What is new is the case $0 < S_1 < S_2 < T$, in which staging occupies an interior window and the direct route is used at both ends.

The extension cutoff of Proposition 1 generalizes to

$$\bar{\Delta}(\eta, \pi) \equiv \begin{cases} \frac{1}{\lambda_S} \log \left[\frac{(1+\eta)\pi - 1}{\eta\pi - 1} \right], & \text{if } \eta\pi > 1, \\ \infty, & \text{otherwise,} \end{cases} \quad (5.2)$$

which reduces to the cutoff of Proposition 1 at $\eta = 1$.

Alongside (5.2), define a second cutoff on the extension,

$$\Delta^*(\eta, \pi) \equiv \begin{cases} \frac{1}{\lambda_S} \left\{ \log \left[\frac{(1+\eta)((1+\eta)\pi - 1)}{\eta(\pi - 1)} \right] + \eta \log \left[\frac{1 - \eta^2}{\eta^2(\pi - 1)} \right] \right\}, & \text{if } \eta^2\pi > 1, \\ \bar{\Delta}(\eta, \pi), & \text{otherwise,} \end{cases} \quad (5.3)$$

so that $\Delta^*(\eta, \pi) \leq \bar{\Delta}(\eta, \pi)$, with equality if and only if $\eta^2\pi \leq 1$.

Proposition 3 (Optimal policy under efficiency loss). *Suppose that staging is less efficient ($0 < \eta < 1$). There exist $\hat{T}$ and $\check{T}$ with $0 \leq \hat{T} \leq \check{T} \leq \infty$, both independent of the deadline T , such that for every deadline T the two-switch policy with $S_1 = \max\{T - \check{T}, 0\}$ and $S_2 = \max\{T - \hat{T}, 0\}$ is optimal, with $S_i = 0$ when the corresponding threshold is infinite. The extension determines the thresholds:*

- (a) (**Short extension**) if $\Delta \leq \Delta^*(\eta, \pi)$, then $\hat{T} = \check{T} = \infty$;
- (b) (**Intermediate extension**) if $\Delta^*(\eta, \pi) < \Delta \leq \bar{\Delta}(\eta, \pi)$, then $0 \leq \hat{T} < \check{T} < \infty$;
- (c) (**Long extension**) if $\Delta > \bar{\Delta}(\eta, \pi)$, then $\hat{T} = 0 < \check{T} < \infty$.

The thresholds depend on the extension as well as on η and π . This section holds Δ fixed and suppresses that dependence; Section 5.2 restores it, since the extension is there pinned at $1/\lambda_S$ by binding incentive conditions.

The two cutoffs are amounts of extension, and they order the regimes by how much staging is used: never, in an interior window, or from the start. Read this way, Δ^* is the

extension staging must earn to be used at any deadline and $\bar{\Delta}$ the extension it must earn to be used at every one. The first may be negative, in which case the project return is high enough that staging is used even with no extension at all and case (a) is empty. The two coincide when $\eta \leq 1/\sqrt{\pi}$: the efficiency loss is then large relative to the return, the intermediate regime disappears, and staging is either never used or used from the start.

Because $\hat{T}$ and $\check{T}$ are amounts of time remaining rather than calendar dates, the policy can be read off the clock: the direct route while fewer than $\hat{T}$ units remain, staging between $\hat{T}$ and $\check{T}$, and the direct route again beyond $\check{T}$. In case (a) neither threshold is ever reached and the milestone is never sought. In case (b) a deadline of $T \leq \hat{T}$ gives the direct-only policy, a deadline in $(\hat{T}, \check{T}]$ the one-switch policy with intermediate deadline $T - \hat{T}$, and a longer one a two-switch policy with both switches interior. In case (c) the inner direct phase is absent, so staging begins at once: the staged-only policy is optimal for $T \leq \check{T}$, and for longer deadlines the direct route is used early and the staged route late.

The outer phase is what is new. It is absent under technologically innocuous staging because the two no-deadline surpluses then coincide; formally, $\check{T} \rightarrow \infty$ as $\eta \uparrow 1$, and Proposition 3 converges to Proposition 1.

Case (c) records a second departure from Proposition 1. When the extension is generous, staging is attractive near the deadline, since delivering the milestone buys Δ additional units of time exactly when time is scarce. The direct route is then used early rather than late, reversing the ordering of the one-switch policy in Proposition 1(b)—an ordering that cannot arise when staging is technologically innocuous.

5.2 The Optimal Contract under Efficiency Loss

The argument of Section 4 goes through unchanged. None of it uses $\lambda_S = 2\lambda_D$: binding incentive conditions still deliver $u_t = \phi(T - t)$ and a jump of ϕ/λ_S upon delivery, and the route recommended still leaves promised utility untouched, so the route choice under agency

is the same as under observable effort. Proposition 2 therefore holds as stated,

$$\mathcal{V}(u) = \mathcal{W}^* \left(\frac{u}{\phi}, \frac{1}{\lambda_S} \right) - u, \quad (5.4)$$

with $\mathcal{W}^*$ now supplied by Proposition 3 in place of Proposition 1. The contracting problem again inherits the structure of the benchmark; what changes is the benchmark itself.

What it inherits is now a two-switch policy. Measuring the thresholds of Proposition 3 in units of promised utility, at the extension that binding incentives purchase,

$$\hat{u}_1 \equiv \phi \hat{T} \left(\frac{1}{\lambda_S} \right) \quad \text{and} \quad \hat{u}_2 \equiv \phi \check{T} \left(\frac{1}{\lambda_S} \right), \quad (5.5)$$

the direct route is recommended when $u < \hat{u}_1$, the staged route when $\hat{u}_1 < u < \hat{u}_2$, and the direct route again when $u > \hat{u}_2$. Theorem 1 is the case in which the upper threshold is absent: as $\eta \uparrow 1$ we have $\hat{u}_2 \rightarrow \infty$, and the outer region is pushed beyond every attainable promised utility.

Theorem 2 (Optimal contract under efficiency loss). *Suppose staging is less efficient ($0 < \eta < 1$), and let $\hat{u}_1$ and $\hat{u}_2$ be as in (5.5). Then:*

- (a) (**At most two switches**) *the optimal contract changes the recommended route at most twice, at every efficiency level $\eta \in (0, 1)$;*
- (b) (**Taxonomy**) *it takes one of the six forms in Table 1, and which one obtains is determined by the position of $\bar{u}$ relative to $\hat{u}_1$ and $\hat{u}_2$.*

In every case the reward schedules are those of Definition 2 and the deadline is extended by $1/\lambda_S$ upon delivery of the interim milestone.

Part (a) deserves emphasis. Definition 3 restricted attention to policies with at most two switches, and Proposition 3 showed that the restriction costs nothing: whatever the efficiency loss, the staged advantage crosses zero at most twice, so no third switch can be

Table 1: Forms of the optimal contract under efficiency loss

Condition	Route sequence	Contract
$\bar{u} = 0$	—	infeasible
$0 < \bar{u} \leq \hat{u}_1$	direct	direct-only
$\hat{u}_1 = 0, \bar{u} \leq \hat{u}_2$	staged	staged-only
$\hat{u}_1 > 0, \hat{u}_1 < \bar{u} \leq \hat{u}_2$	staged $\rightarrow$ direct	one-switch
$\hat{u}_1 = 0, \bar{u} > \hat{u}_2$	direct $\rightarrow$ staged	reversed one-switch
$\hat{u}_1 > 0, \bar{u} > \hat{u}_2$	direct $\rightarrow$ staged $\rightarrow$ direct	two-switch

worthwhile. That property is inherited here. The bound is therefore universal rather than a tractability assumption—it holds at every $\eta \in (0, 1)$, and it is what makes the taxonomy in Table 1 finite.

Two of the forms in Table 1 are new. The two-switch contract is the one anticipated by Definition 3: the principal begins on the direct route because the horizon is long and efficiency dominates, turns to staging once the deadline is near enough that the concentration of staged completion times is worth its cost, and returns to the direct route as the deadline approaches. Staging occupies the middle of the relationship and neither end. The reversed one-switch contract requires an extension long enough that staging remains attractive at the deadline ($\hat{u}_1 = 0$) together with a return high enough that the principal buys more time than the staged route can profitably use ($\bar{u} > \hat{u}_2$); it then begins on the direct route and turns to staging as the deadline approaches, inheriting the ordering of Proposition 3(c).

Locating the thresholds Theorem 2 classifies the possible contracts but leaves $\hat{u}_1$, $\hat{u}_2$ and $\bar{u}$ implicit. At the two extremes of the efficiency loss they can be located in terms of the primitives, and each mechanism appears on its own. Define

$$\bar{\eta} \equiv \max \left\{ \sqrt{\frac{c}{c + \phi}}, \frac{1}{e - 1} \right\} \quad \text{and} \quad \underline{\eta} \equiv \min \left\{ \frac{c}{c + \phi}, \frac{1}{e - 1} \right\}.$$

We treat the two extremes separately—small efficiency loss ($\eta > \bar{\eta}$) and large efficiency loss ($\eta < \underline{\eta}$)—since the optimal contract admits a clean characterization in each. In the intermediate band, $\eta \in [\underline{\eta}, \bar{\eta}]$, Theorem 2 continues to apply: the contract still switches at

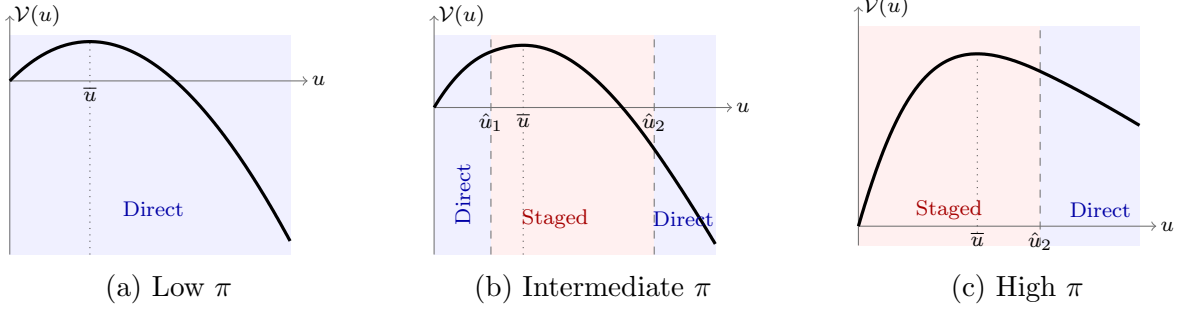


Figure 5: Value functions when the efficiency loss is small ($\lambda_D = 2$, $\lambda_S = 3.8$, $c = 1$, $\phi = 0.5$)

most twice and still takes one of the forms in Table 1. A full characterization there would require working through many subcases, so we close the section with a numerical example instead, exhibiting a form that neither extreme produces.

Small efficiency loss When the loss is small the outer threshold is never reached: $\hat{u}_2$ exceeds $\bar{u}$ at every project return (Lemma OA.17), so the principal never buys enough time for the efficiency effect to bite, and at most one switch occurs. The characterization of Section 4 is then intact.

Proposition 4 (Small efficiency loss). *Suppose $\eta \in (\bar{\eta}, 1)$. There exist thresholds $\tilde{\pi}_S(\eta) > \tilde{\pi}_D(\eta) > \pi_F$ such that a staged-only contract is optimal when $\pi > \tilde{\pi}_S(\eta)$, a one-switch contract is optimal when $\tilde{\pi}_S(\eta) > \pi > \tilde{\pi}_D(\eta)$, and a direct-only contract is optimal when $\tilde{\pi}_D(\eta) > \pi > \pi_F$.*

Figure 5 shows this. The outer direct region is visible in panels (b) and (c) as the band to the right of $\hat{u}_2$, but $\bar{u}$ lies to its left in both. Panel (a) is a direct-only contract, panel (b) a one-switch contract, and panel (c) a staged-only contract—exactly the three cases of Theorem 1.

Large efficiency loss When the efficiency loss is large the picture changes qualitatively. The direct route is now preferred not only near the deadline but also at the beginning of a long project, so for most project returns the contract never stages at all. Only when the



Figure 6: Value functions when the efficiency loss is large ($\lambda_D = 2$, $\lambda_S = 3.1$, $c = 1$, $\phi = 0.5$)

return is high enough to sustain a long deadline does staging appear, and then only in an interior window.

Proposition 5 (Large efficiency loss). *Suppose $\eta < \underline{\eta}$. There exists a threshold $\tilde{\pi}_M(\eta) > \pi_F$ such that a two-switch contract is optimal when $\pi > \tilde{\pi}_M(\eta)$, and a direct-only contract is optimal when $\tilde{\pi}_M(\eta) > \pi > \pi_F$.*

Figure 6 shows both cases. In panel (a) the staged region has disappeared entirely: staging is now slow enough that it is never worth its cost, whatever the remaining time. In panel (b) the return is high enough that staging reappears as an interior window—the principal buys enough time that $\bar{u}$ lies beyond $\hat{u}_2$, so the contract begins in the outer direct region, crosses into staging, and crosses back. Comparing the two figures makes the efficiency effect concrete: it leaves the ordering of the deadline and milestone effects near the deadline untouched, adds a third region far from it, and, when large enough, both widens that region and pushes $\bar{u}$ into it.

Between the two extremes lies $\eta \in [\underline{\eta}, \bar{\eta}]$, where the thresholds admit no closed form and the mapping from parameters to contract forms must be computed. Nothing qualitative changes there: Theorem 2 still applies, the contract still switches at most twice, and it still takes one of the forms in Table 1. What the band adds is the reversed one-switch contract, which arises within it and in neither proposition.

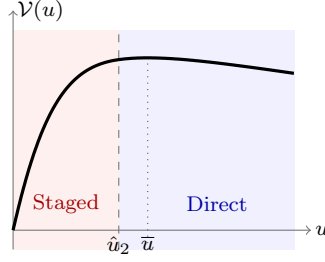


Figure 7: The reversed one-switch contract at an intermediate efficiency loss ($\lambda_D = 2$, $\lambda_S = 3.4$, $c = 1$, $\phi = 0.3$, $\pi = 10$)

Intermediate efficiency loss Five of the six forms in Table 1 have now appeared. The two propositions above deliver the direct-only, staged-only, one-switch and two-switch contracts, and the infeasible case needs no illustration. The reversed one-switch contract arises in neither, and it is the intermediate band that produces it.

Figure 7 shows an instance, at $\eta = 0.7$. The extension is long enough that staging remains attractive at the deadline, so $\hat{u}_1 = 0$ and the inner direct region is absent; but the project return is high enough that the principal buys more time than the staged route can profitably use, so $\bar{u}$ falls beyond $\hat{u}_2$. The contract therefore begins on the direct route and turns to staging as the deadline approaches. Neither extreme permits this combination: a small loss keeps $\bar{u}$ below $\hat{u}_2$, and a large one keeps $\hat{u}_1$ strictly positive.

6 Discussion

This section examines how far the analysis extends beyond the assumptions used to obtain it. We take up in turn the observability of the interim milestone, discounting, the arrival technology, and the class of contracts.

6.1 Contractual Forms

The optimal contract is stated as a deadline and a declining schedule of completion payments, which is convenient for the analysis but is not how such arrangements are written in

practice. This subsection re-expresses it in three ways, each closer to an observed contractual form. Nothing new is derived; the point is that the object the model produces is one that practitioners already use.

A buyout of the right to divert Splitting (IC_D) at equality,

$$R_t^D = \underbrace{\phi(T-t)}_{\text{remaining diversion}} + \underbrace{\frac{\phi}{\lambda_D}}_{\text{completion bonus}},$$

the agent holds a right to divert at rate ϕ until the deadline, and completing the project surrenders it. The principal must therefore buy that right out at face value and add a fixed bonus worth one expected attempt's diversion. The post-milestone schedule is the same two components measured against the extended deadline, $R_t^S = \phi(T + 1/\lambda_S - t) + \phi/\lambda_S$, with the bonus set at the rate of the route being worked.

The buyout is profitable because completion also ends the operating cost. Against a counterfactual in which the agent diverts until T , completion at t delivers the principal

$$\Pi - \frac{\phi}{\lambda_D} + (c - \phi)(T - t) :$$

she purchases the remaining time at ϕ and saves c on it. This is why she prefers early completion even though neither party discounts, and it is where $\phi \leq c$ does its work—were diversion efficient, the buyout would cost more than the burn it saves.

The milestone can then be read as a trade. Delivering it extends the deadline by $1/\lambda_S$, raising the eventual buyout by ϕ/λ_S , and simultaneously lowers the bonus from ϕ/λ_D to ϕ/λ_S , a reduction of $\phi\eta/\lambda_S$. Under technologically innocuous staging the two exactly offset: the agent buys a longer runway with a smaller bonus, at par. This is the sense in which the extension of one expected stage is free, and why neither Π nor c enters it. When staging is less efficient the trade is no longer even and the milestone must also be paid for in cash, by $\phi(1 - \eta)/\lambda_S$.

Probation and up-or-out Read as an employment arrangement, a one-switch contract is a probation period followed by an up-or-out decision. The intermediate deadline S is a tenure clock: the agent is put on the staged route and must show verifiable progress by S . If he does, his clock is extended and he continues on the post-milestone schedule; if he does not, the arrangement is not terminated outright but converted—he is moved onto the direct route and given until T to deliver the finished project. This is the pattern documented in academic and professional employment, and the model delivers it as an optimum rather than assuming it. The reversed one-switch contract of Section 5 inverts the sequence: the agent is given a free hand first and placed under milestone review as the deadline nears.

Runway and milestone-contingent funding Read as a financing arrangement, the same contract is a burn rate and a runway. The principal advances an account $M \equiv \phi T + \phi/\lambda_D$ that depletes at the diversion rate, since $R_t^D = M - \phi t$, and the project is abandoned when what remains falls to the bonus ϕ/λ_D —the minimum premium that still sustains effort. Delivering the milestone does not add cash to the account at $\eta = 1$; it lowers the threshold at which the account is deemed exhausted, from ϕ/λ_D to ϕ/λ_S , and so lengthens the runway.

This is the form in which staged financing is written. In venture capital contracts the release of committed funds is routinely conditioned on interim performance, and [Kaplan and Strömberg \(2003\)](#) document that withholding funds shortens the time until the venture must return for further financing—so that a milestone-contingent release of capital is, in substance, a milestone-contingent extension of the horizon. Their evidence also shows that both forms are used: in first financing rounds the whole commitment is released at closing in the majority of deals, while in the remainder roughly half is held back against milestones. Our results speak to which of the two a principal should choose.

6.2 What the Principal Cannot Observe

Two things are observable throughout the analysis: the route the agent is asked to work, and the interim milestone when it arrives. Each is worth relaxing, and they come apart. An unobservable route turns out to cost the principal nothing, for a reason that is already implicit in the structure of the optimal contract. An unobservable milestone costs her nothing either, provided the agent cannot fabricate one; if he can, the problem becomes that of [Green and Taylor \(2016\)](#) and we do not solve it. We take the three cases in turn.

(i) The milestone is observed by neither party Neither the principal nor the agent learns whether the first stage is complete. Nothing can be conditioned on the milestone, so no extension is possible and the contract is a single deadline with a declining reward—what [Green and Taylor \(2016\)](#) call a *simple contract*. This is the case $\Delta = 0$, which Section 3 has already characterized: by Lemma 1 the deadline effect operates on its own, so the direct route is preferred at short horizons and the staged route at long ones even with no monitoring at all.

The decomposition is worth stating plainly. The ordering of the routes along the horizon is a property of the completion-time distributions, illustrated in Figure 2, and owes nothing to monitoring. What monitoring adds is the extension $1/\lambda_S$, and with it the range of project returns over which the staged route is used at all: without it, staging must earn its place on the shape of its completion time alone.

(ii) The route is unobservable Suppose next that the milestone remains observable but the principal cannot verify which route the agent is working. He may then depart from the recommendation, and—since he controls when he presents an output—he may also hold one back until the contract asks for it. An agent told to stage while the contract will later turn to the direct route could work that route now, sit on the finished project, and present it at the switch; an agent told to work directly while staging comes later could bank a milestone

and report it when the staged phase opens.

Neither pays, and the reason is the property that organizes the whole analysis. Binding incentive conditions price a delivery on route j at $u_t + \phi/\lambda_j$, so the flow value of working that route is $\lambda_j \cdot \phi/\lambda_j = \phi$, whatever j is; and shirking pays ϕ by construction. The premium and the arrival rate move together, and the agent is left indifferent.

Proposition 6 (Route choice is incentive-irrelevant). *Fix any contract of Theorem 2 and suppose the agent privately chooses which route to work at each date and may withhold an output until a later date. Consider a date t at which route j is recommended, and let $k \neq j$.*

- (a) *If the contract calls for route k 's output at some later date, the agent is indifferent between working route j , working route k and withholding its output until it is called for, and shirking.*
- (b) *If it never does, working route k is weakly worse than working route j , and strictly worse whenever $\eta < 1$.*

In neither case is a deviation profitable, and the termination penalty for route-inconsistent deliveries is not needed for incentive compatibility.

Proof of Proposition 6. (a) Under binding incentive conditions $\dot{u}_t = -\phi$, and delivering on route i at date τ , while the contract is in phase i , is worth $u_\tau + \phi/\lambda_i$. Producing route k 's output at s and withholding it until τ therefore yields $\phi(\tau - s)$ from shirking in the interim plus $u_\tau + \phi/\lambda_k = u_s - \phi(\tau - s) + \phi/\lambda_k$, that is $u_s + \phi/\lambda_k$: withholding is exactly free. The flow value of working route k at t is then $\lambda_k(u_t + \phi/\lambda_k - u_t) = \phi$, which is also the flow value of working route j and of shirking.

(b) Route k 's output is now worthless on its own, so it is useful only in so far as it carries the agent to completion. Suppose first that k is the direct route and j the staged one. A completed project delivered in a staged phase is never paid for, so working k has flow value $0 < \phi$. Suppose instead that k is the staged route and j the direct one, and work in remaining time z , so that the reward on completion is $R(z) = \phi(z + 1/\lambda_D)$. Let $M(z)$ be

the agent's value holding an unreported milestone. Since the milestone can be cashed only by completing the project, M solves

$$M'(z) = \max \{ \lambda_S (R(z) - M(z)), \phi \}, \quad M(0) = 0, \quad (6.1)$$

and, the first branch attaining the maximum throughout,

$$M(z) = \phi \left(z + \frac{1}{\lambda_D} - \frac{1}{\lambda_S} \right) - \phi \left(\frac{1}{\lambda_D} - \frac{1}{\lambda_S} \right) e^{-\lambda_S z}. \quad (6.2)$$

The flow value of working route k is therefore

$$\lambda_S (M(z) - \phi z) = \phi \eta (1 - e^{-\lambda_S z}),$$

which is below ϕ for every z when $\eta \leq 1$, and strictly below when $\eta < 1$. $\square$

Part (a) says that the switching structure is not vulnerable to being anticipated: the premium the contract must pay for a delivery is inversely proportional to the rate at which that delivery arrives, so producing the wrong output early and banking it is exactly break-even. Part (b) is the deadline effect running in the agent's direction. Where the staged route's output will never be asked for, the agent can profit from it only by completing the project himself, and that requires two arrivals rather than one. Because the reward declines in calendar time, he wants completion concentrated early; the direct route's greater dispersion is exactly what he wants, and the staged route's concentration around the same mean is worth strictly less to him. The inequality reverses only at $\eta > 1$, outside the range maintained throughout.

(iii) The milestone is privately observed by the agent Finally, suppose the agent knows when the first stage is complete and the principal does not. What matters here is not observability itself but what a report can contain: whether the agent can merely *conceal* a

milestone, or can also *fabricate* one. The two lead to opposite conclusions, and separating them locates exactly where our analysis ends and that of [Green and Taylor \(2016\)](#) begins.

Suppose first that the milestone can be concealed but not fabricated: the agent chooses when to reveal it, but a false claim is detectable. Then private observation costs the principal nothing.

Proposition 7 (Unfalsifiable milestones). *Suppose the agent privately observes the milestone and chooses when, if ever, to reveal it, but cannot fabricate one. Then the contracts of Theorem 2, with the extension granted upon revelation, remain incentive compatible, and the principal's value function is unchanged.*

Proof of Proposition 7. Revelation is the only new decision, so it suffices to show that revealing a milestone as soon as it arrives is optimal for the agent.

Delay is free but not profitable. Suppose the agent holds a milestone at s and reveals it at $\tau > s$, shirking in between. Revelation at τ delivers $u_\tau + \phi/\lambda_S$, and $\dot{u}_t = -\phi$ gives $u_\tau = u_s - \phi(\tau - s)$, so the total is $u_s + \phi/\lambda_S$, the value of revealing at s . Delay therefore neither gains nor loses, and it is enough to compare revealing with never revealing.

Concealment. An agent who never reveals holds a milestone the contract will not pay for, so it is worth something only in so far as it carries him to completion. This is case (b) of Proposition 6: by (6.2), concealment earns him

$$M(z) - \phi z = \phi \frac{\eta}{\lambda_S} (1 - e^{-\lambda_S z})$$

over the shirk-out value, against ϕ/λ_S from revealing. Since $\eta(1 - e^{-\lambda_S z}) < 1$ for every z when $\eta \leq 1$, revelation is strictly better. On-path behavior is therefore unchanged, and so is the principal's payoff. $\square$

The contrast with [Kim and Poggi \(2026\)](#) is instructive. There the same interim discovery is routinely concealed, because the party who would learn of it is a rival who re-optimizes

against the discoverer. Concealment is a response to a strategic audience, and commitment is precisely what removes it.

Fabrication is another matter. If the agent can falsely claim a milestone, the binding constraint is no-false-progress, and it binds for a reason with no counterpart above: the low type’s deviation is to *lie and then shirk*. Under binding incentives, shirking pays ϕ per unit of time—the same rate at which promised utility declines—so a false report converts the revelation jump into shirk-out value at par. The lie costs nothing to execute and pays exactly ϕ/λ_S .

It might seem that the second route makes matters worse, since a false reporter can use it to complete the project rather than merely running out the clock. It does not. The post-report schedule prices a delivery at the staged premium ϕ/λ_S , so working the direct route earns a false reporter $\lambda_D \cdot \phi/\lambda_S = \phi/(1 + \eta)$ per unit of time, strictly less than the ϕ he collects by shirking; working the staged route is worse still, since without a genuine first-stage success he needs two further arrivals. The route choice is available to a liar but underpriced for him, and the binding deviation remains lie-and-shirk.

To deter it, the principal must cap what a reporting agent can collect by the clock a liar could shirk out, so the extension after a report is limited to u_1/ϕ rather than the $1/\lambda_S$ that binding incentive conditions would otherwise purchase. The milestone effect is thereby weakened while the deadline effect is untouched; the reduction in (4.3), which prices the extension at $1/\lambda_S$ independently of the state, no longer applies; and the problem acquires the structure of [Green and Taylor \(2016\)](#): a pair of promised utilities as the state variable, a no-false-progress constraint that binds, and—in their setting—a soft deadline with random termination once the agent runs out of steam.

A route choice is likely to change that last feature, though we do not attempt a characterization. Their agent runs out of steam because, near a hard deadline, completing *two* stages is second-order in the time remaining, so no bounded reward sustains effort; random termination exists to pause the clock at that point. The direct route removes the pathology

at its source: a single arrival is first-order in remaining time, so effort can be sustained up to a hard deadline with bounded rewards. We conjecture that with a route choice the optimal contract replaces probation with a route switch—a hard deadline, the staged route with progress reports early, and the direct route, requiring no communication, once the clock runs short. Verifying this requires solving a free-boundary problem in two promised utilities with the no-false-progress rent, which we leave to future work. What the present analysis establishes is the division of labor between the two informational assumptions: concealment is disciplined by the same binding incentive conditions that organize our contract, at no cost to the principal, while fabrication is the source of information rents and of everything in [Green and Taylor \(2016\)](#) that our observable-milestone analysis does not capture.

6.3 Discounting

Both parties are patient throughout the analysis, as in [Green and Taylor \(2016\)](#). Introducing a common discount rate $r > 0$ leaves the incentive conditions intact but breaks the reduction that organizes the argument, so we characterize the optimal contract numerically. What emerges is that discounting operates on the same margin as an efficiency loss, and the taxonomy of Section 5 describes it without amendment.

What changes Incentive compatibility is unaffected. Working route j over $[t, t + dt)$ still delivers $\lambda_j(R_t - u_t) dt$ against ϕdt from shirking, so (IC_D) and (IC_S) hold as stated and the premium for a delivery remains ϕ/λ_j . What changes is the law of motion for promised utility. Discounting adds a return on the outstanding promise, so binding incentives now give

$$\dot{u}_t = r u_t - \phi, \tag{6.3}$$

and the deadline at which promised utility reaches zero is

$$T(u) = -\frac{1}{r} \log \left(1 - \frac{ru}{\phi} \right),$$

which is finite only for $u < \phi/r$ and converges to u/ϕ as $r \rightarrow 0$. Promised utility remains a clock, but one that runs at a rate depending on how much is owed.

The consequence is that the extension is no longer a constant. Delivering the milestone still raises promised utility by ϕ/λ_S , but the time that buys is

$$\Delta(u) = T\left(u + \frac{\phi}{\lambda_S}\right) - T(u) = -\frac{1}{r} \log \left[1 - \frac{r/\lambda_S}{1 - ru/\phi} \right], \quad (6.4)$$

which is increasing in u and tends to $1/\lambda_S$ as $r \rightarrow 0$. Because (4.3) prices the extension at $1/\lambda_S$ independently of the state, it no longer holds, and the principal's problem cannot be read off the observable-effort benchmark. We therefore solve the Hamilton–Jacobi–Bellman equations for $\mathcal{V}$ and V_M directly, with the route chosen pointwise as in (HJB $_V$).

Discounting as an efficiency loss Fix $\lambda_S = 2\lambda_D$, so that staging is technologically innocuous and Theorem 1 applies at $r = 0$. There the outer threshold is absent: $\hat{u}_2 = \infty$, and the contract takes one of three forms. Any positive discount rate makes $\hat{u}_2$ finite.

Figure 8 plots $\hat{u}_2(r)$ against the optimal promised utility $\bar{u}(r)$. Both decline in r —a more impatient principal buys less time—but the outer threshold falls much faster, and the two cross at $r^* \approx 0.335$. Below r^* the outer direct region exists but is never reached: the principal does not buy enough time for the efficiency margin to bite, and the contract looks exactly as it does under patience. Above r^* the contract begins on the direct route and turns to staging as the deadline approaches. This is the reversed one-switch contract of Table 1, a form that Theorem 1 excludes.

The mechanism is the one identified in Section 5. Two arrivals take longer than one in expectation, so discounting bears more heavily on the staged route, and it bears most heavily where completion is furthest away—far from the deadline. That is precisely where the efficiency effect operates when $\eta < 1$. Impatience and inefficiency therefore act on the same margin, and produce the same third region.

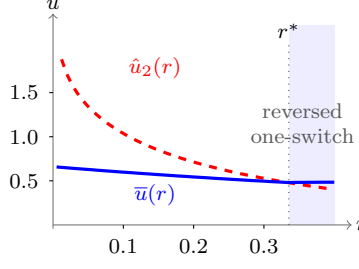


Figure 8: The outer threshold and the optimal promised utility as functions of the discount rate ($\lambda_D = 1$, $\lambda_S = 2$, $c = 1$, $\phi = 0.5$, $\pi = 3$)

The taxonomy is unchanged Table 2 records the form of the optimal contract across project returns and discount rates, again at $\lambda_S = 2\lambda_D$. Every cell is one of the six forms of Table 1: discounting relabels the parameter regions but generates nothing new. The two-switch and reversed one-switch contracts, which require $\eta < 1$ under patience, arise here at $\eta = 1$ once the principal is impatient enough.

Table 2: Form of the optimal contract under discounting ($\lambda_S = 2\lambda_D$, $c = 1$, $\phi = 0.5$)

π	Discount rate r			
	0	0.10	0.20	0.40
1.8	one-switch	one-switch	one-switch	direct-only
2.0	one-switch	one-switch	one-switch	two-switch
2.4	staged-only	one-switch	one-switch	two-switch
3.0	staged-only	staged-only	staged-only	reversed one-switch
4.0	staged-only	staged-only	staged-only	reversed one-switch

Two comparative statics are worth noting. Staging survives impatience best when the project return is high: at $\pi = 2.4$ a discount rate of 0.10 is already enough to reintroduce the direct route near the deadline, while at $\pi = 3$ the contract is staged-only up to r^* . And the direction of the switch reverses as r grows: at moderate returns the contract stages first and turns direct, as under patience, whereas at high returns and high r it does the opposite. The reason is that a high return buys a long deadline, and a long deadline is where impatience penalizes staging most.

Scope These are numerical results at $\lambda_S = 2\lambda_D$, not a characterization. Two features of the discounted problem stand in the way of one. The extension (6.4) depends on the state,

so the thresholds are no longer determined by a benchmark problem in remaining time; and the domain of $\mathcal{V}$ is bounded by ϕ/r , so the comparative statics in r cannot be taken far without also raising ϕ . What the exercise establishes is the qualitative claim: impatience is a third route to the outer direct region, alongside efficiency loss, and it produces no contract that Theorem 2 has not already classified.

6.4 The Arrival Technology

Poisson arrivals Memorylessness is what makes calendar time a sufficient public state and reduces the contracting problem to one dimension. It is not what drives the economics. The deadline effect requires only that staged completion time be a mean-preserving contraction of direct completion time—the content of Lemma 1—and the milestone effect requires only that the interim delivery be informative about remaining work. The first holds well beyond the exponential case: for any scale family with log-concave density, the sum of two independent draws with half the mean is a mean-preserving contraction of a single draw, so the routes’ completion probabilities cross once and the ordering along the horizon is unchanged. Log-concavity is not innocuous, however. It fails for stage-time distributions with a decreasing hazard rate, and there the completion probabilities can cross twice, so that the direct route is preferred both at short and at long horizons for reasons unrelated to efficiency. Whether the switching structure survives that case is a separate question.

Efficiency gains from staging If $\lambda_S > 2\lambda_D$, staging completes the project sooner in expectation, and it is then preferred far from the deadline on efficiency grounds. It does not follow that the direct route disappears. Whether staging is also preferred *at* the deadline is the comparison in Lemma OA.10: with the extension fixed at $1/\lambda_S$, the staged route wins the last instant precisely when $\pi \geq \tilde{\pi}_S(\eta) = (e - 1)/(\eta(e - 1) - 1)$. Since feasibility requires

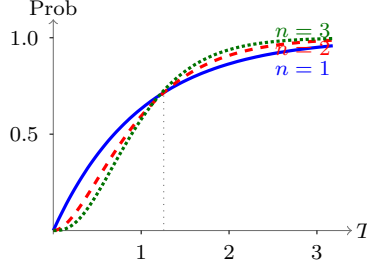


Figure 9: Completion probability under stage decompositions, holding the expected completion time fixed at λ_D

$\pi > \pi_F > 1$, the direct route drops out for every project return only once $\tilde{\pi}_S(\eta) \leq 1$, that is,

$$\eta \geq \frac{e}{e-1} \approx 1.582, \quad \text{equivalently} \quad \lambda_S \geq \frac{2e-1}{e-1} \lambda_D \approx 2.582 \lambda_D.$$

Between $\lambda_S = 2\lambda_D$ and this threshold the trade-off survives intact: staging is efficient, yet the deadline effect still delivers the direct route at short horizons whenever the return is below $\tilde{\pi}_S(\eta)$. We maintain $\lambda_S \leq 2\lambda_D$ to keep the two forces in genuine competition, not because the analysis breaks down above it; what does break down, above $2.582\lambda_D$, is the competition itself, and the problem reduces to the single-route analysis of [Green and Taylor \(2015\)](#).

More than two stages The project admits exactly two decompositions here: one stage or two. In practice the number of interim deliverables is itself a design variable, and the forces we identify are monotone in it. Figure 9 plots completion probabilities for $n = 1, 2, 3$ stages with the expected completion time held fixed, so that only the shape of the distribution varies. The deadline effect strengthens with n : at a short deadline, more stages means a lower completion probability, and the gap widens as n grows. The crossing point, by contrast, moves slightly *in*—finer staging is worse at short horizons but begins to pay marginally sooner—and the curves converge quickly, so that most of the gain from staging is realized at $n = 2$. Characterizing the optimal n -stage contract requires n promised utilities and is left to future work.

6.5 The Contract Space

Effort and recommendations The agent’s effort is a continuous intensity $\beta_t \in [0, 1]$ and the principal’s recommendation could in principle be randomized. Neither enlargement changes anything. The agent’s payoff is affine in his effort, so he optimizes at a corner; the principal’s objective is affine in the recommendation for fixed contractual terms, so her value is convex in it and is maximized by recommending a single route (Lemma 6).

Randomized contracts The principal may also randomize over the terms of the contract without disclosing the realization. Remark 2 states the result and Appendix OA.1 gives the argument; what is worth adding here is its scope. The argument turns on the likelihood of a public history factoring into a part governed by effort and a part governed by the randomization device, which requires the device to be drawn independently of the agent’s actions. It therefore covers soft deadlines, randomized extensions and randomized reward schedules, but not a principal who conditions the randomization on her own private information about the agent’s progress—a possibility that does not arise here, since she observes no more than he does.

Restarts and repeated hiring The contract ends at the deadline, whereas a principal who values the project at Π today presumably still values it tomorrow. Restarting is nested in the analysis: a restart at T delivers the principal $\mathcal{V}(\bar{u})$ afresh, so with no discounting and no friction her value at the deadline is $\max\{0, \mathcal{V}(\bar{u})\}$ rather than zero, and the deadline loses its bite—it is optimal to restart immediately and repeatedly. What restores a non-trivial deadline is any cost of restarting: a search or replacement cost, a delay before a new agent is productive, or discounting, each of which makes the continuation worth strictly less than $\mathcal{V}(\bar{u})$. The deadline in our model should accordingly be read as the point at which the principal abandons *this* attempt, with the outside option normalized to zero; the route-switching structure is unaffected, since it is determined by the shape of $\mathcal{V}$ on $[0, \bar{u}]$ and not

by the level of the terminal payoff.

7 Conclusion

In this article, we study the economic tradeoffs between execution and training in solving a problem in the presence of agency frictions. The optimal contract is determined by the interplay of three effects: efficiency, milestone, and endogenous deadlines. We show that the form of the optimal contract crucially depends on the project return. When the efficiency loss from training does not exist or is small, the optimal contract involves at most one switch. Specifically, if the project return is low, the principal always recommends the agent to solve the problem with a basic skill, whereas if the project return is high, the principal always recommends the agent to acquire the advanced skill, then extend the deadline upon skill acquisition. If the project return is intermediate, it is optimal to begin with training and then switch to execution upon lack of skill acquisition. When the efficiency loss is large, the principal generally recommends the agent to solve a problem with the more efficient basic skill route. However, if the project return is above a certain cutoff, for a short period of time in the middle of the contract, the principal recommends the agent to acquire the advanced skill to mitigate moral hazard (i.e., there may be two switches).

There are numerous avenues open for further research. For example, the principal may be able to design the approaches directly. In this article, we assume that the two approaches are exogenously given and the principal chooses between them. However, in practice, a project manager often designs how many milestones to partition the main project into and how difficult each subproject is. We could also introduce ‘learning by doing’ into the model. If we assume that the agent learns from early errors, the arrival rate of project completion would increase over time.¹³ We leave these intriguing questions—and others—for future work.

¹³This possibility contrasts with the setting considered by [Carnehl and Schneider \(2023\)](#), where learning causes the expected arrival rate to fall.

References

- Anat R Admati and Paul Pfleiderer. Robust financial contracting and the role of venture capitalists. *The Journal of Finance*, 49(2):371–402, 1994.
- Dirk Bergemann and Ulrich Hege. Venture capital financing, moral hazard, and learning. *Journal of Banking & Finance*, 22(6-8):703–735, 1998.
- Dimitri P Bertsekas. *Dynamic programming and optimal control*, volume 1. Athena scientific Belmont, MA, 1995.
- Kevin A Bryan and Jorge Lemus. The direction of innovation. *Journal of Economic Theory*, 172:247–272, 2017.
- Christoph Carnehl and Johannes Schneider. On risk and time pressure: When to think and when to do. *Journal of the European Economic Association*, 21(1):1–47, 2023.
- Mingliu Chen, Peng Sun, and Yongbo Xiao. Optimal monitoring schedule in dynamic contracts. *Operations Research*, 68(5):1285–1314, 2020.
- Francesca Cornelli and Oved Yosha. Stage financing and the role of convertible securities. *The Review of Economic Studies*, 70(1):1–32, 2003.
- Peter M DeMarzo and Yuliy Sannikov. Optimal security design and dynamic capital structure in a continuous-time agency model. *The journal of Finance*, 61(6):2681–2724, 2006.
- Francesc Dilmé and Daniel F Garrett. Residual deterrence. *Journal of the European Economic Association*, 17(5):1654–1686, 2019.
- Richard J Gilbert and Michael L Katz. Efficient division of profits from complementary innovations. *International Journal of Industrial Organization*, 29(4):443–454, 2011.
- Paul A Gompers. Optimal investment, monitoring, and the staging of venture capital. *Journal of Finance*, 50(5):1461–1489, 1995.

- Brett Green and Curtis R Taylor. On breakthroughs, deadlines, and the nature of progress: Contracting for multistage projects. 2015. URL https://www.researchgate.net/publication/282014265_On_Breakthroughs_Deadlines_and_the_Nature_of_Progress_Contracting_for_Multistage_Projects.
- Brett Green and Curtis R Taylor. Breakthroughs, deadlines, and self-reported progress: contracting for multistage projects. *American Economic Review*, 106(12):3660–99, 2016.
- Jerry R Green and Suzanne Scotchmer. On the division of profit in sequential innovation. *The Rand journal of economics*, pages 20–33, 1995.
- Zehao Hu. Financing innovation with unobserved progress. 2014.
- Steven N Kaplan and Per Strömberg. Financial contracting theory meets the real world: An empirical analysis of venture capital contracts. *The review of economic studies*, 70(2): 281–315, 2003.
- Yonggyun Kim and Francisco Poggi. Strategic concealment in innovation races. *Available at SSRN 4638677*, 2026.
- Iván Marinovic and Martin Szydlowski. Monitoring with career concerns. *The RAND Journal of Economics*, 53(2):404–428, 2022.
- Iván Marinovic and Martin Szydlowski. Monitor reputation and transparency. *American Economic Journal: Microeconomics*, 15(4):1–67, 2023.
- Sofia Moroni. Experimentation in organizations. *Theoretical Economics*, 17(3):1403–1450, 2022.
- Darwin V Neher. Staged financing: an agency perspective. *The Review of Economic Studies*, 66(2):255–274, 1999.
- Dmitry Orlov. Frequent monitoring in dynamic contracts. *Journal of Economic Theory*, 206:105550, 2022.

- Tomasz Piskorski and Mark M Westerfield. Optimal dynamic contracts with moral hazard and costly monitoring. *Journal of Economic Theory*, 166:242–281, 2016.
- Francisco Poggi. The timing of complementary innovations. 2021.
- Debraj Ray. The time structure of self-enforcing agreements. *Econometrica*, 70(2):547–582, 2002.
- Stephen E Spear and Sanjay Srivastava. On repeated moral hazard with discounting. *The Review of Economic Studies*, 54(4):599–617, 1987.
- Felipe Varas, Iván Marinovic, and Andrzej Skrzypacz. Random inspections and periodic reviews: Optimal dynamic monitoring. *The Review of Economic Studies*, 87(6):2893–2937, 2020.
- Christoph Wolf. Informative milestones in experimentation. *University of Mannheim, Working Paper*, 2018. URL <https://carnehl.github.io/IME.pdf>.
- Yu Fu Wong. Dynamic monitoring design. *Available at 4466562*, 2023.

Appendix

A Proofs for Section 3

A.1 Expected Surpluses of Benchmark Policies

In this section, we provide formal representations of the expected social surpluses under the direct-only and staged-only policies.

Suppose that the principal employs the direct-only policy. Recall that the completion probability $\mathcal{P}^D(T)$ is derived in (3.1). Next, the expected duration of project completion is

$$\mathcal{D}^D(T) \equiv \int_0^T \tau \cdot \lambda_D \cdot e^{-\lambda_D \tau} d\tau + T \cdot e^{-\lambda_D T} = \frac{1}{\lambda_D} (1 - e^{-\lambda_D T}). \quad (\text{A.1})$$

Then, the expected social surplus of the direct-only policy is

$$\mathcal{W}^D(T) \equiv \Pi \cdot \mathcal{P}^D(T) - c \cdot \mathcal{D}^D(T) = \left(\Pi - \frac{c}{\lambda_D} \right) \cdot (1 - e^{-\lambda_D T}). \quad (\text{A.2})$$

Now suppose that the principal employs the staged-only policy. Recall that the completion probability $\mathcal{P}^S(T, \Delta)$ is derived in (3.2). Conditional on milestone delivery at τ_m , the expected duration is

$$\mathcal{D}^1(T, \Delta, \tau_m) \equiv \int_{\tau_m}^{T+\Delta} \tau \cdot \lambda_S e^{-\lambda_S(\tau-\tau_m)} d\tau + (T + \Delta) \cdot e^{-\lambda_S(T+\Delta-\tau_m)}.$$

Then, the total expected duration of the project can be derived as follows:

$$\begin{aligned} \mathcal{D}^S(T, \Delta) &\equiv \int_0^T \mathcal{D}^1(T, \Delta, \tau_m) \cdot \lambda_S e^{-\lambda_S \tau_m} d\tau_m + T \cdot e^{-\lambda_S T} \\ &= \frac{2}{\lambda_S} (1 - e^{-\lambda_S T}) - T \cdot e^{-\lambda_S(T+\Delta)}. \end{aligned} \quad (\text{A.3})$$

The expected social surplus of the staged-only policy is thus

$$\begin{aligned}\mathcal{W}^S(T, \Delta) &\equiv \Pi \cdot \mathcal{P}^S(T, \Delta) - c \cdot \mathcal{D}^S(T, \Delta) \\ &= \left(\Pi - \frac{2c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S T}) - \lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) \cdot T \cdot e^{-\lambda_S (T+\Delta)}.\end{aligned}\tag{A.4}$$

A.2 Proof of Lemma 1

Proof of Lemma 1. Observe that $\mathcal{P}^D(T, 0) \geq \mathcal{P}^S(T, 0)$ is equivalent to:

$$(1 + \lambda_S T) \geq e^{(\lambda_S - \lambda_D)T}.$$

Equality holds at $T = 0$. The left-hand side linearly increases with slope λ_S , while the right-hand side exponentially increases with an initial slope at $T = 0$ of $\lambda_S - \lambda_D \in (0, \lambda_S)$. Therefore, for sufficiently small T , $\mathcal{P}^D(T, 0) > \mathcal{P}^S(T, 0)$, but there exists a unique cutoff $\check{T} > 0$ equating the two sides. Thus, $\mathcal{P}^D(T, 0) > \mathcal{P}^S(T, 0)$ for all $T < \check{T}$, and $\mathcal{P}^D(T, 0) < \mathcal{P}^S(T, 0)$ for all $T > \check{T}$. $\square$

A.3 Proof of Proposition 1

Value functions and the HJB equation Let W_x^1 denote the expected surplus once the interim milestone has been delivered and x units of time remain. By the same steps that deliver (A.2),

$$W_x^1 \equiv \left(\Pi - \frac{c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S x}).\tag{A.5}$$

If the milestone arrives with z units of time remaining, the deadline extends by Δ , so the continuation surplus is $W_{z+\Delta}^1$.

Before the milestone arrives, the optimal expected surplus $W_z^{0,\Delta}$ with z units of time

remaining satisfies

$$\dot{W}_z^{0,\Delta} = \max_{a_z \in \{0,1\}} [\lambda_D a_z \cdot (\Pi - W_z^{0,\Delta}) + \lambda_S (1 - a_z) \cdot (W_{z+\Delta}^1 - W_z^{0,\Delta}) - c], \quad (\text{HJB}_W)$$

with $W_0^{0,\Delta} = 0$, since the project is abandoned at the deadline. By standard verification arguments (e.g., Proposition 3.2.1 in Bertsekas, 1995), if $w : [0, T] \rightarrow \mathbb{R}$ is differentiable, satisfies (HJB_W) , and has $w(0) = 0$, then $w(z) = W_z^{0,\Delta}$.

The staged advantage For a continuation value W , define the net flow advantage of the staged route over the direct route,

$$\Lambda(z; W) \equiv \lambda_S (W_{z+\Delta}^1 - W) - \lambda_D (\Pi - W) = \lambda_S W_{z+\Delta}^1 - \lambda_D \Pi - (\lambda_S - \lambda_D)W. \quad (\text{A.6})$$

The staged route attains the maximum in (HJB_W) at z if and only if $\Lambda(z; W_z^{0,\Delta}) \geq 0$. Two observations organize the proof. First, because $\lambda_S > \lambda_D$, the map $W \mapsto \Lambda(z; W)$ is strictly decreasing: a higher continuation value makes the direct route relatively more attractive, since the direct route delivers Π outright whereas the staged route only advances the project by one stage. Second, and consequently, Λ must be evaluated at the value function that actually prevails at z . Substituting the direct-only value $\mathcal{W}^D(z)$ is legitimate only where the direct route is in fact used, and it *overstates* the staged advantage elsewhere. Lemma 3 locates the switch point using $\mathcal{W}^D$, and Lemma 4 then propagates the comparison beyond it using the correct value function.

The one-switch value function Fix a switching threshold $Z \in [0, T]$ in remaining time: the direct route is used when $z < Z$ and the staged route when $z \geq Z$, which is the one-switch policy with intermediate deadline $S = T - Z$. Let $\mathcal{W}_Z^{SD}(z, \Delta)$ denote its expected

surplus with z units of time remaining. It satisfies

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_D \cdot (\Pi - \mathcal{W}_Z^{SD}(z, \Delta)) - c \quad \text{for } z < Z, \quad (\text{A.7})$$

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_S \cdot (W_{z+\Delta}^1 - \mathcal{W}_Z^{SD}(z, \Delta)) - c \quad \text{for } z \geq Z, \quad (\text{A.8})$$

and solving with $\mathcal{W}_Z^{SD}(0, \Delta) = 0$ gives

$$\mathcal{W}_Z^{SD}(z, \Delta) = \begin{cases} \mathcal{W}^D(z), & \text{if } z \leq Z, \\ \left(\Pi - \frac{2c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S(z-Z)}) + \mathcal{W}^D(Z) \cdot e^{-\lambda_S(z-Z)} & \text{if } z > Z. \\ -\lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) \cdot (z - Z) \cdot e^{-\lambda_S(z+\Delta)}, & \end{cases} \quad (\text{A.9})$$

The two benchmark policies are the endpoints of this family: $\mathcal{W}_0^{SD}(z, \Delta) = \mathcal{W}^S(z, \Delta)$ is the staged-only surplus and $\mathcal{W}_T^{SD}(z, \Delta) = \mathcal{W}^D(z)$ is the direct-only surplus. We prove the proposition by exhibiting a $Z \in [0, T]$ for which $\mathcal{W}_Z^{SD}(\cdot, \Delta)$ solves (HJB_W).

Optimal route at the deadline Since $W_0^{0,\Delta} = 0$, the staged route is preferred at the deadline if and only if $\Lambda(0; 0) \geq 0$, that is,

$$\lambda_D \Pi \leq (\lambda_S \Pi - c) \cdot (1 - e^{-\lambda_S \Delta}). \quad (\text{A.10})$$

Under $\lambda_S = 2\lambda_D$, dividing by c and rearranging shows that (A.10) holds if and only if $\Delta \geq \bar{\Delta}$.

Lemma 3 (Locating the switch point). *Suppose $\lambda_S = 2\lambda_D$ and $\Delta < \bar{\Delta}$, and set $\hat{T} \equiv 2(\bar{\Delta} - \Delta)$. Then $\hat{T} > 0$ and (i) $\Lambda(z; \mathcal{W}^D(z)) < 0$ for all $z < \hat{T}$; and (ii) $\Lambda(\hat{T}; \mathcal{W}^D(\hat{T})) = 0$.*

Proof of Lemma 3. Substituting (A.5) and $\mathcal{W}^D(z) = (\Pi - c/\lambda_D)(1 - e^{-\lambda_D z})$ into (A.6) and

writing $x \equiv e^{-\lambda_D z} \in (0, 1]$,

$$\begin{aligned} \Lambda_D(x) \equiv \Lambda(z; \mathcal{W}^D(z)) &= (\lambda_S \Pi - c) (1 - e^{-\lambda_S \Delta} x^{\lambda_S/\lambda_D}) - \lambda_D \Pi \\ &\quad - (\lambda_S - \lambda_D) \left(\Pi - \frac{c}{\lambda_D} \right) (1 - x). \end{aligned} \quad (\text{A.11})$$

Under $\lambda_S = 2\lambda_D$ the exponent λ_S/λ_D equals 2 and the constant terms cancel, so (A.11) reduces to the quadratic

$$\tilde{\Lambda}(x) \equiv (\lambda_D \Pi - c) \cdot x - (\lambda_S \Pi - c) \cdot e^{-\lambda_S \Delta} \cdot x^2 = x [(\lambda_D \Pi - c) - (\lambda_S \Pi - c)e^{-\lambda_S \Delta}]. \quad (\text{A.12})$$

Since $\pi > 1$ by (2.1), both $\lambda_D \Pi - c > 0$ and $\lambda_S \Pi - c > 0$, so $\tilde{\Lambda}$ is strictly concave with roots 0 and

$$\hat{x} \equiv \frac{\lambda_D \Pi - c}{\lambda_S \Pi - c} \cdot e^{\lambda_S \Delta}. \quad (\text{A.13})$$

Equality in (A.10) defines $\bar{\Delta}$, and under $\lambda_S = 2\lambda_D$ it reads $e^{-\lambda_S \bar{\Delta}} = (\lambda_D \Pi - c)/(\lambda_S \Pi - c)$, which is precisely the constant multiplying $e^{\lambda_S \Delta}$ in (A.13). Hence

$$\hat{x} = e^{-\lambda_S(\bar{\Delta} - \Delta)}, \quad (\text{A.14})$$

so $\hat{x} < 1$ precisely when $\Delta < \bar{\Delta}$. Thus $\tilde{\Lambda}(\hat{x}) = 0$ and $\tilde{\Lambda}(x) < 0$ for all $\hat{x} < x \leq 1$.

Taking logarithms in (A.14) and using $\lambda_S = 2\lambda_D$,

$$\hat{T} \equiv -\frac{\log(\hat{x})}{\lambda_D} = \frac{\lambda_S}{\lambda_D} (\bar{\Delta} - \Delta) = 2 (\bar{\Delta} - \Delta) > 0.$$

Because $x = e^{-\lambda_D z}$ is strictly decreasing in z , $z < \hat{T}$ is equivalent to $x > \hat{x}$, which yields (i); and $\tilde{\Lambda}(\hat{x}) = 0$ yields (ii). $\square$

Condition (ii) is precisely the indifference condition delivered by smooth pasting: at $\hat{T}$ the two branches of (HJB_W) have equal flow value, so the candidate value function is

continuously differentiable there.

Lemma 4 (Propagating past the switch point). *Suppose $\lambda_S = 2\lambda_D$ and $\Lambda(Z; \mathcal{W}^D(Z)) \geq 0$ for some $Z \geq 0$. Then $\Lambda(z; \mathcal{W}_Z^{SD}(z, \Delta)) \geq 0$ for all $z > Z$.*

Proof of Lemma 4. For $z > Z$ write $y \equiv e^{-\lambda_S(z-Z)} \in (0, 1)$ and $\kappa \equiv (\Pi - c/\lambda_S) e^{-\lambda_S(Z+\Delta)} > 0$. Rearranging (A.9) and (A.5) gives

$$\begin{aligned}\mathcal{W}_Z^{SD}(z, \Delta) &= \mathcal{W}^D(Z) + \left(\Pi - \frac{2c}{\lambda_S} - \mathcal{W}^D(Z) \right) (1 - y) + \kappa \log(y) y, \\ W_{z+\Delta}^1 &= W_{Z+\Delta}^1 + \left(\Pi - \frac{c}{\lambda_S} - W_{Z+\Delta}^1 \right) (1 - y).\end{aligned}$$

Substituting these into (A.6) and collecting terms, $\Lambda(z; \mathcal{W}_Z^{SD}(z, \Delta)) = \Lambda_{SD}(y)$, where

$$\Lambda_{SD}(y) \equiv \Lambda(Z; \mathcal{W}^D(Z)) + h \cdot (1 - y) - (\lambda_S - \lambda_D) \kappa \log(y) y, \quad (\text{A.15})$$

$$h \equiv \lambda_S \left(\Pi - \frac{c}{\lambda_S} - W_{Z+\Delta}^1 \right) - (\lambda_S - \lambda_D) \left(\Pi - \frac{2c}{\lambda_S} - \mathcal{W}^D(Z) \right).$$

Note that $\mathcal{W}_Z^{SD}$ enters (A.6) with coefficient $-(\lambda_S - \lambda_D)$, which is what the final term of (A.15) records. Differentiating twice,

$$\Lambda_{SD}''(y) = -\frac{(\lambda_S - \lambda_D) \kappa}{y} < 0,$$

so Λ_{SD} is strictly concave on $(0, 1)$. At the upper endpoint, $\Lambda_{SD}(1) = \Lambda(Z; \mathcal{W}^D(Z)) \geq 0$ by hypothesis. At the lower endpoint, $\log(y) y \rightarrow 0$ and

$$\lim_{y \rightarrow 0} \Lambda_{SD}(y) = \Lambda(Z; \mathcal{W}^D(Z)) + h = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c = 0, \quad (\text{A.16})$$

where the final equality uses $\lambda_S = 2\lambda_D$. Strict concavity together with $\Lambda_{SD}(1) \geq 0$ and $\lim_{y \rightarrow 0} \Lambda_{SD}(y) = 0$ implies $\Lambda_{SD}(y) \geq 0$ for all $y \in (0, 1)$, which is the claim. $\square$

Equation (A.16) is where technologically innocuous staging does its work: the two end-

point values of Λ_{SD} are non-negative only because the efficiency wedge $\lambda_S - 2\lambda_D$ vanishes.

Proof of Proposition 1. (a) **Long extension** ($\Delta \geq \bar{\Delta}$). Since $\mathcal{W}^D(0) = 0$, (A.10) gives $\Lambda(0; \mathcal{W}^D(0)) \geq 0$. Applying Lemma 4 with $Z = 0$ yields $\Lambda(z; \mathcal{W}^S(z, \Delta)) \geq 0$ for all $z > 0$, so the staged branch attains the maximum in (HJB_W) at every z . By (A.8), $\mathcal{W}^S(\cdot, \Delta) = \mathcal{W}_0^{SD}(\cdot, \Delta)$ therefore solves (HJB_W) on $\mathbb{R}_+$ with the correct boundary condition, and the staged-only policy is optimal.

(b) **Short extension** ($\Delta < \bar{\Delta}$). Let $\hat{T}$ be as in Lemma 3.

If $T < \hat{T}$, then Lemma 3(i) gives $\Lambda(z; \mathcal{W}^D(z)) < 0$ for all $z \in [0, T]$, so the direct branch attains the maximum throughout. By (A.7), $\mathcal{W}^D$ solves (HJB_W) on $[0, T]$ and the direct-only policy is optimal.

If $T \geq \hat{T}$, consider $\mathcal{W}_{\hat{T}}^{SD}$. For $z \leq \hat{T}$ it coincides with $\mathcal{W}^D(z)$, and Lemma 3(i) shows the direct branch attains the maximum there. At $z = \hat{T}$, Lemma 3(ii) delivers both value matching and smooth pasting, so $\mathcal{W}_{\hat{T}}^{SD}$ is continuously differentiable at $\hat{T}$. That same condition supplies the hypothesis of Lemma 4 at $Z = \hat{T}$, whence $\Lambda(z; \mathcal{W}_{\hat{T}}^{SD}(z, \Delta)) \geq 0$ and the staged branch attains the maximum for all $z > \hat{T}$. Hence $\mathcal{W}_{\hat{T}}^{SD}$ is differentiable, satisfies (HJB_W) on $[0, T]$, and vanishes at $z = 0$; by the verification argument it is the value function, and the one-switch policy with intermediate deadline $S = T - \hat{T}$ is optimal.

□

B Proofs for Section 4

This appendix follows the structure of Section 4. Appendix B.1 sets the problem up recursively: it computes the two parties' expected payoffs, shows that the pointwise condition (IC) suffices for incentive compatibility, and derives the Hamilton–Jacobi–Bellman equation together with a verification lemma. These are the formal counterparts of the constraints

introduced in Section 4.2. Appendix B.2 proves Proposition 2 of Section 4.3 in two steps: Appendix B.2.1 constructs the candidate value function and exhibits, for each promised utility, a contract implementing it, and Appendix B.2.2 shows that no deviation to slack incentives improves upon the candidate. Appendix B.3 derives the thresholds in (4.1) and assembles them into a proof of Theorem 1, completing the argument of Section 4.4.

B.1 Recursive Formulation

B.1.1 Expected Payoffs and Incentive Compatibility

In this section, we formally present the expected payoffs of the principal and the agent, conditional on the agent's effort schedule.

Post-milestone payoffs We begin with the subgame where the interim milestone is delivered at time t . Let u_M^t denote the agent's continuation utility—the expected payoff when the agent works until the project is completed or the deadline T^t is reached.

Since this subgame only requires one more success, it is identical to the single-stage benchmark of Green and Taylor (2016). They show that the principal's value function V_M is characterized as follows:

$$V_M(u_M^t) = W_{u_M^t/\phi}^1 - u_M^t = \left(\Pi - \frac{c}{\lambda_S} \right) \left(1 - e^{-\frac{\lambda_S}{\phi} u_M^t} \right) - u_M^t. \quad (\text{B.1})$$

Expected payoffs at time 0 Now consider the problem at time 0. The agent's admissible effort schedule (prior to either completion or delivery of the interim milestone) is $\mathcal{B} \equiv \{\{\beta_t\}_{0 \leq t \leq T} : \beta_t \in \{0, 1\}\}$. In this case, any arrival depends not only on the agent's effort (β) but also the principal's route choice ($a = \{a_t\}_{0 \leq t \leq T}$). Given (a, β) , the probability that neither completion nor the interim milestone has arrived by time τ is $f(a, \beta; 0, \tau)$, where

$$f(a, \beta; t, \tau) \equiv e^{-\lambda_D \int_t^\tau a_s \beta_s ds - \lambda_S \int_t^\tau (1 - a_s) \beta_s ds}.$$

Accordingly, the probability density of completion at time τ is $\lambda_D a_\tau \beta_\tau \cdot f(a, \beta; 0, \tau)$ and that of the delivery of the interim milestone at time τ is $\lambda_S(1-a_\tau)\beta_\tau \cdot f(a, \beta; 0, \tau)$. If the agent follows the recommendation, the above expressions simplify to: $\tilde{f}(a; t, \tau) \equiv e^{-\lambda_D \int_t^\tau a_s ds - \lambda_S \int_t^\tau (1-a_s) ds}$, $\lambda_D a_\tau \cdot \tilde{f}(a; t, \tau)$, and $\lambda_S(1-a_\tau) \cdot \tilde{f}(a; t, \tau)$.

Given a contract Γ , an effort schedule β and under the assumption that the agent follows the recommendation in the updated contract, the principal's expected payoff at time 0 is:

$$P_0(\beta, \Gamma) \equiv \int_0^T \{(\Pi - R_\tau) \cdot \lambda_D a_\tau \beta_\tau + V_M(u_M^\tau) \cdot \lambda_S(1-a_\tau)\beta_\tau - c\} \cdot f(a, \beta; 0, \tau) d\tau.$$

while the agent's expected payoff at time t is:

$$U_0(\beta, \Gamma) \equiv \int_0^T \{R_\tau \cdot \lambda_D a_\tau \beta_\tau + u_M^\tau \cdot \lambda_S(1-a_\tau)\beta_\tau + \phi(1-\beta_\tau)\} \cdot f(a, \beta; 0, \tau) d\tau.$$

Similarly, the expected payoffs of the principal and the agent when the agent follows the recommendation can be obtained by evaluating the general expressions using $\tilde{f}(a; 0, \tau)$ in place of $f(a, \hat{\beta}^t; 0, \tau)$. We denote these by $\tilde{P}_0(\Gamma)$ and $\tilde{U}_0(\Gamma)$, respectively.

Incentive Compatibility Using the terms defined above, we define incentive compatibility (IC) of contracts as follows.

Definition 4. A contract $\Gamma = \{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \leq t \leq T}\}$ is *incentive compatible* if $\tilde{U}_0(\Gamma) \geq U_0(\beta, \Gamma)$ for all $\beta \in \mathcal{B}$.

The objective of the principal is to find a contract Γ that maximizes her ex ante expected payoff $P_0(\Gamma)$ subject to the incentive compatibility constraint and the limited liability constraints $R_\tau \geq 0$ for all $\tau \in [0, T]$ and $R_\tau^t \geq 0$ for all $t \in [0, T]$ and $\tau \in [t, T^t]$. Designate such a contract as an *optimal contract*.

B.1.2 The Agent's Problem

Given a contract Γ , define the continuation utility that the interim milestone is delivered at t as $u_M^t = \tilde{U}^t(\hat{\Gamma}^t)$ and the continuation utility at t when neither the project is completed nor the interim milestone is delivered as

$$u_t = \tilde{U}_t(\Gamma) \equiv \int_t^T \{R_\tau \cdot \lambda_D a_\tau + u_M^t \cdot \lambda_S(1 - a_\tau)\} \cdot \tilde{f}(a; t, \tau) d\tau.$$

Observe that

$$0 = \dot{u}_t + (R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S(1 - a_t) \quad (\text{HJB}_{PK})$$

where $\dot{u}_t \equiv \frac{du_t}{dt}$.

Also note that (IC_S) for $a_t = 0$ and (IC_D) for $a_t = 1$ can be written together as follows:

$$(R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S(1 - a_t) \geq \phi. \quad (\text{IC})$$

The following lemma shows that this condition serves as a sufficient condition for incentive compatibility defined in Definition 4.

Lemma 5. *Given a contract Γ , suppose that there exists a continuous and differentiable process $\{u_t\}_{0 \leq t \leq T}$ satisfying (HJB_{PK}) and $u_T = 0$, and (IC) holds for $0 \leq t \leq T$. Then, Γ is incentive compatible.*

Proof of Lemma 5. The proof is inspired by Proposition 3.2.1 in Bertsekas (1995).

Consider an arbitrary admissible action $\beta \in \mathcal{B}$. Using (HJB_{PK}) , (IC_S) and (IC_D) , we can derive that

$$0 \geq \dot{u}_t + (R_t - u_t)\lambda_D a_t \beta_t + (u_M^t - u_t)\lambda_S(1 - a_t)\beta_t + \phi(1 - \beta_t),$$

or equivalently,

$$-\dot{u}_t + (\lambda_D a_t + \lambda_S(1 - a_t))\beta_t \cdot u_t \geq R_t \cdot \lambda_D a_t \beta_t + u_M^t \cdot \lambda_S(1 - a_t)\beta_t + \phi(1 - \beta_t).$$

It is further equivalent to

$$\frac{d}{dt} [-u_t \cdot f(a, \beta; 0, t)] \geq [(R_t \lambda_D a_t + u_M^t \lambda_S(1 - a_t)) \beta_t + \phi(1 - \beta_t)] \cdot f(a, \beta; 0, t).$$

By integrating the above inequality from 0 to T and using $u_T = 0$, we can derive that

$$u_0 \geq \int_0^T [(R_t \lambda_D a_t + u_M^t \lambda_S(1 - a_t)) \beta_t + \phi(1 - \beta_t)] \cdot f(a, \beta; 0, t) dt = U_0(\beta, \Gamma)$$

for all $\beta \in \mathcal{B}$. Furthermore, the equality holds when $\beta_t = 1$ for all $t \in [0, T]$ from (HJB_{PK}), i.e., u_0 is equal to $\tilde{U}_0(\Gamma)$. Thus, $\tilde{U}_0(\Gamma) \geq U_0(\beta, \Gamma)$ for all $\beta \in \mathcal{B}$, which implies incentive compatibility. $\square$

B.1.3 The Principal's Problem

We now explore how the principal's value function $V(u_t)$ evolves. Note that $V(0) = 0$ since the agent will not participate in the contract when the continuation utility is zero. This will serve as a boundary condition. The value function $V(u_t)$ can be heuristically written as follows:

$$V(u_t) = \max_{R_t, u_M^t, a_t} \begin{aligned} & - c dt + (\Pi - R_t) \lambda_D a_t dt + V_M(u_M^t) \lambda_S(1 - a_t) dt \\ & + \{1 - \lambda_D a_t dt - \lambda_S(1 - a_t) dt\} \cdot V(u_{t+dt}) \end{aligned}$$

subject to (IC).

By using $V(u_{t+dt}) = V(u_t) + V'(u_t) \dot{u}_t dt + o(dt)$, canceling $V(u_t)$ on both sides, taking

the limit as $dt \rightarrow 0$ and plugging (HJB_{PK}) in, we obtain an HJB equation:

$$0 = \max_{R, u_M, a} \mathcal{J}(V(\cdot), R, u_M, a). \quad (\text{HJB}_V)$$

where

$$\begin{aligned} \mathcal{J}(V(\cdot), R, u_M, a) \equiv & -c + (\Pi - R - V(u))\lambda_D a + (V_M(u_M) - V(u))\lambda_S(1 - a) \\ & - \{(R - u)\lambda_D a + (u_M - u)\lambda_S(1 - a)\} \cdot V'(u). \end{aligned} \quad (\text{B.2})$$

Lemma 6 (Bang-bang). *Suppose the agent's effort may take any value $\beta_t \in [0, 1]$, and the principal's recommendation any value $a_t \in [0, 1]$, interpreted as the intensity with which the direct route is worked, so that arrivals occur at rate $\lambda_D a_t \beta_t$ on the direct route and $\lambda_S(1 - a_t)\beta_t$ on the staged route. Then:*

- (a) *given any contract, the agent has an optimal effort process taking values in $\{0, 1\}$; and*
- (b) *the principal's value is unchanged if recommendations are restricted to $a_t \in \{0, 1\}$, and public randomization over recommendations does not improve upon it.*

Proof of Lemma 6. **(a)** Deriving the agent's Hamilton–Jacobi–Bellman equation as in the proof of Lemma 5, but with the supremum taken over $[0, 1]$,

$$0 = \dot{U}_t + \sup_{\beta \in [0, 1]} \{ \phi(1 - \beta) + \beta [(R_t - U_t)\lambda_D a_t + (u_M^t - U_t)\lambda_S(1 - a_t)] \}.$$

The maximand is affine in β , so the supremum is attained at $\beta = 0$ or $\beta = 1$; when the bracket equals ϕ every β attains it, and $\beta = 1$ is among them.

(b) Fix u and V . By (B.2), $\mathcal{J}(V(\cdot), R, u_M, a)$ is affine in a for fixed (R, u_M) , and the constraint (IC) is affine in a as well. Hence

$$g(a) \equiv \sup \{ \mathcal{J}(V(\cdot), R, u_M, a) : (R, u_M) \text{ satisfying (IC)} \}$$

is a supremum of affine functions of a and is therefore convex on $[0, 1]$, so that $\sup_{a \in [0, 1]} g(a) = \max\{g(0), g(1)\}$: the maximum in (HJB_V) is attained at a pure recommendation. Finally, a public randomization placing probability ρ on the recommendation a' and $1 - \rho$ on a'' delivers $\rho g(a') + (1 - \rho) g(a'') \leq \max\{g(0), g(1)\}$, so randomizing over recommendations cannot improve upon the best pure one. $\square$

Then, the principal's problem is to solve (HJB_V) subject to (IC) with the boundary condition $V(0) = 0$. The following lemma shows that the solution of the problem maximizes the principal's expected payoff subject to a promise keeping constraint $U_0(\Gamma) = u$.

Lemma 7 (Verification Lemma). *Suppose that a differentiable and concave function $\tilde{V}$ solves (HJB_V) subject to (IC) with the boundary condition $\tilde{V}(0) = 0$. Then, for any incentive-compatible contract Γ with $U_0(\Gamma) = u$,*

$$\tilde{V}(u) \geq \tilde{P}_0(\Gamma).$$

Proof of Lemma 7. Consider an arbitrary (deterministic) incentive-compatible contract Γ where the agent's expected payoff is given by u_t . The payoff to the principal under Γ is

$$\begin{aligned} \tilde{P}_0(\Gamma) &= \int_0^T (\Pi - R_t - c \cdot t) \cdot \lambda_D a_t \tilde{f}(a; 0, t) dt \\ &\quad + \int_0^T (V_M(u_M^t) - c \cdot t) \cdot \lambda_S (1 - a_t) \tilde{f}(a; 0, t) dt - c \cdot T \cdot \tilde{f}(a; 0, T) \\ &= \int_0^T ((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_S (1 - a_t) - c) \tilde{f}(a; 0, t) dt. \end{aligned}$$

Since $\tilde{V}$ solves the HJB equation, we have

$$\begin{aligned} 0 &\geq -c + (\Pi - R_t - \tilde{V}(u_t)) \lambda_D a_t + (V_M(u_M^t) - \tilde{V}(u_t)) \lambda_S (1 - a_t) \\ &\quad - \{(R_t - u_t) \cdot \lambda_D a_t + (u_M^t - u_t) \cdot \lambda_S (1 - a_t)\} \tilde{V}'(u_t). \end{aligned}$$

By using (HJB_{PK}), rearranging, and multiplying by $\tilde{f}(a; 0, t)$, we can obtain that

$$\begin{aligned} & (\lambda_D a_t + \lambda_S(1 - a_t)) \tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) - \tilde{f}(a; 0, t) \cdot \tilde{V}'(u_t) \dot{u}_t \\ & \geq \tilde{f}(a; 0, t) \left((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_S(1 - a_t) - c \right) \end{aligned} \quad (\text{B.3})$$

Note that

$$\frac{d}{dt} \left(-\tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) \right) = (\lambda_D a_t + \lambda_S(1 - a_t)) \tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) - \tilde{f}(a; 0, t) \cdot \tilde{V}'(u_t) \dot{u}_t.$$

Then, by integrating (B.3) over $[0, T]$ and noting that $\tilde{f}(a; 0, 0) = 1$, $u_T = 0$ and $\tilde{V}(0) = 0$, we have

$$\begin{aligned} \tilde{V}(u_0) &= \tilde{V}(u_0) - \tilde{f}(a; 0, T) \tilde{V}(u_T) \\ &\geq \int_0^T \tilde{f}(a; 0, t) \cdot \left((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_S(1 - a_t) - c \right) dt = \tilde{P}_0(\Gamma). \end{aligned}$$

Therefore, $\tilde{V}(u_0)$ is at least the payoff of any deterministic incentive compatible contract delivering the agent u_0 . Proposition 8 in the Online Appendix extends this bound to contracts that randomize. $\square$

B.2 Value Function Verification

B.2.1 Value Function Candidates and Implementation

Section 4.3 argued that route choice under agency is the same as under observable effort, because promised utility evolves in the same way whichever route is recommended. The following lemma is the formal counterpart. Parts (a)–(c) show that a contract with deadline u/ϕ delivers the agent exactly u regardless of the route it follows, so that the principal's payoff from any of them is the corresponding benchmark surplus of Section 3 net of u . Part (d) records the differential equations these payoffs satisfy, which are used throughout what

follows.

Lemma 8. *The following statements hold.*

- (a) *A direct-only contract with the deadline u/ϕ implements a pair of expected payoffs of the principal and the agent $(V^D(u), u)$ where*

$$V^D(u) \equiv \mathcal{W}^D\left(\frac{u}{\phi}\right) - u. \quad (\text{B.4})$$

- (b) *When $0 < u_1 < u$, a one-switch contract with the intermediate deadline $(u - u_1)/\phi$ and the final deadline u/ϕ implements $(V^{SD}(u|u_1), u)$ where*

$$V^{SD}(u|u_1) \equiv \mathcal{W}_{u_1/\phi}^{SD}\left(\frac{u}{\phi}, \frac{1}{\lambda_S}\right) - u. \quad (\text{B.5})$$

- (c) *A staged-only contract with the deadline u/ϕ implements $(V^{SD}(u|0), u)$.*

- (d) *The following differential equations hold:*

$$\phi V^{D'}(u) = \lambda_D \left(\Pi - \frac{\phi}{\lambda_D} - u - V^D(u) \right) - c, \quad (\text{B.6})$$

$$\phi V^{SD'}(u|u_1) = \lambda_S \left(V_M(u + \frac{\phi}{\lambda_S}) - V^{SD}(u|u_1) \right) - c. \quad (\text{B.7})$$

Proof of Lemma 8. (a) Let $\Gamma_D(T)$ denote a direct-only contract with the deadline T . The agent's expected payoff is

$$\begin{aligned} U_0(\Gamma_D(T)) &= \int_0^T R_\tau \lambda_D e^{-\lambda_D \tau} d\tau = \int_0^T \phi \left(T - \tau + \frac{1}{\lambda_D} \right) \lambda_D e^{-\lambda_D \tau} d\tau \\ &= -\phi (T - \tau) e^{-\lambda_D \tau} \Big|_0^T = \phi T. \end{aligned}$$

Therefore, $U_0(\Gamma_D(u/\phi)) = u$.

Also note that the sum of the expected payoffs of the principal and the agent should

equal to the expected surplus from the direct-only policy with a deadline of T :

$$P_0(\Gamma_D(T)) + U_0(\Gamma_D(T)) = \mathcal{W}^D(T).$$

Therefore,

$$P_0(\Gamma_D(u/\phi)) = \mathcal{W}^D(u/\phi) - u = V^D(u).$$

- (b) Let $\Gamma_{SD}(T_1, T)$ denote a contract with a switch from the staged route to the direct route at T_1 and the deadline T . The subcontract at time $t \leq T_1$ is denoted by $\hat{\Gamma}_{SD}(t|T_1, T)$. Then, the agent's expected payoff for the subcontract $\hat{\Gamma}_{SD}(t|T_1, T)$ at time t is

$$\begin{aligned} U_t(\hat{\Gamma}_{SD}(t|T_1, T)) &= \int_t^{T+\frac{1}{\lambda_S}} \phi\left(T + \frac{1}{\lambda_S} - \tau + \frac{1}{\lambda_S}\right) \lambda_S e^{-\lambda_S(\tau-t)} d\tau \\ &= -\phi\left(T + \frac{1}{\lambda_S} - \tau\right) e^{-\lambda_S(\tau-t)} \Big|_t^{T+\frac{1}{\lambda_S}} = \phi\left(T + \frac{1}{\lambda_S} - t\right). \end{aligned}$$

Also note that

$$\begin{aligned} \int_0^{T_1} U_\tau(\hat{\Gamma}_{SD}(\tau|T_1, T)) \lambda_S e^{-\lambda_S \tau} d\tau &= \int_0^{T_1} \phi\left(T + \frac{1}{\lambda_S} - \tau\right) \lambda_S e^{-\lambda_S \tau} d\tau \\ &= -\phi(T - \tau) e^{-\lambda_S \tau} \Big|_0^{T_1} = \phi T - \phi(T - T_1) e^{-\lambda_S T_1}. \end{aligned}$$

Then, the agent's expected payoff at time 0 is

$$\begin{aligned} U_0(\Gamma_{SD}(T_1, T)) &= \int_0^{T_1} U_\tau(\hat{\Gamma}_{SD}(\tau|T_1, T)) \lambda_S e^{-\lambda_S \tau} d\tau \\ &\quad + e^{-\lambda_S T_1} \int_{T_1}^T \phi\left(T + \frac{1}{\lambda_D} - \tau\right) \lambda_D e^{-\lambda_D(\tau-T_1)} d\tau \\ &= \phi T - \phi(T - T_1) e^{-\lambda_S T_1} - e^{-\lambda_S T_1} \left[\phi(T - \tau) e^{-\lambda_D(\tau-T_1)} \Big|_{T_1}^T \right] = \phi T. \end{aligned}$$

Thus, $U_0(\Gamma_{SD}(T_1, u/\phi)) = u$.

As in the previous case, the sum of the expected payoffs of the principal and the agent is equal to the one-switch policy with the intermediate deadline T_1 , the deadline T , and the extension $1/\lambda_S$:

$$P_0(\Gamma_{SD}(T_1, T)) + U_0(\Gamma_{SD}(T_1, T)) = \mathcal{W}_{T-T_1}^{SD}(T, 1/\lambda_S).$$

By plugging in $T = u/\phi$ and $T_1 = (u - u_1)/\phi$, (B.5) holds.

(c) A staged-only contract with deadline T is the case $T_1 = T$ of part (b), so it implements $(V^{SD}(u|0), u)$.

(d) By the construction of $\mathcal{W}^D$,

$$\dot{\mathcal{W}}^D(T) = \lambda_D(\Pi - \mathcal{W}^D(T)) - c, \quad (\text{B.8})$$

for all $T \geq 0$. Similarly,

$$\dot{\mathcal{W}}_{\hat{T}}^{SD}(T, 1/\lambda_S) = \lambda_S(W_{T+1/\lambda_S}^1 - \mathcal{W}_{\hat{T}}^{SD}(T, 1/\lambda_S)) - c, \quad (\text{B.9})$$

for all $T \geq \hat{T}$.

Using the definitions of V^D , V^{SD} and V_M , (B.6) and (B.7) follow.

□

Together with Proposition 1, this lemma implies that $(\mathcal{V}(u), u)$ —defined in (4.3)—can be implemented by one of the above three contracts. Moreover, there exists $\hat{u}_1 \geq 0$ such that $\mathcal{V}$ can be rewritten as follows:

$$\mathcal{V}(u) = \begin{cases} V^D(u), & \text{if } u < \hat{u}_1, \\ V^{SD}(u|\hat{u}_1), & \text{if } u \geq \hat{u}_1. \end{cases} \quad (\text{B.10})$$

Specifically, $\hat{u}_1$ is chosen to be equal to $\phi\hat{T}$ if $1/\lambda_S < \bar{\Delta}$, and 0 if $1/\lambda_S \geq \bar{\Delta}$. The following lemma provides useful properties of $\mathcal{V}$ and $\hat{u}_1$.

Lemma 9. *Suppose that $\lambda_S = 2\lambda_D$. The following statements hold.*

(a) *if $\hat{u}_1 > 0$, $V^{SD'}(\hat{u}_1|\hat{u}_1) = V^{D'}(\hat{u}_1)$ and $V^{SD'}(u|u) < V^{D'}(u)$ for all $u < \hat{u}_1$, and if $\hat{u}_1 = 0$, $V^{SD'}(0|0) \geq V^{D'}(0)$.*

(b) *$\mathcal{V}'(u) \geq -1$ for all $u \geq 0$.*

(c) *$\mathcal{V}$ is strictly concave, and hence $\bar{u}$ is unique.*

Proof of Lemma 9. (a) Suppose that $\hat{u}_1 > 0$, which implies that $1/\lambda_S < \bar{\Delta}$. Now set

$\Delta = 1/\lambda_S$. In Lemma 3, $\hat{T}$ is chosen to satisfy $\lambda_D(\Pi - \mathcal{W}^D(\hat{T})) = \lambda_S(W_{\hat{T}+1/\lambda_S}^1 - \mathcal{W}^D(\hat{T}))$ and $\lambda_D(\Pi - \mathcal{W}^D(z)) > \lambda_S(W_{z+1/\lambda_S}^1 - \mathcal{W}^D(z))$ for all $z < \hat{T}$.

Using Lemma 8 (d) and $\mathcal{W}^D(\hat{T}) = \mathcal{W}_{\hat{T}}^{SD}(\hat{T}, 1/\lambda_S)$, we can derive that $V^{SD'}(\hat{u}_1|\hat{u}_1) = V^{D'}(\hat{u}_1)$ and $V^{SD'}(u|u) < V^{D'}(u)$ for all $u < \hat{u}_1$.

When $\hat{u}_1 = 0$, and thereby $1/\lambda_S \geq \bar{\Delta}$, it follows from (A.10) that $V^{SD'}(\hat{u}_1|\hat{u}_1) \geq V^{D'}(\hat{u}_1)$.

(b) Since a longer deadline increases the expected social surplus, $\mathcal{W}^*(T, \Delta)$ is increasing in T . Therefore,

$$\mathcal{V}'(u) = \frac{\dot{\mathcal{W}}^*(u/\phi, 1/\lambda_S)}{\phi} - 1 \geq -1.$$

(c) If $u \leq \hat{u}_1$,

$$\mathcal{V}''(u) = V^{D''}(u) = -\left(\Pi - \frac{c}{\lambda_D}\right) \frac{\lambda_D^2}{\phi^2} e^{-\frac{\lambda_D}{\phi}u} < 0.$$

Now assume that $u > \hat{u}_1$. By using (B.8), (B.9) and $V^{D'}(\hat{u}_1) \leq V^{SD'}(\hat{u}_1|\hat{u}_1)$, we can derive that

$$\mathcal{W}^D(\hat{u}_1/\phi) \leq \Pi - \frac{c}{\lambda_S - \lambda_D} - \frac{\lambda_S}{\lambda_S - \lambda_D} \left(\Pi - \frac{c}{\lambda_S}\right) e^{-\frac{\lambda_S}{\phi}\hat{u}_1-1}. \quad (\text{B.11})$$

Using (A.9), we can derive the followings though some algebra:

$$V^{SD''}(u|\hat{u}_1) = \left(\frac{\lambda_S}{\phi}\right)^2 e^{\frac{\lambda_S}{\phi}(\hat{u}_1-u)} \left[\mathcal{W}^D\left(\frac{\hat{u}_1}{\phi}\right) - \left(\Pi - \frac{2c}{\lambda_S}\right) + 2\left(\Pi - \frac{c}{\lambda_S}\right) e^{-\frac{\lambda_S}{\phi}\hat{u}_1-1} \right] \\ - \left(\Pi - \frac{c}{\lambda_S}\right) \left(\frac{\lambda_S}{\phi}\right)^3 (u - \hat{u}_1) e^{-\frac{\lambda_S}{\phi}(u+\frac{1}{\lambda_S})}$$

By plugging (B.11) in, we have

$$V^{SD''}(u|\hat{u}_1) \leq \left(\frac{\lambda_S}{\phi}\right)^2 e^{\frac{\lambda_S}{\phi}(\hat{u}_1-u)} \left[\frac{\lambda_S - 2\lambda_D}{\lambda_S(\lambda_S - \lambda_D)} c + \frac{\lambda_S - 2\lambda_D}{\lambda_S - \lambda_D} \left(\Pi - \frac{c}{\lambda_S}\right) e^{-\frac{\lambda_S}{\phi}\hat{u}_1-1} \right] \\ - \left(\Pi - \frac{c}{\lambda_S}\right) \left(\frac{\lambda_S}{\phi}\right)^3 (u - \hat{u}_1) e^{-\frac{\lambda_S}{\phi}(u+\frac{1}{\lambda_S})}. \quad (\text{B.12})$$

Then, from $\lambda_S = 2\lambda_D$, $\mathcal{V}''(u) = V^{SD''}(u|\hat{u}_1) \leq 0$ for all $u \geq \hat{u}_1$.

□

B.2.2 Proof of Proposition 2

We show that the value function defined in the previous section solves (HJB_V) subject to (IC). To this end we introduce functions that specify potential deviations and establish their properties as a lemma.

First, define

$$L^D(u, R) \equiv \mathcal{J}(\mathcal{V}(\cdot), R, \cdot, 1) = \lambda_D(\Pi - R - \mathcal{V}(u)) - c - \lambda_D(R - u)\mathcal{V}'(u), \quad (\text{B.13})$$

and

$$L^S(u, w) \equiv \mathcal{J}(\mathcal{V}(\cdot), \cdot, w, 0) = \lambda_S(V_M(w) - \mathcal{V}(u)) - c - \lambda_S(w - u)\mathcal{V}'(u). \quad (\text{B.14})$$

Given u , maximizing $L^D(u, R)$ over $R \geq u + \phi/\lambda_D$, and $L^S(u, w)$ over $w \geq u + \phi/\lambda_S$, is equivalent to maximizing the right-hand side of (HJB_V) under (IC) at $a = 1$ and $a = 0$ respectively.

Lemma 10. Suppose that $\Pi > c/\lambda_D$ and $\lambda_S = 2\lambda_D$. Then, for any $u \geq 0$, $L^D(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$, and $L^S(u, w) \leq 0$ for all $w \geq u + \phi/\lambda_S$.

Proof of Lemma 10. We begin by showing $L^D(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$. Observe that

$$\frac{\partial L^D}{\partial R} = -\lambda_D(1 + \mathcal{V}'(u)) \leq 0.$$

from Lemma 9 (b). Also note that

$$\begin{aligned} L^D(u, u + \phi/\lambda_D) &= \lambda_D(\Pi - u - \mathcal{V}(u)) - c - \phi(\mathcal{V}'(u) + 1) \\ &= \lambda_D(\Pi - \mathcal{W}^*(u/\phi, 1/\lambda_S)) - c - \mathcal{W}^*(u/\phi, 1/\lambda_S) \leq 0. \end{aligned}$$

The last inequality is due to (HJB_W). Therefore, $L^D(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$.

For L^S , observe that

$$\frac{\partial L^S}{\partial u} = -\lambda_S(w - u)\mathcal{V}''(u) \geq 0$$

by the concavity of $\mathcal{V}$. Therefore, it is sufficient to check whether $L^S(u, u + \phi/\lambda_S) \leq 0$ for all $u \geq 0$.

Note that for all $u \geq \hat{u}_1$, $L^S(u, u + \phi/\lambda_S) = 0$ holds by (B.7). Now, suppose that $u < \hat{u}_1$, thereby $\mathcal{V}(u) = V^D(u)$. Using (B.7) and Lemma 9 (a), we have $L^S(u, u + \phi/\lambda_S) = \phi(V^{SD'}(u|u) - V^{D'}(u)) < 0$ for all $u < \hat{u}_1$. $\square$

Proof of Proposition 2. We begin by showing that $\mathcal{J}$ becomes zero when the value function $\mathcal{V}$ is utilized alongside contractual terms with binding ICs.

When $u \geq \hat{u}_1$, $\mathcal{V}(u) = V^{SD}(u|\hat{u}_1)$. Then, by (B.7),

$$\mathcal{J}(V^{SD}(u|\hat{u}_1), \cdot, u + \phi/\lambda_S, 0) = 0.$$

Likewise, when $u < \hat{u}_1$, $\mathcal{V}(u) = V^D(u)$, and by (B.6),

$$\mathcal{J}(V^D(u), u + \phi/\lambda_D, \cdot, 1) = 0.$$

Lemma 10 shows that $\mathcal{J}$ is nonpositive for any feasible deviations. Therefore, $\mathcal{V}$ solves (HJB $_{\mathcal{V}}$) subject to (IC).

The concavity of $\mathcal{V}$ is shown in Lemma 9 (c). If $\hat{u}_1 = 0$, $\mathcal{V}(u) = V^{SD}(u|0)$ is differentiable for all $u \geq 0$. If $\hat{u}_1 > 0$, $V^D(u)$ is differentiable for all $u < \hat{u}_1$ and $V^{SD}(u|\hat{u}_1)$ is differentiable for all $u > \hat{u}_1$. By Lemma 9 (a), $\mathcal{V}$ is differentiable at $\hat{u}_1$ as well. Also note that $\mathcal{V}(0) = 0$. Therefore, by Lemma 7, for any incentive compatible contract promising the agent u units of utility, the principal's expected payoff is lower than or equal to $\mathcal{V}(u)$.

Last, by Lemma 8, there exists a contract implementing $(\mathcal{V}(u), u)$. Therefore, $\mathcal{V}(u)$ is the principal's maximized expected payoff subject to the promise-keeping constraint $U_0(\Gamma) = u$ and the incentive compatibility constraints. $\square$

B.3 Thresholds and Proof of Theorem 1

In this section, we pin down the thresholds π_D and π_S , establish their ordering, and complete the proof of Theorem 1.

First, recall that π_S is the solution of $1/\lambda_S = \bar{\Delta}$. This gives us

$$\frac{1}{\lambda_S} = \frac{1}{\lambda_S} \log \left[\frac{2\pi - 1}{\pi - 1} \right] \quad \Leftrightarrow \quad \pi_S = \frac{e - 1}{e - 2} \approx 2.392. \quad (\text{B.15})$$

Also recall that the threshold is relevant to whether the switching point $\hat{u}_1$ is greater than $\bar{u}$ or not. Since $\mathcal{V}$ is concave, $\hat{u}_1 \leq \bar{u}$ if and only if $\mathcal{V}'(\hat{u}_1) \geq 0$. Observe that by using the formula of V^D and (A.13), we can derive that

$$\mathcal{V}'(\hat{u}_1) = V^{D'}(\hat{u}_1) = \frac{(\lambda_D \Pi - c)^2}{\phi \cdot (\lambda_S \Pi - c)} e - 1.$$

Setting the above equal to zero and discarding the smaller root, which lies below 1, it follows that $\mathcal{V}'(\hat{u}_1) \geq 0$ if and only if

$$\pi \geq \pi_D \equiv \frac{c \cdot e + \phi + \sqrt{\phi(c \cdot e + \phi)}}{c \cdot e}. \quad (\text{B.16})$$

Lemma 11. *When $\lambda_S = 2\lambda_D$, $\pi_D \in (\pi_F, \pi_S)$.*

Proof of Lemma 11. From $\phi \leq c$, we have

$$\pi_D \leq \frac{(e+1) + \sqrt{e+1}}{e} \approx 2.077.$$

Therefore, by (B.15), $\pi_D < \pi_S$.

Next, observe that

$$\pi_D - \pi_F = \frac{-\phi(e-1) + \sqrt{\phi(c \cdot e + \phi)}}{c \cdot e} \geq \frac{\phi(\sqrt{e+1} - (e-1))}{c \cdot e}.$$

Since $\sqrt{e+1} - (e-1) \approx .21 > 0$, $\pi_D > \pi_F$. □

Proof of Theorem 1. The ordering $\pi_F < \pi_D < \pi_S$ is Lemma 11. By Proposition 2, $\mathcal{V}$ is concave, so (MP) admits a solution $\bar{u}$, and by Lemma 8 some contract implements $(\mathcal{V}(\bar{u}), \bar{u})$. It remains to locate $\bar{u}$ relative to 0 and to the switching point $\hat{u}_1$.

Feasibility. Since $\mathcal{V}$ is concave with $\mathcal{V}(0) = 0$, we have $\bar{u} > 0$ if and only if $\mathcal{V}'(0) > 0$. By (4.3) and $\dot{\mathcal{W}}^*(0) = \lambda_D \Pi - c$,

$$\mathcal{V}'(0) = \frac{\dot{\mathcal{W}}^*(0)}{\phi} - 1 = \frac{\lambda_D \Pi - c - \phi}{\phi},$$

which is positive if and only if $\pi > 1 + \phi/c = \pi_F$. For $\pi < \pi_F$ we have $\bar{u} = 0$ and no contract is offered, which is part ((d)).

Staging throughout. By Proposition 1 the benchmark uses the direct route only when the time remaining is below $\hat{T} = 2(\bar{\Delta} - \Delta)$, and binding incentive conditions fix $\Delta = 1/\lambda_S$.

Hence

$$\hat{u}_1 = \phi \hat{T} = 2\phi \left(\bar{\Delta} - \frac{1}{\lambda_S} \right)^+,$$

which vanishes precisely when $\bar{\Delta} \leq 1/\lambda_S$, that is, by (B.15), when $\pi \geq \pi_S$. For such π the staged region is all of $\mathbb{R}_+$ and the implementing contract is staged-only, which is part ((a)).

The intermediate cases. Let $\pi_F < \pi < \pi_S$, so that $\bar{u} > 0$ and $\hat{u}_1 > 0$. Concavity of $\mathcal{V}$ gives $\bar{u} > \hat{u}_1$ if and only if $\mathcal{V}'(\hat{u}_1) > 0$, which by (B.16) holds if and only if $\pi > \pi_D$. When $\pi \in (\pi_D, \pi_S)$ the contract begins in the staged region and crosses into the direct region at $\hat{u}_1$, so it is a one-switch contract, which is part ((b)). When $\pi \in (\pi_F, \pi_D)$ the contract never leaves the direct region and the milestone is never sought, which is part ((c)).

The four cases are mutually exclusive, and by the ordering of the thresholds they exhaust $\pi > 0$. □

Online Appendix for

“Staging as a Monitoring Device under Endogenous Deadlines”

Yonggyun Kim Curtis Taylor

OA.1 Randomized Contracts

This appendix establishes Theorem 1 against a contract space in which the principal may randomize and need not disclose the realization to the agent. Only the upper bound requires the enlarged space: the contract exhibited in Appendix B.2.1 is deterministic and delivers $\mathcal{V}(u)$, so attainment is unaffected.

Contracts. At $t = 0$ the principal offers a contract and commits fully to its terms; if the agent rejects, both parties receive zero. We restrict attention to contracts that recommend full effort at every date, which is without loss of generality: because $c > \phi$, any interval on which the agent is meant to shirk can be excised, advancing all subsequent events and saving $c - \phi$ per unit of time.

To allow randomization we endow the contract with a *randomization device*: a random variable ω , drawn at $t = 0$ from a probability space $(\Omega, \mathcal{F}, \mu)$ independently of everything the agent does, on which the terms may depend. A *contract* is an object

$$\Gamma \equiv \left\{ \mu, \left\{ T(\omega), \left\{ a_t(\omega), R_t(\omega), \hat{\Gamma}^t(\omega) \right\}_{0 \leq t \leq T(\omega)} \right\}_{\omega \in \Omega} \right\}, \quad (\text{OA.1.1})$$

whose components are defined at calendar time t , for each realization ω , as follows.

1. $T(\omega) \in \mathbb{R}_+$: the deadline at which the project is abandoned if no delivery has occurred.

2. $a_t(\omega) \in \{0, 1\}$: the route recommended at t , where $a_t = 1$ denotes the *direct* route and $a_t = 0$ the *staged* route.
3. $R_t(\omega) \geq 0$: the payment to the agent if he completes the project at t without having delivered an interim milestone.¹⁴
4. $\hat{\Gamma}^t(\omega) \equiv \{T^t(\omega), \{R_s^t(\omega)\}_{t \leq s \leq T^t(\omega)}\}$: the continuation contract taking effect if the milestone arrives at t , consisting of a revised deadline $T^t(\omega) \geq t$ and payments $R_s^t(\omega) \geq 0$ for completing the project at s .

Any delivery inconsistent with the recommended route—a milestone under the direct route, or a completed project with no milestone on record under the staged route—terminates the contract without payment, so the agent follows the recommendation. A contract is *deterministic* if μ is degenerate, in which case (OA.1.1) reduces to $\Gamma = \{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \leq t \leq T}\}$.

The formulation is more permissive than it may appear. A soft deadline—termination at a random date, as in Green and Taylor (2016)—is the case in which $T(\omega)$ is non-degenerate and the agent learns its realization only by being terminated; randomization over routes, payments or the length of the extension is likewise a special case. Because ω is drawn once and for all at $t = 0$, devices that unfold gradually are covered too: a sequence of later randomizations is a single draw of the whole path, revealed as the contract proceeds. Nothing requires the principal to disclose ω , and this is what makes the enlarged space necessary rather than decorative: an agent who does not know which contract he faces cannot tailor a deviation to it, so a mixture of contracts that are individually not incentive compatible may be incentive compatible as a whole.¹⁵

The agent need not observe ω . He observes the route he is asked to work on and whether the contract is still running; beyond that, what he learns is whatever those observations reveal. Write $\mathcal{H}_t$ for the public history of deliveries, recommendations and survival up to t ,

¹⁴Since both parties are risk neutral and do not discount, all payments may be backloaded to the moment of a delivery without loss of generality (see, e.g., Ray, 2002). Nonnegativity is limited liability.

¹⁵We are grateful to a referee for pressing this point.

and $\mu_t \equiv \mu(\cdot \mid \mathcal{H}_t)$ for his posterior over the device. The argument has three parts: Appendix OA.1.1 shows that μ_t is unaffected by the agent's own deviations, Appendix OA.1.2 uses this to show that incentive compatibility constrains only conditional means, and Appendix OA.1.3 extends the verification argument, the new content being two terms that concavity signs.

OA.1.1 Beliefs and Deviations

An *effort strategy* is an $\mathcal{H}$ -adapted process $\beta = \{\beta_s\}$ together with a choice of route at each date; write $\mathbb{P}^\beta$ for the law it induces jointly with μ .

Lemma OA.1 (Beliefs are deviation-proof). *For every effort strategy β and every t , the posterior μ_t is the same under $\mathbb{P}^\beta$ as under the recommended strategy, $\mathbb{P}^1$, on every history that has positive probability under both.*

Proof. Fix a public history h_t on which no delivery has occurred, the observed recommendation path is $a_{[0,t]}$, and the contract is still running.¹⁶ Its likelihood under (ω, β) is

$$\mathbb{P}(h_t \mid \omega, \beta) = \underbrace{\mathbf{1}\{a_{[0,t]}(\omega) = a_{[0,t]}, T(\omega) \geq t\}}_{\equiv g(h_t, \omega)} \cdot \underbrace{\exp\left(-\int_0^t \lambda(a_s) \beta_s ds\right)}_{\equiv f(h_t, \beta)},$$

where $\lambda(a_s) \in \{\lambda_D, \lambda_S\}$ is the arrival rate of the route worked at s . The first factor depends on ω but not on β ; the second depends on β but not on ω , because the recommendation path is part of h_t and hence known to the agent. By Bayes' rule,

$$\mu_t(\omega) = \frac{g(h_t, \omega) f(h_t, \beta) \mu(d\omega)}{\int_{\Omega} g(h_t, \omega') f(h_t, \beta) \mu(d\omega')} = \frac{g(h_t, \omega) \mu(d\omega)}{\int_{\Omega} g(h_t, \omega') \mu(d\omega')},$$

and f cancels. □

¹⁶Histories on which a delivery has occurred are terminal for the pre-milestone problem: completion ends the game, and delivery of the milestone starts the continuation contract $\hat{\Gamma}^t$, to which the same argument applies verbatim.

The lemma does not say the agent learns nothing about ω : survival rules out realizations with $T(\omega) < t$, and each recommendation rules out those inconsistent with it. It says he learns nothing *extra* by deviating. Arrival rates depend on his effort but not on the device, contractual terms depend on the device but not on his effort, and the one channel through which the device could affect arrival rates—the recommendation—is observed. Shirking therefore slows the project without revealing which contract he is in, which is what closes the pooling channel above.

OA.1.2 Incentive Compatibility in Conditional Means

Define the agent's promised utility and the conditional means of the contractual terms,

$$u_t \equiv \mathbb{E}[\text{continuation utility} \mid \mathcal{H}_t], \quad \bar{R}_t \equiv \mathbb{E}[R_t(\omega) \mid \mathcal{H}_t], \quad \bar{u}_M^t \equiv \mathbb{E}[u_M^t(\omega) \mid \mathcal{H}_t],$$

all expectations under the recommended strategy.

Lemma OA.2 (Incentive compatibility). *A contract Γ is incentive compatible if and only if, at every t and every history reached with positive probability,*

$$\bar{R}_t \geq u_t + \frac{\phi}{\lambda_D} \quad \text{when } a_t = 1, \quad \text{and} \quad \bar{u}_M^t \geq u_t + \frac{\phi}{\lambda_S} \quad \text{when } a_t = 0. \quad (\text{OA.1.2})$$

Proof. Consider a one-shot deviation at t on the direct route. By Lemma OA.1 the agent's posterior at the moment of completion is still μ_t , since completion is driven by his own effort and carries no information about the device; his expected payment on completing is therefore $\bar{R}_t$. Given this, the comparison is the one in Section 4.2 with $\bar{R}_t$ in place of R_t , and yields the first inequality; the second is identical with λ_S and $\bar{u}_M^t$.

That one-shot deviations suffice follows from Lemma OA.1: the agent's beliefs after any deviation coincide with his beliefs on path, so the continuation problem he faces after deviating is the one he faces without deviating. The requisite integrability is automatic—payments

are nonnegative and the agent's total expected payoff is $u_0 < \infty$, so each component of $U_0(\Gamma)$ is bounded in expectation by u_0 , and contracts with $\mathbb{E}[T(\omega)] = \infty$ may be excluded as they deliver the principal $-\infty$. Route deviations are deterred by the off-path punishment specified above, realization by realization and hence in expectation. $\square$

Randomization thus enters only through conditional means, and (OA.1.2) is linear in $(u_t, \bar{R}_t, \bar{u}_M^t)$, so the set of incentive-compatible continuations is convex.

OA.1.3 Verification

Because u_t is a conditional expectation adapted to $\mathcal{H}$, its dynamics on the pre-milestone path are those of the deterministic problem in conditional means, plus a martingale:

$$du_t = - [(\bar{R}_t - u_t)\lambda_D a_t + (\bar{u}_M^t - u_t)\lambda_S(1 - a_t)] dt + dN_t, \quad (\text{OA.1.3})$$

where the drift is (HJB_{PK}) evaluated at $\bar{R}_t$ and $\bar{u}_M^t$, and N is a mean-zero martingale collecting revisions to the agent's beliefs about ω and the jump to zero on termination. When μ is degenerate, N vanishes and (OA.1.3) reduces to $\dot{u}_t = -\phi$ under binding incentives.

Proposition 8 (Randomization does not help). *Let $\mathcal{V}$ be as in Proposition 2. For every contract Γ in (OA.1.1) satisfying (OA.1.2) and delivering the agent u_0 ,*

$$P_0(\Gamma) \leq \mathcal{V}(u_0).$$

Proof. Let M_t denote the principal's accumulated payoff up to t plus $\mathcal{V}(u_t)$ if the contract is still running at t , and zero thereafter, so that $M_0 = \mathcal{V}(u_0)$ and $\mathbb{E}[M_\infty] = P_0(\Gamma)$. We show that M is a supermartingale; the optional stopping theorem then gives $\mathbb{E}[M_\infty] \leq M_0$, which is the claim. Decompose the drift of M using (OA.1.3) into three terms.

(i) *The deterministic drift.* Together with the flow cost and the jumps at deliveries, this is exactly the expression appearing in the proof of Lemma 7, evaluated at $\bar{R}_t$ and $\bar{u}_M^t$. By

Lemma OA.2 those means satisfy the same constraints as in the deterministic problem, so that argument applies unchanged and the term is non-positive.

(ii) *Belief revisions.* Let Δ be an increment of N arising from a revision of the agent's beliefs, so that $\mathbb{E}[\Delta \mid \mathcal{H}_t] = 0$. By the tangent-line characterization of concavity (Lemma 9),

$$\mathbb{E}[\mathcal{V}(u_t + \Delta) \mid \mathcal{H}_t] - \mathcal{V}(u_t) \leq \mathcal{V}'(u_t) \cdot \mathbb{E}[\Delta \mid \mathcal{H}_t] = 0.$$

(iii) *Termination.* Suppose the contract terminates at hazard $h_t \geq 0$. Termination sends both continuation values to zero, and compensating the resulting jump adds $h_t u_t$ to the drift of u . The net contribution is

$$h_t [u_t \mathcal{V}'(u_t) + \mathcal{V}(0) - \mathcal{V}(u_t)] \leq 0,$$

because $\mathcal{V}(0) = 0$ and concavity gives $\mathcal{V}(0) \leq \mathcal{V}(u_t) - u_t \mathcal{V}'(u_t)$.

All three terms are non-positive, so M is a supermartingale. □

Corollary OA.3 (Soft deadlines). *No contract that terminates the agent at a random date improves upon the optimal contract of Theorem 1.*

Proof. A soft deadline is the case of (OA.1.1) in which $T(\omega)$ is non-degenerate, and is covered by term (iii) above. □

OA.2 Proofs for Section 5

This appendix follows the structure of Section 5. Appendix OA.2.1 treats the observable-effort problem of Section 5.1: it establishes the crossing property on which the two-switch bound rests, locates the two thresholds, and proves Proposition 3. Appendix OA.2.2 constructs the principal's value function under an efficiency loss and Appendix OA.2.3 proves Theorem 2, which together discharge the first part of Section 5.2. Appendix OA.2.4 locates the thresholds in terms of the primitives at the two extremes of the efficiency loss and proves Propositions 4 and 5.

OA.2.1 Route Choice under an Exogenous Deadline

Throughout this subsection we assume $\lambda_D < \lambda_S < 2\lambda_D$, i.e. $0 < \eta < 1$. For a candidate surplus function w it is convenient to abbreviate the gap defined in (A.6) as

$$\Lambda_z[w] \equiv \Lambda(z; w(z)) = \lambda_S (W_{z+\Delta}^1 - w(z)) - \lambda_D (\Pi - w(z)), \quad (\text{OA.2.1})$$

the difference between the two branches of (HJB_W): the staged route attains the maximum in (HJB_W) at z if and only if $\Lambda_z[w] \geq 0$. The two substituted forms used in the proof of Proposition 1 are $\Lambda_D(x)$ of (A.11), obtained by setting $w = \mathcal{W}^D$ and $x = e^{-\lambda_D z}$, and $\Lambda_{SD}(y)$ of (A.15), obtained by setting $w = \mathcal{W}_Z^{SD}$ and $y = e^{-\lambda_S(z-Z)}$.

The proof of Proposition 1 draws its conclusions from $\lambda_S = 2\lambda_D$ in exactly two places, and they are the same place: the two substituted forms satisfy

$$\lim_{x \rightarrow 0} \Lambda_D(x) = \frac{\lambda_S - 2\lambda_D}{\lambda_D} \cdot c \quad \text{and} \quad \lim_{y \rightarrow 0} \Lambda_{SD}(y) = \frac{\lambda_S - 2\lambda_D}{\lambda_S} \cdot c, \quad (\text{OA.2.2})$$

and both vanish at $\lambda_S = 2\lambda_D$, which is what allows each of Lemmas 3 and 4 to conclude that a sign holds on an entire half-line.¹⁷ When $\lambda_S < 2\lambda_D$ both limits are strictly negative,

¹⁷The knife-edge is also what reduces Λ_D to the quadratic (A.12) and hence delivers $\hat{x}$ in closed form. Under an efficiency loss we retain the general form (A.11) and characterize the thresholds implicitly.

and each conclusion weakens in the same way, from a sign to a single crossing. The following lemma converts these two single crossings into a policy.

Lemma OA.4 (Crossing rate). *Let w be differentiable on an interval $I \subseteq \mathbb{R}_+$ and satisfy either (A.7) or (A.8) on I . Then at every $z \in I$ with $\Lambda_z[w] = 0$,*

$$\frac{d}{dz}\Lambda_z[w] = \Phi(z) \equiv \lambda_S(\lambda_S - \lambda_D) (\Pi - W_{z+\Delta}^1) - \lambda_D c, \quad (\text{OA.2.3})$$

irrespective of which of the two equations w satisfies. Moreover Φ is strictly decreasing, with $\lim_{z \rightarrow \infty} \Phi(z) = (\lambda_S - 2\lambda_D)c < 0$.

Proof of Lemma OA.4. From (A.5), $\frac{d}{dz}W_{z+\Delta}^1 = \lambda_S(\Pi - \frac{c}{\lambda_S} - W_{z+\Delta}^1)$, so

$$\frac{d}{dz}\Lambda_z[w] = \lambda_S^2 \left(\Pi - \frac{c}{\lambda_S} - W_{z+\Delta}^1 \right) - (\lambda_S - \lambda_D)\dot{w}(z).$$

At a point with $\Lambda_z[w] = 0$ we have $(\lambda_S - \lambda_D)w(z) = \lambda_S W_{z+\Delta}^1 - \lambda_D \Pi$, and therefore

$$\begin{aligned} (\lambda_S - \lambda_D) (\Pi - w(z)) &= \lambda_S (\Pi - W_{z+\Delta}^1), \\ (\lambda_S - \lambda_D) (W_{z+\Delta}^1 - w(z)) &= \lambda_D (\Pi - W_{z+\Delta}^1). \end{aligned}$$

Substituting the first identity into (A.7) and the second into (A.8) gives, in both cases,

$$(\lambda_S - \lambda_D)\dot{w}(z) = \lambda_S \lambda_D (\Pi - W_{z+\Delta}^1) - (\lambda_S - \lambda_D)c,$$

and (OA.2.3) follows. Since $W_{z+\Delta}^1$ is strictly increasing in z and converges to $\Pi - c/\lambda_S$, the function Φ is strictly decreasing with limit $\lambda_S(\lambda_S - \lambda_D)\frac{c}{\lambda_S} - \lambda_D c = (\lambda_S - 2\lambda_D)c$, which is negative because $\lambda_S < 2\lambda_D$. $\square$

Corollary OA.5 (No return). *Suppose w satisfies (A.7) on $[z_0, \infty)$, with $\Lambda_{z_0}[w] = 0$ and $\Phi(z_0) < 0$. Then $\Lambda_z[w] < 0$ for all $z > z_0$.*

Proof of Corollary OA.5. Since Φ is strictly decreasing, $\Phi(z) < 0$ for all $z \geq z_0$. By Lemma OA.4 the derivative of $\Lambda_z[w]$ at z_0 equals $\Phi(z_0) < 0$, so $\Lambda_z[w] < 0$ for z slightly above z_0 . Suppose $\Lambda_z[w] = 0$ for some $z > z_0$ and let z_1 be the smallest such point. Then $\Lambda_z[w] < 0$ on (z_0, z_1) and $\Lambda_{z_1}[w] = 0$, so the derivative at z_1 is non-negative; but Lemma OA.4 gives $\Phi(z_1) < 0$, a contradiction. $\square$

The next two lemmas are Lemmas 3 and 4 with $\lambda_S = 2\lambda_D$ relaxed. Neither of the two concavity properties they rely on uses the knife-edge restriction.

Lemma OA.6 (Locating the inner switch). *Suppose that $\lambda_D < \lambda_S < 2\lambda_D$ and $\Delta \leq \bar{\Delta}$. Then Λ_D is strictly concave on $(0, 1]$ and attains its maximum at*

$$x^* \equiv \left[\frac{\lambda_D(\lambda_S - \lambda_D) \left(\Pi - \frac{c}{\lambda_D} \right)}{\lambda_S(\lambda_S \Pi - c)} \cdot e^{\lambda_S \Delta} \right]^{\frac{\lambda_D}{\lambda_S - \lambda_D}}. \quad (\text{OA.2.4})$$

Moreover:

- (i) if $\Delta \leq \Delta^*(\eta, \pi)$, then $\Lambda_z[\mathcal{W}^D] \leq 0$ for all $z \geq 0$, with equality at most at one point;
- (ii) if $\Delta > \Delta^*(\eta, \pi)$, there is a unique $\hat{x} \in (x^*, 1]$ with $\Lambda_D(\hat{x}) = 0$, and, setting $\hat{T} \equiv -\log(\hat{x})/\lambda_D \geq 0$, we have $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$, and $\Phi(\hat{T}) > 0$.

Proof of Lemma OA.6. Differentiating (A.11) twice,

$$\Lambda_D''(x) = -(\lambda_S \Pi - c) e^{-\lambda_S \Delta} \frac{\lambda_S}{\lambda_D} \left(\frac{\lambda_S}{\lambda_D} - 1 \right) x^{\frac{\lambda_S}{\lambda_D} - 2} < 0,$$

where the sign uses $\lambda_S > \lambda_D$ and $\lambda_S \Pi > \lambda_D \Pi > c$ from (2.1). Setting $\Lambda_D'(x) = 0$ and solving gives (OA.2.4); since $\lambda_S > \lambda_D$ we have $\Lambda_D'(x) \rightarrow C \equiv (\lambda_S - \lambda_D)(\Pi - \frac{c}{\lambda_D}) > 0$ as $x \rightarrow 0$, so $x^* > 0$ is the global maximizer. Evaluating (A.11) at the two ends of the domain,

$$\Lambda_D(0) \equiv \lim_{x \rightarrow 0} \Lambda_D(x) = \frac{\lambda_S - 2\lambda_D}{\lambda_D} c < 0 \quad \text{and} \quad \Lambda_D(1) = (\lambda_S \Pi - c) (1 - e^{-\lambda_S \Delta}) - \lambda_D \Pi,$$

and $\Lambda_D(1) \leq 0$ is precisely (A.10) reversed, i.e. $\Delta \leq \bar{\Delta}$ with $\bar{\Delta}$ as in (5.2).

The value at the maximum. Write $\Lambda_D(x) = \Lambda_D(0) + Cx - (\lambda_S\Pi - c)e^{-\lambda_S\Delta}x^{\lambda_S/\lambda_D}$. Stationarity at x^* gives $(\lambda_S\Pi - c)e^{-\lambda_S\Delta}(x^*)^{\lambda_S/\lambda_D} = \frac{\lambda_D}{\lambda_S}Cx^*$, so the power term collapses into the linear one and

$$\Lambda_D(x^*) = \Lambda_D(0) + \frac{\lambda_S - \lambda_D}{\lambda_S} C x^* = \frac{\lambda_S - 2\lambda_D}{\lambda_D} c + \frac{(\lambda_S - \lambda_D)^2(\lambda_D\Pi - c)}{\lambda_S\lambda_D} x^*. \quad (\text{OA.2.5})$$

Consequently $\Lambda_D(x^*) \leq 0$ if and only if

$$x^* \leq \frac{\lambda_S(2\lambda_D - \lambda_S)c}{(\lambda_S - \lambda_D)^2(\lambda_D\Pi - c)} = \frac{1 - \eta^2}{\eta^2(\pi - 1)}. \quad (\text{OA.2.6})$$

Case (i). Suppose first that $x^* \geq 1$. Then Λ_D is strictly increasing on $(0, 1)$, so $\Lambda_D(x) < \Lambda_D(1) \leq 0$ there. Moreover $x^* \geq 1$ and $\Delta \leq \bar{\Delta}$ together force $\eta^2\pi \leq 1$: substituting η and π into (OA.2.4) and (5.2), the two inequalities read

$$\frac{(1 + \eta)((1 + \eta)\pi - 1)}{\eta(\pi - 1)} \leq e^{\lambda_S\Delta} \leq \frac{(1 + \eta)\pi - 1}{\eta\pi - 1},$$

which reduces to $(1 + \eta)(\eta\pi - 1) \leq \eta(\pi - 1)$, that is $\eta^2\pi \leq 1$. (If $\eta\pi \leq 1$ then $\bar{\Delta} = \infty$ and $\eta^2\pi \leq \eta\pi \leq 1$ directly.) By (5.3) we then have $\Delta^* = \bar{\Delta} \geq \Delta$, so this case lies in (i).

Suppose instead that $x^* < 1$ and $\Lambda_D(x^*) \leq 0$. Then $\Lambda_D(x) \leq \Lambda_D(x^*) \leq 0$ for all x , with equality only possible at x^* . By (OA.2.6), raising both sides to the power η and solving for Δ , the inequality $\Lambda_D(x^*) \leq 0$ is exactly $\Delta \leq \Delta^*(\eta, \pi)$ with Δ^* as in (5.3). In either case $\Lambda_z[\mathcal{W}^D] \leq 0$ for all z , which is (i).

Case (ii). If $\Delta > \Delta^*$ then, by the previous paragraph, $x^* < 1$ and $\Lambda_D(x^*) > 0 \geq \Lambda_D(1)$. Since Λ_D is strictly decreasing on $[x^*, 1]$, it has a unique zero $\hat{x}$ there, and $\hat{x} > x^*$. Strict concavity and $\Lambda'_D(x^*) = 0$ give $\Lambda'_D(\hat{x}) < 0$, and $\Lambda_D(x) < 0$ for all $x \in (\hat{x}, 1]$. Because $x = e^{-\lambda_D z}$ is strictly decreasing in z , this says $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, with equality at $\hat{T}$.

Finally $\mathcal{W}^D$ satisfies (A.7) on $\mathbb{R}_+$, so Lemma OA.4 applies at $\hat{T}$ and the chain rule gives

$$\Phi(\hat{T}) = \frac{d}{dz} \Lambda_z[\mathcal{W}^D] \Big|_{z=\hat{T}} = -\lambda_D \hat{x} \Lambda'_D(\hat{x}) > 0. \quad \square$$

Lemma OA.7 (Locating the outer switch). *Suppose that $\lambda_D < \lambda_S < 2\lambda_D$ and that $Z \geq 0$ satisfies either $\Lambda_Z[\mathcal{W}^D] > 0$, or $\Lambda_Z[\mathcal{W}^D] = 0$ and $\Phi(Z) > 0$. Then Λ_{SD} is strictly concave on $(0, 1]$ and there is a unique $\check{y} \in (0, 1)$ with $\Lambda_{SD}(\check{y}) = 0$. Setting $\check{T} \equiv Z - \log(\check{y})/\lambda_S > Z$, we have $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for all $z \in (Z, \check{T})$, $\Lambda_{\check{T}}[\mathcal{W}_Z^{SD}] = 0$, and $\Phi(\check{T}) < 0$.*

Proof of Lemma OA.7. Strict concavity is $\Lambda''_{SD}(y) = -(\lambda_S - \lambda_D)\kappa/y < 0$, and the limit in (OA.2.2) is $\Lambda_Z[\mathcal{W}^D] + h = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c$, which is strictly negative because $\lambda_S < 2\lambda_D$. Neither computation uses the knife-edge restriction.

We first show that $\Lambda_{SD}(y) > 0$ for y slightly below 1. If $\Lambda_Z[\mathcal{W}^D] > 0$ this is immediate from $\Lambda_{SD}(1) = \Lambda_Z[\mathcal{W}^D]$ and continuity. If instead $\Lambda_Z[\mathcal{W}^D] = 0$, then $\mathcal{W}_Z^{SD}$ satisfies (A.8) on $[Z, \infty)$, so Lemma OA.4 gives a derivative of $\Phi(Z) > 0$ at Z and hence $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for z slightly above Z , which is the same statement since $y = e^{-\lambda_S(z-Z)}$ is strictly decreasing in z .

Because Λ_{SD} is concave, the superlevel set $\{y \in (0, 1] : \Lambda_{SD}(y) \geq 0\}$ is an interval. It contains a left neighborhood of 1 and, by the limit above, excludes a right neighborhood of 0; hence it equals $[\check{y}, 1]$ for a unique $\check{y} \in (0, 1)$, with $\Lambda_{SD} > 0$ on $(\check{y}, 1)$ and $\Lambda_{SD} < 0$ on $(0, \check{y})$. Translating back, $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for $z \in (Z, \check{T})$ and vanishes at $\check{T}$.

Finally, Λ_{SD} is positive somewhere on $(\check{y}, 1)$, so $\check{y}$ lies strictly below the maximizer of Λ_{SD} and strict concavity gives $\Lambda'_{SD}(\check{y}) > 0$. Applying Lemma OA.4 at $\check{T}$ and the chain rule,

$$\Phi(\check{T}) = \frac{d}{dz} \Lambda_z[\mathcal{W}_Z^{SD}] \Big|_{z=\check{T}} = -\lambda_S \check{y} \Lambda'_{SD}(\check{y}) < 0. \quad \square$$

It remains to extend the candidate beyond $\check{T}$. Let $\mathcal{W}_{Z_1, Z_2}^{DSD}(z, \Delta)$ denote the expected surplus of the two-switch policy that uses the direct route for $z < Z_1$, the staged route for $Z_1 \leq z \leq Z_2$, and the direct route again for $z > Z_2$. Solving (A.7) on (Z_2, ∞) with the value

$\mathcal{W}_{Z_1}^{SD}(Z_2, \Delta)$ at Z_2 ,

$$\mathcal{W}_{Z_1, Z_2}^{DSD}(z, \Delta) = \begin{cases} \mathcal{W}_{Z_1}^{SD}(z, \Delta), & \text{if } z \leq Z_2, \\ \left(\Pi - \frac{c}{\lambda_D} \right) - \left[\left(\Pi - \frac{c}{\lambda_D} \right) - \mathcal{W}_{Z_1}^{SD}(Z_2, \Delta) \right] e^{-\lambda_D(z-Z_2)}, & \text{if } z > Z_2. \end{cases} \quad (\text{OA.2.7})$$

Lemma OA.8 (No third switch). *Under the hypotheses of Lemma OA.7, $\Lambda_z[\mathcal{W}_{Z, \check{T}}^{DSD}] < 0$ for all $z > \check{T}$.*

Proof of Lemma OA.8. By (OA.2.7) the function $\mathcal{W}_{Z, \check{T}}^{DSD}$ agrees with $\mathcal{W}_Z^{SD}$ at $\check{T}$, so $\Lambda_{\check{T}}[\mathcal{W}_{Z, \check{T}}^{DSD}] = 0$ by Lemma OA.7, and it satisfies (A.7) on $[\check{T}, \infty)$. That same lemma gives $\Phi(\check{T}) < 0$, so Corollary OA.5 applies. $\square$

Proof of Proposition 3. We construct $\hat{T}$ and $\check{T}$ and verify that $\mathcal{W}_{\hat{T}, \check{T}}^{DSD}$ solves (HJB_W) on $\mathbb{R}_+$ with value 0 at $z = 0$. Neither threshold depends on T , because (HJB_W) is stated in remaining time and its boundary condition is imposed at $z = 0$.

- (a) Suppose $\Delta \leq \Delta^*$. Since $\Delta^* \leq \bar{\Delta}$, Lemma OA.6 applies and puts us in its case (i): $\Lambda_z[\mathcal{W}^D] \leq 0$ for all z , so $\mathcal{W}^D$ solves (HJB_W) on $\mathbb{R}_+$ by (A.7). Put $\hat{T} = \check{T} = \infty$.
- (b) Suppose $\Delta^* < \Delta \leq \bar{\Delta}$. Lemma OA.6 case (ii) delivers $\hat{T} \geq 0$ with $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$ and $\Phi(\hat{T}) > 0$, which is the second hypothesis of Lemma OA.7 at $Z = \hat{T}$; that lemma then delivers $\check{T} \in (\hat{T}, \infty)$.
- (c) Suppose $\Delta > \bar{\Delta}$. Set $\hat{T} = 0$. Since $\mathcal{W}^D(0) = 0$, we have $\Lambda_0[\mathcal{W}^D] = \lambda_S W_\Delta^1 - \lambda_D \Pi > 0$ by (A.10), so Lemma OA.7 applies at $Z = 0$ and delivers $\check{T} \in (0, \infty)$.

In each case, write $w \equiv \mathcal{W}_{\hat{T}, \check{T}}^{DSD}(\cdot, \Delta)$. On $[0, \hat{T})$, $w = \mathcal{W}^D$ satisfies (A.7) and $\Lambda_z[w] < 0$ by Lemma OA.6, so the direct branch attains the maximum in (HJB_W). On $(\hat{T}, \check{T})$, $w = \mathcal{W}_{\hat{T}}^{SD}$ satisfies (A.8) and $\Lambda_z[w] > 0$ by Lemma OA.7, so the staged branch attains it. On $(\check{T}, \infty)$, w satisfies (A.7) and $\Lambda_z[w] < 0$ by Lemma OA.8, so the direct branch attains it again.

At $\hat{T}$ and at $\check{T}$ we have $\Lambda_z[w] = 0$, so the two branches of (HJB_w) have equal right-hand sides there; consequently the two one-sided derivatives of w agree and w is differentiable at both points. Since $w(0) = \mathcal{W}^D(0) = 0$, the verification argument following (HJB_w) gives $w(z) = W_z^{0,\Delta}$ for all $z \geq 0$.

Fix a deadline T and translate remaining time into calendar time through $t = T - z$. The optimal policy uses the direct route when $z < \hat{T}$ or $z > \check{T}$ and the staged route when $z \in (\hat{T}, \check{T})$, which is the two-switch policy of Definition 3 with $S_1 = \max\{T - \check{T}, 0\}$ and $S_2 = \max\{T - \hat{T}, 0\}$. The three cases above give the threshold values asserted in parts (a), (b) and (c) of the proposition. $\square$

Remark 3. As $\eta \uparrow 1$ the limit $\lim_{y \rightarrow 0} \Lambda_{SD}(y) = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c$ increases to zero. Since Λ_{SD} is concave and remains strictly positive on a fixed interior interval, its lower zero $\check{y}$ must then converge to zero, so $\check{T} = Z - \log(\check{y})/\lambda_S \uparrow \infty$: the outer direct phase is pushed beyond every finite deadline and Proposition 3 converges to Proposition 1.

OA.2.2 The Value Function under Efficiency Loss

Throughout this subsection $0 < \eta < 1$ and $\pi > 1$. The results of Appendix B that do not invoke $\lambda_S = 2\lambda_D$ remain available: in particular the incentive conditions of Lemma 5, the verification bound of Lemma 7, and the implementation results of Lemma 8. What must be redone is the construction of the candidate value function, since the optimal policy now has three route regions rather than two.

Two conventions carry over from Appendix OA.2.1. Under binding incentive conditions promised utility and remaining time are proportional, $u = \phi z$, and the extension is $\Delta = 1/\lambda_S$; the staged advantage Λ_D of (A.11) therefore reappears here, in the variable $x \equiv e^{-\lambda_D u/\phi}$ and evaluated at $\Delta = 1/\lambda_S$, as

$$\Lambda_D(x; \pi) = -\{(\eta + 1)\pi - 1\} \cdot c \cdot e^{-1} x^{\eta+1} + \eta(\pi - 1) \cdot c \cdot x - (1 - \eta) \cdot c, \quad (\text{OA.2.8})$$

where the dependence on π is displayed because the comparative statics of Appendix OA.2.4 vary it. The object governing the outer threshold, $\tilde{\Lambda}_{SD}$ of (OA.2.13) below, measures the advantage of the *direct* route rather than the staged one, so its sign convention is opposite to that of Λ_D ; this is why it is convex where Λ_D is concave.

OA.2.2.1 A third branch

Let $V^{DS D}(u \mid u_1, u_2)$ denote the principal's payoff under a two-switch contract with switching points $u_1 < u_2$ in promised utility.

Lemma OA.9 (Implementation of the two-switch contract). *The following statements hold.*

- (a) *When $0 < u_1 < u_2 < u$, the two-switch contract with intermediate deadlines $(u - u_2)/\phi$ and $(u - u_1)/\phi$ and final deadline u/ϕ implements $(V^{DS D}(u \mid u_1, u_2), u)$, where*

$$V^{DS D}(u \mid u_1, u_2) \equiv \left(\Pi - \frac{c}{\lambda_D} \right) \left(1 - e^{\frac{\lambda_D}{\phi}(u_2 - u)} \right) + (V^{SD}(u_2 \mid u_1) + u_2) e^{\frac{\lambda_D}{\phi}(u_2 - u)} - u. \quad (\text{OA.2.9})$$

- (b) *The following differential equation holds:*

$$\phi V^{DS D'}(u \mid u_1, u_2) = \lambda_D \left(\Pi - \frac{\phi}{\lambda_D} - u - V^{DS D}(u \mid u_1, u_2) \right) - c. \quad (\text{OA.2.10})$$

Proof of Lemma OA.9. (a) Let $\Gamma_{DS D}(T_1, T_2, T)$ denote a contract with two switches at $T_1 = (u - u_2)/\phi$ and $T_2 = (u - u_1)/\phi$ and a deadline $T = u/\phi$. At time T_1 , if the project has not been completed, the remaining contract is $\Gamma_{SD}(T_2 - T_1, T - T_1)$, and $U_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) = \phi(T - T_1)$ by Lemma 8(b). The agent's expected payoff at time 0 is therefore

$$\begin{aligned} U_0(\Gamma_{DS D}(T_1, T_2, T)) &= \int_0^{T_1} \phi(T + 1/\lambda_D - \tau) \lambda_D e^{-\lambda_D \tau} d\tau + e^{-\lambda_D T_1} U_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) \\ &= \phi T - \phi(T - T_1) e^{-\lambda_D T_1} + e^{-\lambda_D T_1} \phi(T - T_1) = \phi T = u. \end{aligned}$$

For the principal,

$$\begin{aligned}
& P_0(\Gamma_{DS D}(T_1, T_2, T)) + U_0(\Gamma_{DS D}(T_1, T_2, T)) \\
&= \int_0^{T_1} (\Pi - c\tau) \lambda_D e^{-\lambda_D \tau} d\tau - cT_1 e^{-\lambda_D T_1} \\
&+ e^{-\lambda_D T_1} (P_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) + U_0(\Gamma_{SD}(T_2 - T_1, T - T_1))).
\end{aligned}$$

By Lemma 8(b), $U_0(\Gamma_{SD}(T'_2, T'_1)) + P_0(\Gamma_{SD}(T'_2, T'_1)) = V^{SD}(\phi T'_1 \mid \phi(T'_1 - T'_2)) + \phi(T'_1 - T'_2)$. Substituting $T'_1 = T - T_1$, $T'_2 = T_2 - T_1$, $\phi T_1 = u - u_2$ and $\phi T_2 = u - u_1$, the right-hand side equals

$$\left(\Pi - \frac{c}{\lambda_D} \right) (1 - e^{-\lambda_D T_1}) + e^{-\lambda_D T_1} \cdot \{V^{SD}(u_2 \mid u_1) + u_2\} = V^{DS D}(u \mid u_1, u_2) + u,$$

so $P_0(\Gamma_{DS D}(T_1, T_2, T)) = V^{DS D}(u \mid u_1, u_2)$.

(b) Differentiating (OA.2.9) and multiplying by ϕ ,

$$\begin{aligned}
\phi V^{DS D'}(u \mid u_1, u_2) &= \lambda_D \left(\Pi - \frac{c}{\lambda_D} \right) e^{\frac{\lambda_D}{\phi}(u_2 - u)} - \lambda_D (V^{SD}(u_2 \mid u_1) + u_2) e^{\frac{\lambda_D}{\phi}(u_2 - u)} - \phi \\
&= \lambda_D \left(\Pi - \frac{\phi}{\lambda_D} - u - V^{DS D}(u \mid u_1, u_2) \right) - c,
\end{aligned}$$

which is (OA.2.10). □

The candidate value function is accordingly

$$\mathcal{V}(u) = \begin{cases} V^D(u), & \text{if } 0 \leq u \leq \hat{u}_1, \\ V^{SD}(u \mid \hat{u}_1), & \text{if } \hat{u}_1 < u \leq \hat{u}_2, \\ V^{DS D}(u \mid \hat{u}_1, \hat{u}_2), & \text{if } \hat{u}_2 < u, \end{cases} \quad (\text{OA.2.11})$$

with $0 \leq \hat{u}_1 \leq \hat{u}_2 \leq \infty$ to be determined. Section 4 is the case $\hat{u}_2 = \infty$, in which the third branch is never reached.

OA.2.2.2 The two switching points

The thresholds are located by smooth pasting, exactly as in Appendix OA.2.1 but now in the promised-utility variable, with $\hat{u}_i = \phi \cdot$ (the corresponding threshold in remaining time).

Proposition 9 (Value function under efficiency loss). *Suppose $0 < \eta < 1$ and $\pi > 1$. There exist $\hat{u}_2 \geq \hat{u}_1 \geq 0$ such that*

- (i) $V^{D'}(u) > V^{SD'}(u | u)$ for all $0 \leq u < \hat{u}_1$;
- (ii) $V^{SD'}(\hat{u}_1 | \hat{u}_1) \geq V^{D'}(\hat{u}_1)$, with $V^{SD''}(\hat{u}_1 | \hat{u}_1) > V^{D''}(\hat{u}_1)$ when the equality holds;
- (iii) $V^{SD'}(u | \hat{u}_1) > V^{DSD'}(u | \hat{u}_1, u)$ for all $\hat{u}_1 < u < \hat{u}_2$;
- (iv) $V^{DSD'}(\hat{u}_2 | \hat{u}_1, \hat{u}_2) = V^{SD'}(\hat{u}_2 | \hat{u}_1)$ and $V^{DSD''}(\hat{u}_2 | \hat{u}_1, \hat{u}_2) > V^{SD''}(\hat{u}_2 | \hat{u}_1)$;
- (v) $V^{DSD'}(u | \hat{u}_1, \hat{u}_2) > \frac{1}{\phi} [\lambda_S (V_M(u + \phi/\lambda_S) - V^{DSD}(u | \hat{u}_1, \hat{u}_2)) - c]$ for all $u > \hat{u}_2$.

Moreover, the function $\mathcal{V}$ defined in (OA.2.11) with these thresholds is the principal's value function.

Conditions (i)–(ii) locate $\hat{u}_1$ and are the promised-utility counterparts of Lemma OA.6; conditions (iii)–(iv) locate $\hat{u}_2$ and correspond to Lemma OA.7; condition (v) says that the direct route remains optimal above $\hat{u}_2$, corresponding to Corollary OA.5. Lemmas OA.10 and OA.11 deliver (i) and (ii) and construct the thresholds $\tilde{\pi}_S(\eta)$ and $\tilde{\pi}_M(\eta)$ used in Appendix OA.2.4; Lemma OA.12 delivers (iii) and (iv); Lemma OA.13 delivers (v). The final claim is Lemma OA.14 together with the verification bound of Lemma 7.

Lemma OA.10 (Staging at the deadline). *If $\eta \leq 1/(e-1)$, the inequality $V^{D'}(0) > V^{SD'}(0|0)$ always holds. If $\eta > 1/(e-1)$, $V^{D'}(0) > V^{SD'}(0|0)$ is equivalent to*

$$\pi < \tilde{\pi}_S(\eta) \equiv \frac{e-1}{\eta(e-1)-1}.$$

Moreover, if $\pi = \tilde{\pi}_S(\eta)$, then $V^{D'}(0) = V^{SD'}(0|0)$ and $V^{D''}(0) < V^{SD''}(0|0)$.

Proof of Lemma OA.10. By $V^D(0) = V^{SD}(0|0) = 0$ and Lemma 8(d),

$$\begin{aligned}\phi V^{D'}(0) &= \lambda_D \Pi - \phi - c, \\ \phi V^{SD'}(0|0) &= \lambda_S V_M(\frac{\phi}{\lambda_S}) - c = \lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) (1 - e^{-1}) - \phi - c.\end{aligned}$$

Therefore $V^{D'}(0) > V^{SD'}(0|0)$ is equivalent to $(\eta(e-1) - 1) \lambda_D \Pi < c(e-1)$. When $\eta \leq 1/(e-1)$ the left-hand side is non-positive and the inequality always holds; when $\eta > 1/(e-1)$ it is equivalent to $\pi < \tilde{\pi}_S(\eta)$.

Next assume $\eta > 1/(e-1)$ and $\pi = \tilde{\pi}_S(\eta)$. With some algebra,

$$\phi^2 V^{SD''}(0|0) - \phi^2 V^{D''}(0) = \lambda_D c \left[\frac{(e-1)\eta^2}{(e-1)\eta - 1} \right],$$

which is positive since $\eta > 1/(e-1)$, so $V^{SD''}(0|0) > V^{D''}(0)$. $\square$

Lemma OA.11 (Staging at some deadline). *There exists $\tilde{\pi}_M(\eta) \geq 2\lambda_D/\lambda_S = 2/(\eta+1)$ with $\tilde{\pi}_M(1) = 1$ such that the following statements hold.*

(a) *If $1 \leq \pi < \tilde{\pi}_M(\eta)$, then $V^{D'}(u) > V^{SD'}(u|u)$ for all $u \geq 0$.*

(b) *Suppose that either (i) $\eta \leq 1/(e-1)$ and $\pi > \tilde{\pi}_M(\eta)$, or (ii) $\eta > 1/(e-1)$ and $\tilde{\pi}_S(\eta) \geq \pi > \tilde{\pi}_M(\eta)$. Then there exists $\hat{u}_1 > 0$ such that $V^{D'}(\hat{u}_1) = V^{SD'}(\hat{u}_1|\hat{u}_1)$, $V^{D''}(\hat{u}_1) < V^{SD''}(\hat{u}_1|\hat{u}_1)$, and $V^{D'}(u) > V^{SD'}(u|u)$ for all $u \in [0, \hat{u}_1)$.*

Proof of Lemma OA.11. By (B.6) and (B.7), $\phi V^{SD'}(u|u) - \phi V^{D'}(u)$ equals $\Lambda_D(x; \pi)$ of (OA.2.8) with $x = e^{-\lambda_D u / \phi}$. Since

$$\frac{\partial^2 \Lambda_D}{\partial x^2}(x; \pi) = -(\eta+1)\eta \{(\eta+1)\pi - 1\} \cdot c \cdot e^{-1} x^{\eta-1} < 0$$

for $\pi \geq 1$, the function Λ_D is strictly concave in x . Let $x^*(\pi)$ solve $\max_{0 \leq x \leq 1} \Lambda_D(x; \pi)$. The

first-order condition gives

$$x^*(\pi) = \left[\frac{\eta(\pi - 1)}{(\eta + 1)\{(\eta + 1)\pi - 1\}e^{-1}} \right]^{\frac{1}{\eta}}. \quad (\text{OA.2.12})$$

¹⁸ Define

$$h(\pi) \equiv \Lambda_D(x^*(\pi); \pi) = K \left(\frac{\pi - 1}{(\eta + 1)\pi - 1} \right)^{\frac{1}{\eta}} (\pi - 1) \cdot c - (1 - \eta) \cdot c,$$

where $K = \frac{\eta^2}{\eta + 1} \left(\frac{\eta e}{\eta + 1} \right)^{\frac{1}{\eta}}$. Then

$$h \left(\frac{2}{\eta + 1} \right) = (1 - \eta)c \left[\frac{\eta^2}{(\eta + 1)^2} \left(\frac{\eta(1 - \eta)e}{(\eta + 1)^2} \right)^{\frac{1}{\eta}} - 1 \right] < 0$$

because $\eta < 1$ and $\eta(1 - \eta)e \leq \frac{e}{4} < 1 \leq (\eta + 1)^2$. Moreover $\lim_{\pi \rightarrow \infty} h(\pi) = \infty$ and

$$h'(\pi) = \frac{K\pi(1 + \eta)c}{\pi - 1} \cdot \left(\frac{\pi - 1}{(\eta + 1)\pi - 1} \right)^{\frac{1}{\eta} + 1} > 0,$$

so there is a unique $\pi \geq 2/(\eta + 1)$ with $h(\pi) = 0$; denote it $\tilde{\pi}_M(\eta)$. At $\eta = 1$ we have $h(2/(\eta + 1)) = h(1) = 0$, so $\tilde{\pi}_M(1) = 1$.

(a) If $1 \leq \pi < \tilde{\pi}_M(\eta)$ then $0 > h(\pi) = \Lambda_D(x^*(\pi); \pi) \geq \Lambda_D(x; \pi)$ for all $0 \leq x \leq 1$, which is

$$V^{D'}(u) > V^{SD'}(u|u) \text{ for all } u \geq 0.$$

(b) Suppose first $\eta \leq 1/(e - 1)$ and $\pi > \tilde{\pi}_M(\eta)$, so $h(\pi) = \Lambda_D(x^*(\pi); \pi) > 0$. By Lemma

[OA.10](#), $\Lambda_D(1; \pi) = \phi(V^{SD'}(0|0) - V^{D'}(0)) < 0$. Concavity of Λ_D and $\frac{\partial \Lambda_D}{\partial x}(x^*(\pi); \pi) = 0$

then give $x_1 \in (x^*(\pi), 1]$ with $\Lambda_D(x_1; \pi) = 0$, $\frac{\partial \Lambda_D}{\partial x}(x_1; \pi) < 0$, and $\Lambda_D(x; \pi) < 0$ for

all $x \in (x_1, 1]$. Setting $\hat{u}_1 \equiv -(\phi/\lambda_D) \log x_1$ translates these into the three stated

conditions.

¹⁸When $\pi \leq \frac{(\eta+1)e^{-1}-\eta}{(\eta+1)^2e^{-1}-\eta}$ the maximizer is $x^*(\pi) = 1$. Since $\frac{(\eta+1)e^{-1}-\eta}{(\eta+1)^2e^{-1}-\eta} < 1$ for every $\eta > 0$, the interior solution applies whenever $\pi \geq 1$.

Suppose instead $\eta > 1/(e-1)$. By the definition of $\tilde{\pi}_S(\eta)$, if $\pi \geq \tilde{\pi}_S(\eta)$ then $\Lambda_D(1; \pi) \geq 0$, whence $h(\pi) \geq \Lambda_D(1; \pi) \geq 0$ and $\pi \geq \tilde{\pi}_M(\eta)$; therefore $\tilde{\pi}_S(\eta) \geq \tilde{\pi}_M(\eta)$. If $\tilde{\pi}_S(\eta) \geq \pi > \tilde{\pi}_M(\eta)$ then $\Lambda_D(x^*(\pi); \pi) > 0 > \Lambda_D(1; \pi)$, and the same argument applies. $\square$

Lemma OA.12 (Locating the outer threshold). *Suppose that $\eta < 1$, $\Pi > c/\lambda_D$, and either (i) $V^{D'}(\hat{u}_1) < V^{SD'}(\hat{u}_1|\hat{u}_1)$, or (ii) $V^{D'}(\hat{u}_1) = V^{SD'}(\hat{u}_1|\hat{u}_1)$ and $V^{D''}(\hat{u}_1) < V^{SD''}(\hat{u}_1|\hat{u}_1)$. Then there exists a unique $\hat{u}_2 > \hat{u}_1$ such that $V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. Moreover $V^{SD'}(u|\hat{u}_1) > V^{DSD'}(u|\hat{u}_1, u)$ for all $u \in (\hat{u}_1, \hat{u}_2)$.*

Proof of Lemma OA.12. Using (OA.2.10), (B.7) and $V^{DSD}(u|\hat{u}_1, u) = V^{SD}(u|\hat{u}_1)$, the difference $\phi V^{DSD'}(u|\hat{u}_1, u) - \phi V^{SD'}(u|\hat{u}_1)$ equals

$$\lambda_D \Pi - \lambda_S \{V_M(u + \phi/\lambda_S) + u + \phi/\lambda_S\} + (\lambda_S - \lambda_D)(V^{SD}(u|\hat{u}_1) + u).$$

Proceeding as in (A.15) and writing $y \equiv e^{-\frac{\lambda_S}{\phi}(u-\hat{u}_1)}$, this becomes

$$\begin{aligned} \tilde{\Lambda}_{SD}(y) \equiv & \frac{1-\eta}{1+\eta}c + (\lambda_S \Pi - c)e^{-1-\frac{\lambda_S}{\phi}\hat{u}_1} \left[1 + \frac{\eta}{1+\eta} \log y \right] y \\ & + \eta \left[\frac{1-\eta}{1+\eta}c - (\lambda_D \Pi - c)e^{-\frac{\lambda_D}{\phi}\hat{u}_1} \right] y. \end{aligned} \tag{OA.2.13}$$

By hypothesis $\tilde{\Lambda}_{SD}(1) = \phi V^{D'}(\hat{u}_1) - \phi V^{SD'}(\hat{u}_1|\hat{u}_1) \leq 0$. Differentiating twice,

$$\tilde{\Lambda}_{SD}''(y) = \frac{\eta}{1+\eta}(\lambda_S \Pi - c)e^{-1-\frac{\lambda_S}{\phi}\hat{u}_1} \frac{1}{y} > 0$$

since $\Pi > c/\lambda_D > c/\lambda_S$, so $\tilde{\Lambda}_{SD}$ is strictly convex; and $\lim_{y \rightarrow 0} \tilde{\Lambda}_{SD}(y) = \frac{1-\eta}{1+\eta}c > 0$. Convexity then yields a unique $y_2 \in (0, 1)$ with $\tilde{\Lambda}_{SD}(y) < 0$ on $(y_2, 1)$, $\tilde{\Lambda}_{SD}(y_2) = 0$, and $\tilde{\Lambda}_{SD}'(y_2) < 0$. Setting $\hat{u}_2 = \hat{u}_1 - \frac{\phi}{\lambda_S} \log y_2$ gives $V^{SD'}(u|\hat{u}_1) > V^{DSD'}(u|\hat{u}_1, u)$ for all $u \in (\hat{u}_1, \hat{u}_2)$ and equality at $\hat{u}_2$; and since y is decreasing in u , $\tilde{\Lambda}_{SD}'(y_2) < 0$ gives $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. $\square$

Lemma OA.13 (No return to staging). *Suppose that $\pi > 1$, $V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. Then $\lambda_S (V_M(u + \phi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)) - \phi V^{DSD'}(u|\hat{u}_1, \hat{u}_2) -$*

$c < 0$ for all $u > \hat{u}_2$.

Proof of Lemma OA.13. Differentiating (B.6) and (B.7),

$$\begin{aligned}\phi V^{SD''}(u|\hat{u}_1) &= \lambda_S (V'_M(u + \phi/\lambda_S) + 1) - \lambda_S (V^{SD'}(u|\hat{u}_1) + 1), \\ \phi V^{DSD''}(u|\hat{u}_1, \hat{u}_2) &= -\lambda_D (V^{DSD'}(u|\hat{u}_1, \hat{u}_2) + 1),\end{aligned}$$

so the hypotheses imply

$$\begin{aligned}(\lambda_S - \lambda_D)(1 + V^{SD'}(\hat{u}_2|\hat{u}_1)) &> \lambda_S(V'_M(\hat{u}_2 + \phi/\lambda_S) + 1) \\ \iff \eta(1 + V^{SD'}(\hat{u}_2|\hat{u}_1)) &> (\eta + 1) \left(\frac{\lambda_S \Pi - c}{\phi} \right) e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1}. \quad (\text{OA.2.14})\end{aligned}$$

Define $\Lambda_{DSD} : [\hat{u}_2, \infty) \rightarrow \mathbb{R}$ by

$$\Lambda_{DSD}(u) \equiv \lambda_S [V_M(u + \phi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)] - \phi V^{DSD'}(u|\hat{u}_1, \hat{u}_2) - c,$$

so that

$$\Lambda_{DSD}(u) = (\eta - 1)c - (\lambda_S \Pi - c)e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1} \cdot e^{\frac{\lambda_S}{\phi}(\hat{u}_2 - u)} + \eta \phi (V^{SD'}(\hat{u}_2|\hat{u}_1) + 1) e^{\frac{\lambda_D}{\phi}(\hat{u}_2 - u)},$$

and

$$\begin{aligned}\Lambda_{DSD}(\hat{u}_2) &= \lambda_S [V_M(\hat{u}_2 + \phi/\lambda_S) - V^{SD}(\hat{u}_2|\hat{u}_1)] - c - \phi V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) \\ &= \phi V^{SD'}(\hat{u}_2|\hat{u}_1) - \phi V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = 0.\end{aligned}$$

Writing $x \equiv e^{\frac{\lambda_D}{\phi}(\hat{u}_2 - u)}$,

$$\tilde{\Lambda}_{DSD}(x) = (\eta - 1)c - (\lambda_S \Pi - c)e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1} x^{\eta+1} + \eta \phi (V^{SD'}(\hat{u}_2|\hat{u}_1) + 1) x,$$

with $\tilde{\Lambda}_{DSD}(1) = \Lambda_{DSD}(\hat{u}_2) = 0$. By (OA.2.14),

$$\tilde{\Lambda}'_{DSD}(1) = -(\eta + 1)(\lambda_S \Pi - c)e^{-\frac{\lambda_S}{\phi}\hat{u}_2 - 1} + \eta\phi \left(V^{SD'}(\hat{u}_2|\hat{u}_1) + 1 \right) > 0,$$

and $\tilde{\Lambda}''_{DSD}(x) = -(\eta + 1)\eta(\lambda_S \Pi - c)e^{-\frac{\lambda_S}{\phi}\hat{u}_2 - 1}x^{\eta-1} < 0$. Hence $\tilde{\Lambda}'_{DSD} > 0$ on $(0, 1)$, and since $\tilde{\Lambda}_{DSD}(1) = 0$ we get $\tilde{\Lambda}_{DSD}(x) < 0$ for all $x \in (0, 1)$, which is the claim. $\square$

Proof of Proposition 9. Conditions (i) and (ii) are Lemmas OA.10 and OA.11: the former settles whether $\hat{u}_1 = 0$, and the latter constructs $\hat{u}_1 > 0$ when it is positive, together with the inequality $V^{D'} > V^{SD'}$ below it. Condition (iii) and the first half of (iv) are Lemma OA.12, whose hypotheses are exactly (ii); the second half of (iv) is the same lemma's second-order conclusion. Condition (v) is Lemma OA.13, whose hypotheses are exactly (iv).

For the final claim, let $\mathcal{V}$ be given by (OA.2.11) with these thresholds. It satisfies $\mathcal{V}(0) = 0$. It is differentiable: each branch is differentiable on its interior, and where two branches meet the one-sided derivatives agree—at $\hat{u}_2$ by (iv), and at $\hat{u}_1$ by (ii), which holds with equality whenever $\hat{u}_1 > 0$ and is vacuous otherwise, since the first branch is then empty. It is concave by Lemma OA.14. By (B.6), (B.7) and (OA.2.10), $\mathcal{J}$ vanishes on each branch when evaluated at the binding-incentive terms, and by Lemma OA.15 it is non-positive at every feasible deviation; hence $\mathcal{V}$ solves (HJB $_{\mathcal{V}}$) subject to (IC). Lemma 7 then bounds the payoff of every incentive-compatible contract delivering u by $\mathcal{V}(u)$, and Lemmas 8 and OA.9 exhibit a contract attaining it. $\square$

OA.2.2.3 Concavity and verification

Two facts complete the argument, and neither uses the knife edge.

Lemma OA.14 (Concavity). *Suppose $\pi > 1$ and $0 < \eta < 1$. Then $\mathcal{V}$ defined in (OA.2.11) is concave.*

Proof of Lemma OA.14. When $u < \hat{u}_1$, $\mathcal{V}''(u) = V^{D''}(u) < 0$ by Lemma 9(c).

When $\hat{u}_1 < u < \hat{u}_2$, the bound (B.12) still applies, and $\lambda_S < 2\lambda_D$ gives $\mathcal{V}''(u) = V^{SD''}(u|\hat{u}_1) \leq 0$.

When $u > \hat{u}_2$, differentiating (OA.2.9) twice,

$$V^{DSD''}(u|\hat{u}_1, \hat{u}_2) = - \left(\frac{\lambda_D}{\phi} \right)^2 \cdot \left(\Pi - \frac{c}{\lambda_D} - (V^{SD}(\hat{u}_2 | \hat{u}_1) + \hat{u}_2) \right) e^{-\frac{\lambda_D}{\phi}(u-\hat{u}_2)},$$

which is negative because the total surplus $V^{SD}(\hat{u}_2 | \hat{u}_1) + \hat{u}_2$ cannot exceed the first-best surplus $\Pi - c/\lambda_D$. Since the three branches are smoothly pasted at $\hat{u}_1$ and $\hat{u}_2$, $\mathcal{V}$ is concave throughout. $\square$

Part of this is worth a remark, because it is the one place where an efficiency loss makes the argument *easier* rather than harder. On the middle branch the bracketed term of (B.12) carries the factor $\lambda_S - 2\lambda_D$. At the knife edge that term vanishes and concavity follows from the remaining term alone; when $\lambda_S < 2\lambda_D$ the factor is strictly negative, so the bracket reinforces rather than offsets.

Lemma OA.15 (No profitable slack). *Suppose $\pi > 1$ and $0 < \eta < 1$. Then for every $u \geq 0$, $L^D(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$, and $L^S(u, w) \leq 0$ for all $w \geq u + \phi/\lambda_S$.*

Proof of Lemma OA.15. As in the case without efficiency loss, $\mathcal{V}(u) + u$ is increasing in u , so $\mathcal{V}'(u) \geq -1$ and $\frac{\partial L^D}{\partial R} \leq 0$. It therefore suffices to show $L^D(u, u + \phi/\lambda_D) \leq 0$ for all $u \geq 0$. From (B.6), (B.7) and (OA.2.10),

$$L^D(u, u + \phi/\lambda_D) = \begin{cases} 0, & \text{if } u \leq \hat{u}_1 \text{ or } u \geq \hat{u}_2, \\ \phi V^{DSD'}(u|\hat{u}_1, u) - \phi V^{SD'}(u|\hat{u}_1), & \text{if } u \in (\hat{u}_1, \hat{u}_2), \end{cases}$$

which is non-positive because $\hat{u}_1$ and $\hat{u}_2$ satisfy $V^{DSD'}(u|\hat{u}_1, u) < V^{SD'}(u|\hat{u}_1)$ on $(\hat{u}_1, \hat{u}_2)$ by Lemma OA.12.

Likewise, concavity of $\mathcal{V}$ gives $\frac{\partial L^S}{\partial u} \geq 0$, so it suffices to show $L^S(u, u + \phi/\lambda_S) \leq 0$ for all

$u \geq 0$. Here

$$L^S(u, u + \phi/\lambda_S) = \begin{cases} \phi V^{SD'}(u|u) - \phi V^{D'}(u), & \text{if } u \leq \hat{u}_1, \\ 0, & \text{if } u \in (\hat{u}_1, \hat{u}_2), \\ \lambda_S(V_M(u + \phi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)) & \text{if } u \geq \hat{u}_2, \\ -c - \phi V^{DSD'}(u|\hat{u}_1, \hat{u}_2), & \end{cases}$$

which is non-positive by Lemma OA.11 on the first branch and Lemma OA.13 on the third. $\square$

OA.2.3 Proof of Theorem 2

Proof of Theorem 2. (a) By Proposition 9 the optimal contract recommends the direct route on $[0, \hat{u}_1)$, the staged route on $(\hat{u}_1, \hat{u}_2)$, and the direct route on $(\hat{u}_2, \infty)$. Under binding incentive conditions promised utility declines at the constant rate ϕ , so along any contract u_t is strictly decreasing and crosses each threshold at most once. The recommended route can therefore change at most twice.

The bound holds at every efficiency level because Proposition 9 produces exactly two thresholds for every $\eta \in (0, 1)$, which in turn rests on Lemma OA.4: the derivative of the staged advantage at any point where it vanishes equals Φ , irrespective of which route is in use, and Φ is strictly decreasing. A third crossing would require Φ to change sign twice.

(b) Fix an optimal contract and let $\bar{u}$ solve (MP). Promised utility declines from $\bar{u}$ to zero, so the set of promised-utility levels visited is exactly $[0, \bar{u}]$, and the routes recommended are those prevailing on that interval. Enumerating the possible positions of $\bar{u}$ relative to the two thresholds:

If $\bar{u} = 0$ the principal does not offer a contract and the project is infeasible.

If $0 < \bar{u} \leq \hat{u}_1$, then $[0, \bar{u}]$ lies in the lower direct region and the contract is direct-only.

If $\hat{u}_1 = 0$ and $\bar{u} \leq \hat{u}_2$, then $[0, \bar{u}]$ lies in the staged region and the contract is staged-only.

If $\hat{u}_1 > 0$ and $\hat{u}_1 < \bar{u} \leq \hat{u}_2$, then $[0, \bar{u}]$ meets the lower direct region and the staged region but not the upper direct region. The contract begins on the staged route and switches to the direct route when u_t falls below $\hat{u}_1$, that is, at calendar time $T - \hat{u}_1/\phi$: a one-switch contract.

If $\hat{u}_1 = 0$ and $\bar{u} > \hat{u}_2$, then $[0, \bar{u}]$ meets the staged region and the upper direct region but not the lower one. The contract begins on the direct route and switches to the staged route when u_t falls below $\hat{u}_2$, and stays there: a reversed one-switch contract.

If $\hat{u}_1 > 0$ and $\bar{u} > \hat{u}_2$, then $[0, \bar{u}]$ meets all three regions and the contract switches twice, at calendar times $T - \hat{u}_2/\phi$ and $T - \hat{u}_1/\phi$: a two-switch contract.

These are the six rows of Table 1, and they exhaust the possibilities since $\bar{u}$ must fall in one of them. In every case the reward schedules follow from binding incentive conditions exactly as in Lemma 8, since those conditions do not involve η . $\square$

Remark 4. The reversed one-switch contract arises only at intermediate efficiency losses. It requires $\hat{u}_1 = 0$ together with $\bar{u} > \hat{u}_2$. When staging is technologically innocuous the second condition fails, since $\hat{u}_2 = \infty$; under a small efficiency loss it fails as well, since Lemma OA.17 gives $\hat{u}_2 \geq \bar{u}$. Under a large efficiency loss the first condition fails instead: $\hat{u}_1 = 0$ requires $V^{SD'}(0|0) \geq V^{D'}(0)$, which by Lemma OA.10 is impossible when $\eta \leq 1/(e-1)$, and $\underline{\eta} \leq 1/(e-1)$ by definition.

OA.2.4 Extreme Efficiency Losses

Theorem 2 classifies the optimal contract by the position of $\bar{u}$ relative to $\hat{u}_1$ and $\hat{u}_2$, but leaves all three quantities implicit. This subsection locates them in terms of the primitives at the two extremes of the efficiency loss. Three lemmas do the work: two compare $\bar{u}$ with $\hat{u}_2$, in opposite directions and under opposite hypotheses, and the third compares $\bar{u}$ with $\hat{u}_1$. Since $\mathcal{V}$ is concave, each comparison reduces to a sign condition on $\mathcal{V}'$ evaluated at a threshold, which is what the three lemmas exploit.

Throughout, $\tilde{\pi}_S(\eta)$ of Lemma OA.10 is the return above which the staged route survives to the deadline ($\hat{u}_1 = 0$), and $\tilde{\pi}_M(\eta)$ of Lemma OA.11 the return above which the staged route is used at all ($\hat{u}_1 < \infty$).

Lemma OA.16 (Reaching the outer region). *Suppose $\pi > \tilde{\pi}_M(\eta)$ and $\eta \leq c/(c + \phi)$. Then $\hat{u}_2 < \bar{u}$.*

Proof of Lemma OA.16. Since $\mathcal{V}$ is strictly concave, $\hat{u}_2 < \bar{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_2) = V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$, and by (OA.2.10) the last inequality is

$$\lambda_D (\hat{u}_2 + V^{SD}(\hat{u}_2|\hat{u}_1)) < \lambda_D \Pi - c - \phi. \quad (\text{OA.2.15})$$

Since $V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{DSD}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = V^{SD}(\hat{u}_2|\hat{u}_1)$, equations (B.6) and (B.7) give

$$\lambda_D (\Pi - \hat{u}_2 - V^{SD}(\hat{u}_2|\hat{u}_1)) = \lambda_S (V_M(\hat{u}_2 + \phi/\lambda_S) + \hat{u}_2 + \phi/\lambda_S) - \lambda_S (V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2).$$

Substituting (B.1),

$$\begin{aligned} (\lambda_S - \lambda_D) (V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2) &= \lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) \left(1 - e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1} \right) - \lambda_D \Pi \\ \iff \eta \lambda_D (V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2) &= \eta \lambda_D \Pi - c - (\lambda_S \Pi - c) e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1}, \end{aligned}$$

so that (OA.2.15) is equivalent to

$$\eta(c + \phi) - c < (\lambda_S \Pi - c) e^{-\frac{\lambda_S}{\phi} \hat{u}_2 - 1}.$$

The right-hand side is strictly positive because $\Pi > c/\lambda_D > c/\lambda_S$, and the left-hand side is non-positive because $\eta \leq c/(c + \phi)$. The inequality therefore holds, and $\hat{u}_2 < \bar{u}$. $\square$

Lemma OA.17 (Not reaching the outer region). *Suppose $\eta > \sqrt{c/(c + \phi)}$. Then $\hat{u}_2 \geq \bar{u}$.*

Proof of Lemma OA.17. Following the computation in the proof of Lemma OA.16, $\hat{u}_2 \geq \bar{u}$ is equivalent to

$$\bar{y} \equiv \frac{(\eta - 1)c + \eta\phi}{(\lambda_S\Pi - c)e^{-\frac{\lambda_S}{\phi}\hat{u}_1 - 1}} \geq e^{\frac{\lambda_S}{\phi}(\hat{u}_1 - \hat{u}_2)} = \hat{y}_2, \quad (\text{OA.2.16})$$

where, by the proof of Lemma OA.12, $\hat{y}_2$ is the root of $\tilde{\Lambda}_{SD}(y) = 0$ other than 1. Since $\hat{u}_2 \geq \hat{u}_1$, the inequality holds trivially if $\bar{y} \geq 1$, so assume $\bar{y} < 1$. Because $\tilde{\Lambda}_{SD}$ is strictly convex with $\tilde{\Lambda}_{SD}(1) \leq 0$, the inequality $\tilde{\Lambda}_{SD}(\bar{y}) \leq 0$ implies (OA.2.16).

Writing $\hat{x}_1 \equiv e^{-\frac{\lambda_D}{\phi}\hat{u}_1}$ and using (OA.2.8), the function $\tilde{\Lambda}_{SD}$ of (OA.2.13) can be rewritten as

$$\tilde{\Lambda}_{SD}(y) = \frac{1 - \eta}{1 + \eta}c - \Lambda_D(\hat{x}_1)y + \left[-\frac{1 - \eta}{1 + \eta}c + \frac{\eta}{1 + \eta}(\lambda_S\Pi - c)e^{-\frac{\lambda_S}{\phi}\hat{u}_1 - 1} \log y \right] y,$$

where $\hat{x}_1$ is chosen so that $\Lambda_D(\hat{x}_1) \geq 0$. Substituting the definition of $\bar{y}$,

$$\tilde{\Lambda}_{SD}(\bar{y}) = \frac{1 - \eta}{1 + \eta}c(1 - \bar{y}) - \Lambda_D(\hat{x}_1)\bar{y} + \frac{\eta}{1 + \eta}((\eta - 1)c + \eta\phi) \log \bar{y}.$$

It therefore suffices to show that

$$G(y) \equiv \frac{1 - \eta}{1 + \eta}c(1 - y) - \Lambda_D(\hat{x}_1)y + \frac{\eta}{1 + \eta}((\eta - 1)c + \eta\phi) \log y$$

satisfies $G(y) \leq 0$ for all $y < 1$. Since $\eta \geq \sqrt{c/(c + \phi)} > c/(c + \phi)$,

$$G''(y) = -\frac{\eta}{1 + \eta} \left(\frac{(\eta - 1)c + \eta\phi}{y^2} \right) < 0,$$

and, using $\eta \geq \sqrt{c/(c + \phi)}$ together with $\Lambda_D(\hat{x}_1) \geq 0$,

$$G'(1) = -\Lambda_D(\hat{x}_1) + \frac{1}{1 + \eta}((\eta^2 - 1)c + \eta^2\phi) < 0.$$

Finally $G(1) = -\Lambda_D(\hat{x}_1) \leq 0$, so concavity gives $G(y) \leq G(1) + G'(1)(1 - y) \leq 0$ for all $y < 1$. Hence $\tilde{\Lambda}_{SD}(\bar{y}) \leq 0$ and $\hat{u}_2 \geq \bar{u}$. $\square$

Lemma OA.18 (Reaching the inner threshold). *Suppose $\eta > \bar{\eta}$. There exists $\tilde{\pi}_D(\eta) \in (\tilde{\pi}_M(\eta), \tilde{\pi}_S(\eta))$ such that $\hat{u}_1 < \bar{u}$ if and only if $\pi > \tilde{\pi}_D(\eta)$.*

Proof of Lemma OA.18. Since $\eta > \bar{\eta} \geq 1/(e-1)$, the threshold $\tilde{\pi}_S(\eta)$ exists. If $\pi \geq \tilde{\pi}_S(\eta)$ then $\hat{u}_1 = 0$ by Lemma OA.10, and

$$\tilde{\pi}_S(\eta) = \frac{e-1}{(e-1)\eta-1} \geq \frac{e-1}{e-2} > 2 \geq \frac{c+\phi}{c} = \pi_F,$$

so the project is feasible, $\bar{u} > 0$, and $\bar{u} > \hat{u}_1$.

Now suppose $\tilde{\pi}_M(\eta) < \pi < \tilde{\pi}_S(\eta)$. Since $\mathcal{V}$ is strictly concave, $\hat{u}_1 < \bar{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_1) = V^{D'}(\hat{u}_1)$, which in turn is equivalent to

$$\frac{\phi}{\lambda_D \Pi - c} < e^{-\frac{\lambda_D}{\phi} \hat{u}_1} = \hat{x}_1, \quad (\text{OA.2.17})$$

where $\hat{x}_1$ is the root of $\Lambda_D(x) = 0$ identified in Lemma OA.11. The hypothesis $\tilde{\pi}_M(\eta) < \pi < \tilde{\pi}_S(\eta)$ gives $\Lambda_D(1) < 0 < \Lambda_D(x^*)$ with x^* as in (OA.2.12).

Two cases satisfy (OA.2.17): (i) $x^* \geq \frac{\phi}{\lambda_D \Pi - c}$; and (ii) $\frac{\phi}{\lambda_D \Pi - c} > x^*$ together with $\Lambda_D(\frac{\phi}{\lambda_D \Pi - c}) < 0$. Case (i) is equivalent to $\Lambda'_D(\frac{\phi}{\lambda_D \Pi - c}) < 0$, which by algebra is

$$\frac{(\eta+1)\pi-1}{(\pi-1)^{\eta+1}} < \frac{\eta e}{\eta+1} \phi^{-\eta} c^\eta, \quad (\text{OA.2.18})$$

and case (ii) is equivalent to $\Lambda'_D(\frac{\phi}{\lambda_D \Pi - c}) \geq 0$ and $\Lambda_D(\frac{\phi}{\lambda_D \Pi - c}) < 0$, that is

$$\frac{\eta e}{\eta+1} \phi^{-\eta} c^\eta \leq \frac{(\eta+1)\pi-1}{(\pi-1)^{\eta+1}} < (\eta(c+\phi)-c) e \phi^{-\eta-1} c^\eta. \quad (\text{OA.2.19})$$

By the proof of Lemma OA.11, the condition $\pi > \tilde{\pi}_M(\eta)$ is equivalent to

$$\frac{(\eta+1)\pi-1}{(\pi-1)^{\eta+1}} < \left(\frac{\eta^2}{1-\eta^2} \right)^\eta \frac{\eta e}{1+\eta}. \quad (\text{OA.2.20})$$

Using $\eta > \sqrt{c/(c+\phi)}$, simple algebra gives

$$\frac{\eta e}{\eta+1} \phi^{-\eta} c^\eta < (\eta(c+\phi) - c) e \phi^{-\eta-1} c^\eta < \left(\frac{\eta^2}{1-\eta^2} \right)^\eta \frac{\eta e}{1+\eta},$$

so the single inequality

$$\frac{(\eta+1)\pi - 1}{(\pi-1)^{\eta+1}} < (\eta(c+\phi) - c) e \phi^{-\eta-1} c^\eta$$

implies each of (OA.2.18), (OA.2.19) and (OA.2.20). Let $\tilde{\pi}_D(\eta)$ be the value of π at which its two sides are equal. Then $\tilde{\pi}_D(\eta) > \tilde{\pi}_M(\eta)$, since $\pi < \tilde{\pi}_M(\eta)$ implies $\pi < \tilde{\pi}_D(\eta)$, and $\hat{u}_1 < \bar{u}$ if and only if $\pi > \tilde{\pi}_D(\eta)$. $\square$

The two bounds on η are what separate the regimes: Lemma OA.17 requires η above $\sqrt{c/(c+\phi)}$ and Lemma OA.16 requires η at or below $c/(c+\phi)$. Since $c/(c+\phi) < 1$, the square root is the larger of the two, and the interval between them is exactly the band that neither proposition covers.

Proof of Proposition 4. Suppose $\eta \in (\bar{\eta}, 1)$. Since $\bar{\eta} \geq \sqrt{c/(c+\phi)}$, Lemma OA.17 gives $\hat{u}_2 \geq \bar{u}$, so the contract never enters the upper direct region and the last two rows of Table 1 are unavailable. The third row is then the only possibility when $\hat{u}_1 = 0$, and the second and fourth the only possibilities when $\hat{u}_1 > 0$; in particular at most one switch occurs.

Since $\bar{\eta} \geq 1/(e-1)$, the threshold $\tilde{\pi}_S(\eta)$ exists. When $\pi > \tilde{\pi}_S(\eta)$ we have $\hat{u}_1 = 0$, so by Theorem 2(b) the contract is staged-only. When $\pi \leq \tilde{\pi}_S(\eta)$ we have $\hat{u}_1 > 0$, and Lemma OA.18 supplies $\tilde{\pi}_D(\eta) < \tilde{\pi}_S(\eta)$ with $\hat{u}_1 < \bar{u}$ precisely when $\pi > \tilde{\pi}_D(\eta)$. The contract is therefore a one-switch contract for $\pi \in (\tilde{\pi}_D(\eta), \tilde{\pi}_S(\eta))$ and direct-only for $\pi \in (\pi_F, \tilde{\pi}_D(\eta))$.

Finally $\tilde{\pi}_D(\eta) \geq \pi_F$: when $\pi < \pi_F$ the project is infeasible, so $\bar{u} = 0 \leq \hat{u}_1$, and Lemma OA.18 then gives $\pi \leq \tilde{\pi}_D(\eta)$. $\square$

Proof of Proposition 5. Suppose $\eta < \underline{\eta}$. Since $\underline{\eta} \leq c/(c+\phi)$, Lemma OA.16 applies whenever $\pi > \tilde{\pi}_M(\eta)$ and gives $\hat{u}_2 < \bar{u}$: the contract does reach the upper direct region.

Since $\underline{\eta} \leq 1/(e-1)$, Lemma OA.10 shows that the staged route is never preferred at the deadline, so $\hat{u}_1 > 0$ whenever the staged route is used at all. Two cases remain.

If $\pi \leq \tilde{\pi}_M(\eta)$, the staged route is never used and $\hat{u}_1 = \hat{u}_2 = \infty$; the second row of Table 1 applies and the contract is direct-only.

If $\pi > \tilde{\pi}_M(\eta)$, then $0 < \hat{u}_1 < \hat{u}_2 < \bar{u}$, which is the sixth row of Table 1: a two-switch contract with intermediate deadlines $T - \hat{u}_2/\phi$ and $T - \hat{u}_1/\phi$. $\square$

Remark 5 (The intermediate band). Propositions 4 and 5 leave $\eta \in [\underline{\eta}, \bar{\eta}]$ uncharacterized, because neither Lemma OA.17 nor Lemma OA.16 applies there and the comparison between $\bar{u}$ and $\hat{u}_2$ can go either way. Theorem 2 continues to hold, so the optimal contract still takes one of the forms in Table 1; what is not available is a closed-form map from (η, π) to forms. The band is not merely a technical residue: by Remark 4 it is the only region in which the reversed one-switch contract arises.